Chapter 5

Analysis

The first task is to define the field of real numbers $\mathbb{R}$, having already constructed the rational numbers $\mathbb{Q} = \mathbf{Frac}(\mathbb{Z})$. Typically this is done by some axiomatisation as an ordered field containing $\mathbb{Q}$ for which every bounded subset has a supremum (**Dedekind Completeness**). We show $\mathbb{R}$ is unique such ordered field.

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5.1 Real Numbers

Definition 5.1.1 (Ordered Ring / Field)
*An **ordered ring** is a ring $(A, +, \cdot)$ such that the additive group $(A, +)$ is an ordered abelian group with ordering $\leq$ which also satisfies*

$$x, y \geq 0 \implies x \cdot y \geq 0$$

*An **ordered field** is a field with the structure of an ordered ring.*

Proposition 5.1.2 (Extension to Field of Fractions)
Let A be an ordered ring with no zero-divisors. The $K = \mathrm{Frac}(A)$ is an ordered ring with the order given by

$$\frac{x}{y} \leq \frac{w}{z} \iff xz \leq wy$$

Example 5.1.3
$\mathbb{Z}$ is an ordered ring and $\mathbb{Q} := \mathrm{Frac}(\mathbb{Z})$ is an ordered field.

Proposition 5.1.4 (Trichotomy Law)
Let G be an ordered abelian group. Then G is the disjoint union of $\{0\}$, G^+ and G^-.

More generally precisely one of $x = y$, $x < y$ and $x > y$ holds.

Lemma 5.1.5 (Elementary Properties)
Let G be an ordered abelian group. Then

a) $x \in G^+ \iff -x \in G^-$
b) $x \in G^+$ and $x \leq y \implies y \in G^+$
c) $x \in G^-$ and $y \leq x \implies x \in G^-$
d) $x, y \in G^+ \implies x + y \in G^+$
e) $x, y \in G^- \implies x + y \in G^-$
f) $x, y \in G^+ \implies xy \in G^+$

Proof.
a) $x \in G^+ \implies 0 \leq x \implies -x \leq 0$. Furthermore $x \neq 0 \implies -x \neq 0$ by assumption so $-x \in G^-$.
b) By transitivity $0 \leq y$. Suppose $y = 0$ then $0 \leq x \leq 0$ so $x = 0$ a contradiction.
c) Similarly
d) Follows from b) since $0 \leq x \implies 0 \leq y \leq x + y$.
e) Follows from c) since $x \leq 0 \implies x + y \leq y \leq 0$.
f) If $xy = 0 \implies x = 0$ or $y = 0$ which is a contradiction. Therefore $xy > 0$.

■

Proposition 5.1.6 (Ordered Rings are Torsion Free)
Let A be a non-zero ordered ring. Then the canonical map $\mathbb{Z} \to A$ is injective, in other words A has characteristic 0.

In particular every ordered field K contains $\mathbb{Q}$ as a subring.

Proof. Follows immediately by induction, since if $n.1 = 0$ then $(n-1) > n \implies 1 < 0 \implies -1 > 0 \implies 1 = -1 * -1 < 0$ a contradiction. ■

Proposition 5.1.7 (Archimedean Field)
Let K be an ordered field. Then the following are equivalent

a) $\mathbb{Q}$ *is dense in K i.e. for all $x, y \in K$ there exists $z \in \mathbb{Q}$ such that $x < z < y$*
b) For all $x, y \in K$ there exists $n \in \mathbb{N}$ such that $ny > x$

*In this case we say K is **Archimedean**.*

In particular for every $\varepsilon \in K^+$ there exists $n \in \mathbb{N}^+$ such that $\frac{1}{n} < \varepsilon$.

Proof. Suppose the second property is satisfied then choose $n > \frac{1}{y-x}$. By definition $0 < \frac{1}{n} < y - x$. Let

$$S = \left\{ m \in \mathbb{Z} \;\middle|\; \frac{m}{n} \leq x \right\}$$

By the Archimedean property (applied to $\frac{1}{n}$ and $-x$) S is non-empty. By the well-ordering principle it has a maximal element. By maximality $\frac{m+1}{n} > x$. Furthermore

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$$x < \frac{m+1}{n} \leq x + \frac{1}{n} < x + (y - x) = y$$

and therefore $z \coloneqq \frac{m+1}{n}$ satisfies the required property.

Conversely suppose K satisfies the first property and assume wlog that $x, y > 0$. Choose $\frac{m}{n}$ such that

$$0 < \frac{m}{n} < \frac{y}{x}$$

which implies $ny > mx \geq x$. ∎

Definition 5.1.8 (Absolute Value)
*Let G be an **ordered abelian group**. Define the absolute value as*

$$|x| \coloneqq \begin{cases} x & x \in G^+ \\ -x & x \in G^- \\ 0 & x = 0 \end{cases}$$

Then this satisfies the following properties

a) $|x| \geq 0$
b) $|-x| = |x|$
c) $|x| = 0 \iff x = 0$
d) $||x| - |y|| \leq |x + y| \leq |x| + |y|$

Proposition 5.1.9 (Topology of Ordered Abelian Group)
Let G be an ordered abelian group. Then $U \subset G$ is open in the order topology iff

$$\forall x \in U \exists \varepsilon \in G^+ \text{ s.t. } |y - x| < \varepsilon \implies y \in U$$

Proof. Let $\mathcal{B}$ be the basis defined in (4.1.120). Recall (...) that U is open iff for all $x \in U$ there exists $V \in \mathcal{B}$ such that $x \in V \subseteq U$. If $U \subset G$ satisfies the given condition holds then evidently $x \in \{g \in G \mid y - \varepsilon < g < y + \varepsilon\}$ which is an element of $\mathcal{B}$. Conversely if $x \in V$ with $V = \{g \in G \mid y < g < z\}$, then we may define $\varepsilon = \min(z - x, x - y)$. ∎

5.1.1 Sequences in an Ordered Field

Proposition 5.1.10
The order topology on an ordered field K is Hausdorff.

Proposition 5.1.11 (Convergent Sequence)
Let K be an ordered field and (a_n) in K a sequence. Then the following are equivalent

a) $a_n \to \alpha$ *in the sense of (4.1.49)*
b) For all $\varepsilon \in K^+$ there exists $N(\varepsilon) \in \mathbb{N}$ such that $n \geq N(\varepsilon) \implies |a_n - \alpha| < \varepsilon$

As K is Hausdorff limits are unique. When K is Archimedean it is sufficient to consider the case $\varepsilon \coloneqq \frac{1}{N}$ for all positive integers N.

Definition 5.1.12 (Convergent Sequence)
*Let K be an ordered field. We say that a sequence $(a_n)_{n \in \mathbb{N}^+}$ is **cauchy** if for all $\varepsilon > 0$ there exists $N(\varepsilon) > 0$ such that*

$$\forall m, n \in \mathbb{N} : (m, n \geq N(\varepsilon) \implies |a_n - a_m| < \varepsilon)$$

If K is Archimedean then it is sufficient to demonstrate this only for ε of the form $\frac{1}{N}$.

Proposition 5.1.13
Let K be an ordered field. Then every convergent sequence is cauchy.

Proof. Given $\varepsilon > 0$ then $\frac{\varepsilon}{2} > 0$. Therefore there exists N such that

$$n \geq N \implies |a_n - \alpha| < \frac{\varepsilon}{2}$$

Then by the triangle inequality

$$n, m \geq N \implies |a_n - a_m| < \varepsilon$$

∎

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Definition 5.1.14 (Sequentially Completeness)
*We say an ordered field K is **sequentially complete** if every cauchy sequence is convergent.*

Proposition 5.1.15 (Completeness)
Let K be an ordered field. The following conditions are equivalent

a) Every non-empty subset $X \subset K$ bounded above admits a supremum $\sup X \in K$
b) Every non-empty subset $X \subset K$ bounded below admits an infimum $\inf X \in K$

*In this case we say that K is **complete**.*

Proof. $\inf X = -\sup(-X)$ and vice versa. ∎

Lemma 5.1.16
*Let K be a complete ordered field. Then K is also **Archimedean***

Proof. Suppose K is not Archimedean then $\mathbb{Z}^+$ is bounded above and in particular has a supremum α. Then for every $n \in \mathbb{Z}^+$ we have $n+1 \leq \alpha \implies n \leq \alpha - 1$, which contradicts maximality. ∎

Lemma 5.1.17
Let $X \subset K$ be a subset of an Archimedean ordered field which is "upwards closed" and bounded below (resp. "downwards closed" and bounded above).

Then for every $\varepsilon \in K^+$ there exists an $x \in X$ such that $x - \varepsilon \notin X$ (resp. $x + \varepsilon \notin X$)

Proof. By (5.1.7) there is some $n \in \mathbb{N}^+$ such that $0 < \frac{1}{n} < \varepsilon$. Consider the set

$$\mathcal{S} \coloneqq \left\{ k \in \mathbb{Z} \;\middle|\; \frac{k}{n} \in X \right\}$$

As X is bounded below then so is $\mathcal{S}$. By the well ordering principle it has a minimal element, say k. Define $x \coloneqq \frac{k}{n}$ and then $x - \varepsilon < x - \frac{1}{n} = \frac{k-1}{n}$ which by minimality is not in X, whence neither is $x - \varepsilon$. ∎

Lemma 5.1.18
Let $X \subset K$ be a non-empty subset for which $x \coloneqq \sup X$ exists. Then there exists a sequence (a_n) in X such that $a_n \uparrow x$.

Proof. For every $n \in \mathbb{N}^+$ there exists some $a_n \in X$ such that $a_n > x - \frac{1}{n}$ (otherwise $x - \frac{1}{n}$ would be a smaller upper bound). Then evidently a_n is increasing. For every $\varepsilon \in K^+$ there exists some N such that $0 < \frac{1}{N} < \varepsilon$ by (5.1.7). Consequently

$$n \geq N \implies a_n > x - \frac{1}{N} \implies x \geq a_n > x - \varepsilon \implies |a_n - x| < \varepsilon$$

∎

Definition 5.1.19 (Extended Real Line)
Let K be an ordered field and define the set

$$K^\sharp \coloneqq K \cup \{-\infty\} \cup \{\infty\}$$

with the obvious total ordering and induced order topology. $K \subset K^\sharp$ has the subspace topology and every open subset is of the form

$$U, U \cup (x, \infty], U \cup [-\infty, x), U \cup [-\infty, x) \cup (y, \infty] \; .$$

If $X \subset K$ is a subset which is unbounded above (resp. below) then define $\sup X$ (resp. $\inf X$) to be ∞ (resp. $-\infty$).

Proposition 5.1.20 (Convergence to infinity)
Let K be an ordered field and $a_n \in K^\sharp$ a sequence. Then the following are equivalent

a) $a_n \to \infty$
b) For all $\lambda \in K$ there exists N such that $n \geq N \implies a_n \geq \lambda$

A similar statement holds for $-\infty$.

Lemma 5.1.21 (Limit of Bounded Sequence)
Let $a_n \leq b_n$ be sequences in $K^\sharp$ converging to α and β respectively. Then $\alpha \leq \beta$.

In particular if $a_n \downarrow \alpha$ then $\alpha \leq a_n$.

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Proof. If $\beta = \infty$ then the result is vacuously true. If $\beta = -\infty$ then evidently $\alpha = -\infty$ also. Therefore we may assume that β is finite without loss of generality.

Suppose $\alpha > \beta$ and define $\varepsilon = (\alpha - \beta)$. By assumption there exists some N such that

$$n \geq N \implies |a_n - \alpha| < \frac{\varepsilon}{2} \wedge |b_n - \beta| < \frac{\varepsilon}{2}$$

Then $a_n > \frac{\alpha+\beta}{2}$ and $b_n < \frac{\alpha+\beta}{2}$ which is a contradiction. ∎

Proposition 5.1.22 (Subsequence of a Convergent Sequence)
Let K be an ordered field, (a_n) a sequence in $K^\sharp$ and $(n_k)_{k\in\mathbb{N}^+}$ a strictly increasing sequence in $\mathbb{N}^+$. If $a_n \to \alpha$ then $a_{n_k} \to \alpha$ as $k \to \infty$.

Proof. By induction $n_k \geq k$ for all k. Let V be a neighbourhood of α in $K^\sharp$, then by (4.1.49) there exists N such that $n \geq N \implies a_n \in V$. Consequently

$$k \geq N \implies n_k \geq k \geq N \implies a_{n_k} \in V$$

and the result follows. ∎

5.1.2 Sequential Completeness

Lemma 5.1.23 (Monotone Convergence Theorem)
Let K be a complete ordered field and (a_n) an increasing (resp. decreasing) sequence in $K^\sharp$. Then $a_n \to \sup\{a_n\}$. This is finite if and only if a_n is bounded above (resp. below).

Proof. Suppose first that (a_n) is bounded above and define $\alpha := \sup\{a_n\}$. Then for every $\varepsilon \in K^+$ there exists N such that $a_N > \alpha - \varepsilon$ (for otherwise $\alpha - \varepsilon$ would be an upper bound). By assumption $n \geq N \implies \alpha - \varepsilon < a_n \leq \alpha$ whence $|\alpha - a_n| < \varepsilon$. Therefore $a_n \to \alpha$.

The case that (a_n) is unbounded above is trivial. ∎

Proposition 5.1.24 (Complete $\implies$ Sequentially Complete)
*Let K be an ordered field which is **complete**. Then it is **sequentially complete**.*

Proof. Let (a_n) be a cauchy sequence. There exists N such that $n \geq N \implies |a_n - a_N| < 1 \implies |a_n| \leq 1 + |a_N|$. Then $|a_n|$ is bounded by $\max(|a_1|, \cdots, |a_N| + 1)$. Consider the sequence

$$a_n^- := \inf_{k \geq n} a_k$$

Then a_n^- is bounded and increasing. Therefore $a_n^- \uparrow \alpha := \sup a_n^- < \infty$ by (5.1.23). For every $\varepsilon \in K^+$ there exists N_1 such that $n \geq N_1 \implies \alpha - a_n^- < \varepsilon$. There exists N_2 such that $n, m \geq N_2 \implies |a_n - a_m| < \varepsilon$. By definition of a_n^-, for every $n \geq N_1$ there exists $m \geq N_1$ such that $|a_m - a_n^-| < \varepsilon$. Consequently every $n \geq \max(N_1, N_2)$ there is some m such that

$$|\alpha - a_n| \leq |\alpha - a_n^-| + |a_n - a_n^-| \leq |\alpha - a_n^-| + |a_n - a_m| + |a_m - a_n^-| < 3\varepsilon$$

As this is independent of m we see that $a_n \to \alpha$ as required. ∎

Proposition 5.1.25 (Sequentially Complete $\implies$ Complete)
*Let K be an Archimedean ordered field which is **sequentially complete**. Then K is **complete**.*

Proof. Let $X \subset K$ be a non-empty subset bounded above, and $\mathcal{U}$ the set of upper bounds for X, which in particular is bounded below. By assumption $\mathcal{U}$ is non-empty. For each $n \in \mathbb{N}^+$ we may choose $a_n \in \mathcal{U}$ such that $a_n - \frac{1}{2^n} \notin \mathcal{U}$ by (5.1.17). We may assume without loss of generality that a_n is decreasing by setting $a_n := \min_{n' \leq n} a_{n'}$. We argue that

$$a_n - a_{n+1} < \frac{1}{2^n}$$

for otherwise we see that $a_{n+1} \leq a_n - \frac{1}{2^n}$ which implies $a_{n+1} \notin \mathcal{U}$. Therefore by the triangle inequality applied repeatedly

$$m \geq n \implies |a_n - a_m| \leq \sum_{i=n}^{m-1} \frac{1}{2^i} < \frac{1}{2^{n-1}} < \frac{1}{n-1}$$

By the Archimedean property we see that the sequence is cauchy and therefore convergences to $\alpha \in K$. By (5.1.21) $\alpha \leq a_n$. We claim that $\alpha \in \mathcal{U}$. For suppose not then there exists $x \in X$ such that $x > \alpha$. Then there exists n such

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that $a_n - \alpha < x - \alpha \implies a_n < x$ which contradicts $a_n \in \mathcal{U}$. Suppose $\exists \alpha' \in \mathcal{U}$ such that $\alpha' < \alpha$. Then for some n we have $\frac{1}{n} < \alpha - \alpha'$ therefore $a_n - \frac{1}{2^n} \geq a_n - \frac{1}{n} \geq \alpha - \frac{1}{n} > \alpha'$. This would imply $a_n - \frac{1}{2^n} \in \mathcal{U}$, a contradiction. Therefore $\alpha' \geq \alpha$ and $\alpha = \sup X$ as required. ∎

5.1.3 Uniqueness of Reals

Lemma 5.1.26
Let K be a complete ordered field. Then every $x \in K$ satisfies

$$x = \sup\left\{\frac{m}{n} \in \mathbb{Q} \;\middle|\; \frac{m}{n} \leq x\right\}$$

Similarly for every $x \in K$ there is a sequence $a_n \in \mathbb{Q}$ such that $a_n \uparrow x$.

Proof. By assumption the supremum α exists. By definition x is an upper bound for the set so $\alpha \leq x$. Suppose $\alpha < x$ then by (5.1.7) there exists $\frac{m}{n}$ such that $\alpha < \frac{m}{n} < x$. On the other hand by definition $\frac{m}{n} \leq \alpha$, which is a contradiction. Therefore $\alpha = x$ as required.

The final statement follows from (5.1.18). ∎

Lemma 5.1.27
Let $\tau : K_1 \hookrightarrow K_2$ be an embedding of ordered fields with K_2 Archimedean. Suppose $x = \sup X$ for some subset $X \subset K_1$. Then $\tau(x) = \sup \tau(X)$.

Proof. Clearly $x' \in X \implies x' \leq x \implies \tau(x') \leq \tau(x) \implies \tau(x) \in \tau(X)^\uparrow$. Suppose $y \in \tau(X)^\uparrow$ and $y < \tau(x)$. Then by (5.1.7) there exists $z \in \mathbb{Q}$ such that $y < z < \tau(x)$. By the order preserving property we have $z \in X^\uparrow$ and $z < x$ which is a contradiction. Therefore $y \in \tau(X)^\uparrow \implies y \geq \tau(x)$ and $\tau(x) = \sup \tau(X)$. ∎

Proposition 5.1.28 (Uniqueness of Reals)
Let K_1, K_2 be complete ordered fields. Then there exists a unique ordered field embedding $K_1 \hookrightarrow K_2$, which is an isomorphism.

Proof. Let $\tau, \tau' : K_1 \to K_2$ be order embeddings. Then they agree on $\mathbb{Q}$. By (5.1.26) every x is the supremum of a set $X \subset \mathbb{Q}$. Then uniqueness follows from (5.1.27). The same result shows it is surjective.

For existence we claim that

$$\tau(x) := \sup\left\{\frac{m}{n} \cdot 1_{K_2} \;\middle|\; \frac{m}{n} \cdot 1_{K_1} \leq x\right\}$$

is a well-defined, order-preserving field isomorphism. Firstly x is bounded by some integer N so the supremum is well-defined. Similarly suppose $x' > x$ then by (5.1.7) we have $x < \frac{m}{n} < x'$. This shows that $\tau(x) \leq \frac{m'}{n'} \leq \tau(x')$ as required, whence τ is order preserving.

By (5.1.26) there exists $a_n, b_n \in \mathbb{Q}$ such that $a_n \uparrow x$, $b_n \uparrow y$ and by the triangle inequality $a_n + b_n \uparrow x + y$. We may show using the definition of τ that $a_n \uparrow \tau(x), b_n \uparrow \tau(y), a_n + b_n \uparrow \tau(x + y)$. Uniqueness of limits shows that $\tau(x + y) = \tau(x) + \tau(y)$. Similarly we may show that $\tau(xy) = \tau(x)\tau(y)$. ∎

5.1.4 Existence of Reals

Proposition 5.1.29 (Construction of the Reals)
There exists a (sequentially) complete ordered field $\mathbb{R}$ which is unique up to a unique order-preserving isomorphism.

*Furthermore it has characteristic zero and the subfield $\mathbb{Q}$ is **order-dense** (5.1.7) and **sequentially dense** (5.1.26).*

*It is **Archimedean** in the sense that for all pairs $x, y \in \mathbb{R}$ there exists $n \in \mathbb{Z}$ such that $nx > y$. Equivalently for all $\varepsilon > 0$ there exists n such that $\frac{1}{n} < \varepsilon$.*

The canonical (order) topology has base the open intervals (a, b), where a, b may be taken to be rational. In particular the order topology on $\mathbb{R}$ is second countable.

Proof. TODO. ∎

5.1.5 n-th Root

For technical reasons it is convenient to introduce the n-th root function at an early stage in an elementary fashion. Other special functions may be defined by similar considerations but it is convenient to wait until there is more machinery available.

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Lemma 5.1.30 (Binomial Theorem)
Let K be a field of characteristic 0 and $x, y \in K$. Then

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$$

where $\binom{n}{k}$ is a positive integer given by the formula

$$\binom{n}{k} := \frac{n(n-1)\cdots(n-k+1)}{k(k-1)\cdots 1}$$

In particular if K is an ordered field and $x \geq 0$ then for every $0 \leq k \leq n$

$$(1+x)^n \geq \binom{n}{k} x^k$$

Lemma 5.1.31
Let K be a field of characteristic zero, $a, b \in K$ and n a positive integer. Then

$$a^n - b^n = (a-b)(a^{n-1}b + a^{n-2}b^2 + \cdots + ab^{n-2} + b^{n-1})$$

In particular if $a > b \geq 0$ then

$$(a-b)nb^{n-1} < a^n - b^n < (a-b)na^{n-1}$$

Proposition 5.1.32 (n-th root over the Reals)
Let n be a positive integer. Then there exists a function

$$\sqrt[n]{\cdot} : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$$

such that

$$\sqrt[n]{x}^n = \sqrt[n]{x^n} = x$$

It is bijective and strictly increasing.

Proof. Let $a > 0$ be a positive real number. There can be at most one solution because $x > y \implies x^n > y^n$. Let

$$X = \{y \in \mathbb{R}_{\geq 0} \mid y^n \leq a\}$$

then we may see that X is bounded by $\max(1, a)$, for $y > \max(1, a) \implies y^n \geq y > a$.

Define $\alpha = \sup X$. Suppose $\alpha^n > a$ then we claim there is an $h > 0$ such that $(\alpha - h)^n > a$. Then $y > \alpha - h \implies y^n > a$, or contrapositively $y^n \leq a \implies y \leq \alpha - h$, and $\alpha - h$ is an upper bound of X, which is a contradiction. By (5.1.31) we have

$$\alpha^n - (\alpha - h)^n \leq hn\alpha^{n-1}$$

therefore we may choose h such that

$$0 < h < \frac{\alpha^n - a}{n\alpha^{n-1}}.$$

Suppose $\alpha^n < a$ then we claim there is $h > 0$ such that $(\alpha + h)^n < a$, which is a contradiction. As before

$$(\alpha + h)^n - \alpha^n \leq hn(\alpha + h)^{n-1}$$

therefore we may choose h such that

$$0 < h < \frac{a - \alpha^n}{n(\alpha + h)^{n-1}}$$

to demonstrate the required property.

By the trichotomy law we deduce that $\alpha^n = a$ as required, and the n-th root function is well-defined and satisfies $\sqrt[n]{x}^n = x$.

Suppose that $a^n = b^n = x > 0$ then evidently $a, b > 0$ and we deduce from (5.1.31) that $a = b$. Therefore the function is unique and injective. It is evidently surjective as $\sqrt[n]{a^n} = a$ by uniqueness. Finally we may deduce from (5.1.31) that for $y > x \geq 0$

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$$\sqrt[n]{y} - \sqrt[n]{x} > \frac{y-x}{n\sqrt[n]{y}^{n-1}} > 0$$

and so the function is strictly increasing. ∎

5.1.6 Limsup and Liminf

Proposition 5.1.33 (Limsup and Liminf)
Let K be a complete ordered field and (a_n) a sequence. Define the associated sequences in $K^\sharp$

$$a_n^+ := \sup_{n' \geq n} a_{n'}$$
$$a_n^- := \inf_{n' \geq n} a_{n'}$$

which are decreasing and increasing sequences respectively. Define

$$\limsup a_n = \lim_{n\to\infty} a_n^+$$
$$\liminf a_n := \lim_{n\to\infty} a_n^-$$

Then

a) a_n is bounded above $\iff a_n^+$ is bounded above $\iff \limsup a_n^+ < \infty$
b) a_n is bounded below $\iff a_n^-$ is bounded above $\iff \liminf a_n^- > -\infty$
c) $\limsup a_n \geq \liminf a_n$
d) $a_n \to \alpha \iff \limsup a_n = \liminf a_n = \alpha$

Proof. Clearly a_n^+ is decreasing and a_n^- is increasing. Therefore by (5.1.23) the definition of $\limsup a_n$ and $\liminf a_n$ is well-defined, the same result also demonstrates a) and b). By definition $a_n^+ \geq a_n \geq a_n^-$ so the inequality c) follows from (5.1.21).

Suppose $\alpha \to \infty$ then for every $L > 0$ there exists N such that

$$n \geq N \implies a_n > L \implies a_n^- \geq L$$

whence $\liminf a_n = \infty$. Conversely suppose $\liminf a_n = \infty$ we see by definition that $a_n \to \infty$ as required. The case of $-\infty$ follows similarly.

Suppose that $\alpha = \limsup a_n = \liminf a_n$ is finite, then there exists N_1 such that

$$n \geq N_1 \implies a_n \geq a_n^- > \alpha - \frac{\varepsilon}{2}$$

and N_2 such that

$$n \geq N_2 \implies a_n \leq a_n^+ < \alpha + \frac{\varepsilon}{2}$$

Then

$$n \geq \max(N_1, N_2) \implies |a_n - \alpha| < \varepsilon$$

and we conclude $\lim_{n\to\infty} a_n = \alpha$.

Conversely suppose $a_n \to \alpha$. Then for every $\varepsilon > 0$ there exists some N such that

$$n \geq N \implies \alpha - \varepsilon < a_n < \alpha + \varepsilon$$

and therefore

$$n \geq N \implies \alpha - \varepsilon \leq a_n^- \leq a_n \leq a_n^+ \leq \alpha + \varepsilon$$

Evidently then

$$\alpha - \varepsilon \leq \liminf a_n \leq \limsup a_n \leq \alpha + \varepsilon$$

As ε was arbitrary then this shows $\alpha = \liminf a_n = \limsup a_n$. ∎

Proposition 5.1.34 (Arithmetic Properties of limsup and liminf)
Let K be a complete ordered field and $(a_n), (b_n)$ bounded sequences. Then

a) $\limsup(a_n + b_n) \leq \limsup a_n + \limsup b_n$

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b) $\liminf(a_n + b_n) \leq \liminf a_n + \liminf b_n$

If $b_n \to b$ *then*

c) $\limsup(a_n + b_n) = \limsup a_n + b$
d) $\liminf(a_n + b_n) = \liminf a_n + b$

and if in addition $b \geq 0$ *then*

e) $\limsup(a_n b_n) = b \limsup a_n$
f) $\liminf(a_n b_n) = b \liminf a_n$

Proof. We prove each in turn

a) By definition $n' \geq n \implies a_{n'} + b_{n'} \leq a_n^+ + b_n^+$, whence $(a_n + b_n)^+ \leq a^+ + b_n^+$. The result follows from (5.1.21).
b) Similarly
c) By a) we have $\limsup(a_n + b_n) \leq \limsup a_n + b$. Furthermore $\limsup(a_n) = \limsup(a_n + b_n + (-b_n)) \leq \limsup(a_n + b_n) + \limsup(-b_n) = \limsup(a_n + b_n) - b$, and the result follows.
d) Similarly

e), f) Suppose first that $b = 0$. Then for sufficiently large n

$$0 \leq b_n \leq \varepsilon \implies -\varepsilon|a_n^-| \leq (a_n b_n)^+ \leq |a_n^+|\varepsilon \implies \left|(a_n b_n)^+\right| \leq \max\left(\left|\sup_k a_k^+\right|, \left|\inf_k a_k^-\right|\right)\varepsilon$$

as (a_n) is assumed bounded then this shows that $(a_n b_n)^+ \to 0$ as required. In this case f) follows similarly. For the general case then

$$a_n b_n = a_n(b_n - b) + a_n b$$

Obviously $b_n - b \to 0$, so by the case already proven we have

$$\limsup\left(a_{n(b_n - b)}\right) = \liminf\left(a_{n(b_n - b)}\right) = 0$$

and in particular $a_n(b_n - b) \to 0$ by (5.1.33).d). Consequently by c)

$$\limsup(a_n b_n) = \limsup(a_n b) = b \limsup(a_n)$$

as required. The case f) follows similarly. ∎

Proposition 5.1.35
Let K *be a complete ordered field and* (a_n) *a bounded sequence. Then every subsequence* $\left(a_{n_k}\right)$ *satisfies*

$$\limsup_k a_{n_k} \leq \limsup_n a_n$$

and there in fact exists a convergent subsequence a_{n_k} *such that*

$$a_{n_k} \downarrow \limsup_n a_n$$

A similar statement applies to $\liminf a_n$.

Proof. Define $b_k := \sup_{k' \geq k} a_{n_{k'}}$, then by definition

$$b_k^+ \leq a_{n_k}^+$$

Taking limits as $k \to \infty$ and considering we find that

$$\limsup_k a_{n_k} \overset{(5.1.21)}{\leq} \lim_{k \to \infty} a_{n_k}^+ \overset{(5.1.22)}{=} \lim_{n \to \infty} a_n^+ =: \limsup_n a_n$$

Recall that $a_n^+ \downarrow \limsup a_n$. Therefore for every $k > 0$ there exists N_k such that $a_{N_k}^+ \leq \limsup a_n + \frac{1}{2k}$. Then by definition there exists some $n_k \geq N_k$ such that $a_{n_k} \leq a_{N_k}^+ + \frac{1}{2k} \leq \limsup a_n + \frac{1}{k}$. We see that a_{n_k} has the required property. ∎

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5.2 Complex Numbers

Proposition 5.2.1 (Existence of Complex Numbers)
Let K be a field for which -1 has no square root (and in particular $\operatorname{char}(K) \neq 2$). Then

$$K[i] \coloneqq K[X]/(X^2+1)$$

is a field with a K-basis $\{1, i\}$ such that $i^2 = -1$. Multiplication in $K[i]$ is defined as follows

$$(a+bi)(c+di) = (ac-bd) + (bc+ad)i$$

and the inverse is given by

$$(a+bi)^{-1} = \frac{a-bi}{a^2+b^2}$$

The field extension $K(i)/K$ is Galois of degree two with

$$\operatorname{Gal}(K(i)/K) = \{1, \bar{\cdot}\}$$

*where $\bar{\cdot}$ is the **complex conjugation** automorphism given by*

$$\overline{a+bi} = a-bi$$

In particular we see that $\bar{\cdot}$ satisfies the following properties

$$\overline{\alpha+\beta} = \overline{\alpha} + \overline{\beta}$$
$$\overline{\alpha\beta} = \overline{\alpha}\,\overline{\beta}$$

Proof. The polynomial X^2+1 is of degree 2 so it is irreducible if and only if has no root. Therefore the quotient ring is a field by (3.17.40). $\mathbb{R}$ satisfies this property by (5.1.5), so $\mathbb{C}$ is a field.

Observe that as polynomials in $K[X]$

$$(a+bX)(c+dX) = ac + (bc+ad)X + bdX^2 = (ac-bd) + (bc+ad)X + bd(X^2+1)$$

from which the multiplication identity follows immediately.

For the inverse we claim that $a \neq 0$ or $b \neq 0 \implies a^2+b^2 \neq 0$. For otherwise either a/b or b/a would be a square root of -1. Therefore the expression for the inverse is well-defined and it may be verified directly to be an inverse.

The complex conjugation operator exists by (3.17.51). Therefore $\operatorname{Aut}(K(i)/K)$ has order at least $[K(i):K] = 2$, and we deduce $K(i)/K$ is Galois (3.17.110). ∎

Definition 5.2.2 (Complex Numbers)
*Define the **complex numbers** $\mathbb{C}$ to be the field $\mathbb{R}[i]$ and identify $\mathbb{R}$ with the subfield $\{a+0i \mid a \in \mathbb{R}\}$.*

*Define the **absolute value** on $\mathbb{C}$ as follows*

$$|a+bi| \coloneqq \sqrt{a^2+b^2} \in \mathbb{R}_{\geq 0}$$

*Define the **real part** and **imaginary part** as follows*

$$\operatorname{Re}(a+bi) = a$$
$$\operatorname{Im}(a+bi) = b$$

Observe that

$$\alpha + \overline{\alpha} = 2\operatorname{Re}(\alpha)$$

Proposition 5.2.3 (Complex Absolute Value)
Let $\alpha, \beta \in \mathbb{C}$ be complex numbers. The following properties hold

$$|\alpha| = |\overline{\alpha}|$$
$$|\alpha|^2 = \alpha\overline{\alpha}$$
$$|\alpha\beta| = |\alpha||\beta|$$
$$|\alpha+\beta| \leq |\alpha| + |\beta|$$

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Furthermore this agrees with the usual notion of absolute value on $\mathbb{R}$.

Proof. The first two relations follow by direct calculation. Then

$$|\alpha\beta|^2 = \alpha\beta\overline{\alpha\beta} = \alpha\overline{\alpha}\beta\overline{\beta} = |\alpha|^2|\beta|^2$$

and the third relation follows by taking square roots. Further

$$\begin{aligned}|\alpha+\beta|^2 &= (\alpha+\beta)\left(\overline{\alpha}+\overline{\beta}\right) = \alpha\overline{\alpha} + \left(\alpha\overline{\beta}+\overline{\alpha}\beta\right) + \beta\overline{\beta} \\ &= |\alpha|^2 + 2\,\mathrm{Re}(\alpha\beta) + |\beta|^2 \\ &\leq |\alpha|^2 + 2|\alpha\beta| + |\beta|^2 \\ &= (|\alpha| + |\beta|)^2\end{aligned}$$

and the result follows by taking square roots (and this being an increasing function). ■

Proposition 5.2.4 (Complex Numbers are Complete)
$\mathbb{C}$ *is complete as a normed vector space.*

Proof. Let (α_n) be a cauchy sequence in $\mathbb{C}$, and consider the decomposition into real and imaginary parts

$$\alpha_n = a_n + b_n i$$

Then

$$|a_n| \leq |\alpha_n|$$

$$|b_n| \leq |\alpha_n|$$

Therefore we see both (a_n) and (b_n) are cauchy sequences of reals, which therefore have limits a and b respectively. We may verify directly that $\alpha_n \to \alpha$ where $\alpha := a + bi$. ■

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5.3 Metric Spaces

Definition 5.3.1 (Metric Space)
*Let X be a set and $d : X \times X \to \mathbb{R}_{\geq 0}$ a function. We say that (X, d) is a **metric space** if the following properties hold*

a) $d(x, y) = 0 \iff x = y$ for all $x, y \in X$
b) $d(x, y) = d(y, x)$
c) $d(x, z) \leq d(x, y) + d(y, z)$ for all $x, y, z \in X$

*Given $x \in X$ and $\varepsilon > 0$ we define the **open ball** at x to be*

$$B(x; \varepsilon) := \{y \in X \mid d(x, y) < \varepsilon\}$$

Proposition 5.3.2
Let (X, d) be a metric space. Then we say that a subset $U \subset X$ is open if

$$\forall x \in X \exists \varepsilon > 0 \text{ s.t. } B(x; \varepsilon) \subset U$$

The open sets form a topology with open balls forming a base.

For each $x \in X$ a local base consists of open balls of the form $B(x; \varepsilon)$. Further we may also consider the countable local base

$$\mathcal{B}_x := \left\{ B\left(x; \frac{1}{n}\right) \;\middle|\; n \in \mathbb{N} \right\}$$

*In particular every metric space is **first countable**.*

Proposition 5.3.3
A metric space is Hausdorff.

Proposition 5.3.4 (Convergent Sequence)
Let (X, d) be a metric space and (x_n) a sequence in X and $x \in X$. Then the following are equivalent

a) For every $\varepsilon > 0$ there exists N such that $n \geq N \implies d(x_n, x) < \varepsilon$
b) The function

$$\begin{aligned} \mathbb{N} \cup \{\infty\} &\to X \\ n &\to x_n \\ \infty &\to x \end{aligned}$$

is continuous at ∞ with respect to the topology on $\mathbb{N} \cup \{\infty\}$ given by open sets of the form $\{N, N+1, \cdots\} \cup \{\infty\}$ and arbitrary subsets of $\mathbb{N}$. Note this function is automatically continuous at every n.

*In this case we say that $x_n \to x$ is a **convergent sequence**.*

Proposition 5.3.5 (Continuous at a Point)
Let $f : (X, d) \to (Y, d')$ be a map of metric spaces. Then for $x \in X$ and $y := f(x)$ the following are equivalent

a) For all $\varepsilon > 0$ there exists $\delta > 0$ such that

$$d(y, x) < \delta \implies d'(f(y), f(x)) < \varepsilon$$

b) $f : X \to Y$ is continuous at x in the sense of topological spaces (4.1.25)
c) For every sequence $x_n \to x$ we have $f(x_n) \to f(x)$

Proposition 5.3.6 (Continuous Criteria)
Let $f : X \to Y$ be a map of metric spaces. Then the following are equivalent

a) f is continuous at every $x \in X$ (5.3.5)
b) f is continuous in the topological sense (4.1.27)
c) For every convergent sequence $x_n \to x$ we have $f(x_n) \to f(x)$

Definition 5.3.7 (Product Metric Space)
Let (X_i, d_i) be metric spaces for $i = 1 \cdots n$. Define the metric space on $X_1 \times \cdots \times X_n$ by

$$d((x_1, \cdots, x_n), (y_1, \cdots, y_n)) = \max\left(d_1(x_1, y_1), \cdots, d_{n(x_n, y_n)}\right)$$

Then the topology induced on (X, d) coincides with the product topology.

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5.3.1 Completeness

Definition 5.3.8 (Cauchy Sequence)
*Let (X,d) be a metric space. A sequence (x_n) in X is said to be **cauchy** if*

$$\forall \varepsilon \exists N \text{ s.t. } n,m \geq N \implies d(x_n,x_m) < \varepsilon$$

Definition 5.3.9
*A metric space (X,d) is said to be **complete** if every cauchy sequence is convergent.*

Proposition 5.3.10
Let (x_n) be a Cauchy sequence in a metric space (X,d) with $x \in X$ a cluster point (equivalently, subsequential limit point). Then $x_n \to x$.

Proposition 5.3.11
Let (X,d) be a complete metric space. A closed subset $Y \subset X$ is complete iff it is closed in X.

Proposition 5.3.12
Let (X_i,d_i) be complete metric spaces for $i = 1 \cdots n$. Then the product metric space is $(X_1 \times \cdots \times X_n, d)$ is complete.

Corollary 5.3.13
The metric space $\mathbb{R}^k$ induced by the product metric is complete.

A subset $X \subset \mathbb{R}^k$ is complete if and only if it is closed.

Proof. This is simply (5.3.12) and (5.3.11). ∎

5.3.2 Compactness

Definition 5.3.14
*We say a metric space (X,d) is **separable** if there exists a countable dense subset X_0.*

Proposition 5.3.15
Suppose X is a separable metric space. Then it is second countable.

Proof. We may show that the family

$$\mathcal{B} \coloneqq \left\{ B\left(x_0; \frac{1}{n}\right) \;\middle|\; x_0 \in X_0, n \in \mathbb{N} \right\}$$

is a countable base. ∎

Proposition 5.3.16 (Precompact)
Let (X,d) be a metric space. The following are equivalent

a) For all $\varepsilon > 0$ there exists a finite covering $X = U_1 \cup \cdots \cup U_n$ such that $\operatorname{diam}(U_i) < \varepsilon$
b) For all $\varepsilon > 0$ there exists a finite set of points $x_1, \cdots, x_n$ such that $X = \bigcup_{i=1}^n B(x_i;\varepsilon)$
c) Every sequence has a cauchy subsequence.

*In this case we say X is **precompact** or **totally bounded**.*

Proof. *a)* $\iff$ *b)* is straightforward. For *c)* $\implies$ *b)* suppose X did not satisfy this property, then we could inductively choose a sequence x_n such that

$$x_n \in X \setminus \bigcup_{i=1}^{n-1} B(x_i;\varepsilon)$$

Then evidently $d(x_n, x_{n+1}) \geq \varepsilon$ for all n and it can have no cauchy subsequence.

For *b)* $\implies$ *c)* Let (x_n) be a sequence. If $\{x_n\}$ is finite then we are done. Otherwise assuming it is infinite, we construct a decreasing sequence of sets

$$A_1 \supset \cdots \supset A_k \supset \cdots$$

such that $\operatorname{diam}(A_k) < \frac{1}{k}$ and $A_k \cap \{x_n\}$ is infinite. For suppose $A_1, \cdots, A_k$ are constructed, then evidently A_k itself satisfies *b)* so we may find $A_k = \bigcup_{i=1}^m B\left(x_i; \frac{1}{k+1}\right)$. For at least one i we must have $B\left(x_i; \frac{1}{k+1}\right) \cap A_k \cap \{x_n\}$ is infinite. Therefore we may define $A_{k+1} \coloneqq B\left(x_i; \frac{1}{k+1}\right) \cap A_k$. Finally choosing an increasing sequence n_k such that $x_{n_k} \in A_k$ yields the required cauchy subsequence. ∎

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Corollary 5.3.17 (Compactness Criteria)
Let (X,d) be a metric space then the following are equivalent

a) X is compact
b) X is countably compact
c) X is sequentially compact
d) X is complete and precompact

Proof. $a), c) \implies b)$ This is (4.1.106).a)

$a), b) \implies c)$ This is (4.1.106).b)

$c) \implies d)$ By (5.3.10) every cauchy sequence is convergent. Therefore X is complete, and precompact by (5.3.16).

$d) \implies a)$ We show that (X,d) is sequentially compact and second countable, and the result follows from (4.1.106).c). Sequential compactness is a consequence of (5.3.16) and (5.3.10). To show that X is second countable we may for each n find a finite cover

$$X = \bigcup_{i=1}^{N_n} B\left(x_{ni}; \frac{1}{n}\right)$$

We claim that

$$\left\{B\left(x_{ni}; \frac{1}{n}\right)\right\}$$

is a base. For given $y \in X$ and a nbhd $U \in \mathcal{U}_y$ we have $B(y;\varepsilon) \subset U$ for some $\varepsilon > 0$. Choose n such that $\frac{1}{n} < \frac{\varepsilon}{2}$. Then for some $1 \leq i \leq N_n$, $y \in B\left(x_{ni}; \frac{1}{n}\right)$. We may show that $B\left(x_{ni}; \frac{1}{n}\right) \subset B(y;\varepsilon) \subset U$ which shows that the given collection is a base. ∎

5.3.3 Uniform Continuity

Definition 5.3.18 (Uniformly Continuous)
*Let (X,d) and (Y,d') be metric spaces and $f : X \to Y$ a function. We say that f is **uniformly continuous** if*

$$\forall \varepsilon > 0 \exists \delta > 0 \text{ s.t. } d(x,y) < \delta \implies d'(f(x), f(y)) < \varepsilon$$

Proposition 5.3.19 (Compact & Continuous $\implies$ Uniformly Continuous)
Let $f : X \to Y$ be a continuous map of metric spaces and suppose X is (sequentially) compact. Then f is uniformly continuous.

Proof 1.. Fix $\varepsilon > 0$ then for every $x \in X$ there exists δ_x such that $f(B(x;\delta_x)) \subseteq B(f(x);\varepsilon)$. Evidently $X = \bigcup_x B\left(x; \frac{\delta_x}{2}\right)$ whence there is a finite subcover associated to $x_1, \cdots, x_n$. Define

$$\delta := \min\left(\frac{\delta_{x_1}}{2}, \cdots, \frac{\delta_{x_n}}{2}\right)$$

Given $y \in X$ then $y \in B\left(x_i; \frac{\delta_{x_i}}{2}\right)$ for some $i = 1 \cdots n$. Furthermore suppose $d(y,z) < \delta$ then

$$d(x_i, z) \leq d(x_i, y) + d(y, z) < \delta_{x_i}$$

whence $y, z \in B\left(x_i; \delta_{x_i}\right)$. Therefore by construction $d(f(y), f(z)) < \varepsilon$, as required. ∎

Proof 2.. By (5.3.17) we know that X is sequentially compact. Suppose that f is not uniformly continuous, then there exists $\varepsilon > 0$ and sequences $\{x_n\}, \{y_n\}$ such that $d(x_n, y_n) < \frac{1}{n}$ and $d(f(x_n), f(y_n)) > \varepsilon$. By assumption there is a subsequence n_k such that $x_{n_k} \to x$ and $y_{n_k} \to y$. Evidently

$$d(x,y) \leq d\left(x, x_{n_k}\right) + d\left(x_{n_k}, y_{n_k}\right) + d\left(y_{n_k}, y\right) \to 0$$

whence $d(x,y) = 0$ and $x = y$. By (...) $f\left(x_{n_k}\right) \to f(x)$ and $f\left(y_{n_k}\right) \to f(x)$ which is a contradiction. ∎

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5.4 Normed Vector Spaces

Definition 5.4.1 (Valued Field)
*Let K be a field. A **valuation** on K is a function*

$$|\cdot| : K \to \mathbb{R}_{\geq 0}$$

such that for all $x, y \in K$ the following relations are satisfied

a) $|x| = 0 \iff x = 0$
b) $|xy| = |x||y|$
c) $|x + y| \leq |x| + |y|$

*We say that this valuation is **non-archimedean** if it satisfies the ultrametric inequality*

$$|x + y| \leq \max(|x|, |y|)$$

*for all $x, y \in K$. Otherwise it is **archimedean**.*

*We say the pair $(K, |\cdot|)$ is a **valued field**.*

Example 5.4.2
Any subfield of $\mathbb{C}$ (and therefore $\mathbb{R}$) is an archimedean valued field, using the complex absolute value $|\cdot|$ (5.2.2).

Definition 5.4.3 (Normed Vector Space)
Let $(K, |\cdot|)$ be a valued field and X a K-vector space. A norm $\|\cdot\|$ is a function

$$\|\cdot\| : X \to \mathbb{R}_{\geq 0}$$

satisfying the following properties

a) $\|x\| = 0 \iff x = 0$
b) $\|\lambda x\| = |\lambda|\|x\|$
c) $\|x + y\| \leq \|x\| + \|y\|$

*We say the pair $(X, \|\cdot\|)$ is a **normed vector space** over the valued field $(K, |\cdot|)$.*

*A normed vector space which is complete as a metric space is called a **Banach space**.*

A Banach space X which is also an associative algebra satisfying

$$\|x \cdot y\| \leq \|x\|\|y\|$$

*is a **Banach Algebra**. Note a valued field is naturally a Banach Algebra.*

Example 5.4.4
A valued field (e.g. $\mathbb{C}$, $\mathbb{R}$, $\mathbb{Q}$) is a normed vector space over itself.

$\mathbb{C}$ is a normed vector space over $\mathbb{R}$.

For any valued field K the vector space K^n is a normed vector space with the following definition of norm

$$\|v\| := \sqrt{v_1^2 + \cdots + v_n^2}$$

(the triangle inequality requires proof).

Proposition 5.4.5
Let $(X, \|\cdot\|)$ a normed vector space. Then it is naturally a metric space with induced metric

$$d(x, y) := \|x - y\|$$

and in particular a Hausdorff topological space with base the open balls

$$B(x; \varepsilon) := \{y \in X \mid \|y - x\| < \varepsilon\}$$

Furthermore the open balls $B(x; \varepsilon)$ and $B\left(x; \frac{1}{n}\right)$ form a local base at x.

The same statement applies to valued fields and in particular $\mathbb{R}$.

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5.4.1 Continuous Functions

Proposition 5.4.6 (Continuous maps of Normed Vector Spaces)
Let $S \subset X$ be a subset of a normed vector space and $f : S \to Y$ be a map to a normed vector space Y and $x \in S$. Then the following are equivalent

a) For all $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x' \in S$, $\|x - x'\| < \delta \implies \|f(x) - f(x')\| < \varepsilon$
b) f is continuous at x with respect to the subspace topology on S

*In this case we say the map f is **continuous** at x.*

Example 5.4.7
For X a normed vector space then $\|\cdot\| : X \to \mathbb{R}$ is evidently continuous.

5.4.2 Product Space

Definition 5.4.8 (Product Norm)
*Let $X_1, \cdots, X_n$ be normed vector spaces over the same valued field K. Then we may define the **product norm** on the vector space $X_1 \times \cdots \times X_n$ by*

$$\|(x_1, \cdots, x_n)\| := \max(\|x_1\|, \cdots, \|x_n\|)$$

Proposition 5.4.9
Let $X_1, \cdots, X_n$ be normed vector spaces over the same valued field K. Then the metric (resp. topology) induced by product norm is the same as product metric (resp. product topology).

Proof. Consider a point $x = (x_1, \cdots, x_n)$ then

$$B(x; \varepsilon) = \{(x_1, \cdots, x_n) \mid \|x_i\| \leq \varepsilon\} = B(x_1; \varepsilon) \times \cdots \times B(x_n; \varepsilon)$$

Therefore the open balls are open in the product topology, and being a base for the norm topology, this is a subset of the product topology (...). To show that the product topology is a subset of the norm topology it is sufficient to show that open sets of the form

$$U := U_1 \times \cdots \times U_n$$

are open in the product norm topology. For given $y \in U$ then by definition $B(y_i; \varepsilon_i) \subset U_i$. Consider $\varepsilon := \min(\varepsilon_1, \cdots, \varepsilon_n)$ and $z \in B(y; \varepsilon)$. Then

$$\|z - y\| < \varepsilon \implies \max_i(\|z_i - y_i\|) < \varepsilon \leq \varepsilon_i \implies z_i \in B(y_i; \varepsilon_i) \implies z \in U$$

in other words $B(y; \varepsilon) \subset U$ as required. ∎

Proposition 5.4.10 (Arithmetic Operations are Continuous)
Let X be a normed vector space over a valued field K. The following maps are continuous

*a) **addition** $+ : X \times X \to X$*
*b) **scalar multiplication** $\cdot : K \times X \to X$*
*c) **inverse scalar** $(-)^{-1} : K \setminus \{0\} \to K$*

Similarly if X is a Banach algebra then the multiplication operator

$$\cdot : X \times X \to K$$

is also continuous.

Proof. By the triangle inequality

$$\|(x_1, x_2) - (y_1, y_2)\| < \varepsilon \implies \|x_1 - y_1\| < \varepsilon \wedge \|x_2 - y_2\| < \varepsilon \implies \|(x_1 + x_2) - (y_1 + y_2)\| < 2\varepsilon$$

Therefore we may apply the criterion given by (5.4.6), and using the product norm.

For scalar multiplication observe

$$\begin{aligned}
\|(\lambda_1, x_1) - (\lambda_2, x_2)\| < \delta &\implies |\lambda_1 - \lambda_2| < \delta \wedge \|x_1 - x_2\| < \delta \\
&\implies \|\lambda_1 x_1 - \lambda_2 x_2\| \leq |\lambda_1| \|x_1 - x_2\| + \|x_2\| |\lambda_1 - \lambda_2| \\
&\implies \|\lambda_1 x_1 - \lambda_2 x_2\| \leq |\lambda_1| \|x_1 - x_2\| + (\|x_1\| + \delta) |\lambda_1 - \lambda_2| \\
&\implies \|\lambda_1 x_1 - \lambda_2 x_2\| \leq \delta^2 + (\|x_1\| + \delta)\delta
\end{aligned}$$

Therefore choose $\delta < \min\left(1, \frac{\varepsilon}{\|x_1\|+1}\right)$ to find $\|\lambda_1 x_1 - \lambda_2 x_2\| < 3\varepsilon$. ∎

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5.4.3 Convergent Sequences

Proposition 5.4.11 (Convergent Sequence)
Let X be a normed vector space and (a_n) a sequence in X. Then the following are equivalent

a) $a_n \to \alpha$ in the topological sense (4.1.49)

b) For all $\varepsilon > 0$ there exists $N(\varepsilon) \in \mathbb{N}^+$ such that

$$n \geq N(\varepsilon) \implies \|a_n - \alpha\| < \varepsilon$$

c) The function

$$\begin{aligned} \mathbb{N} \cup \{\infty\} &\to X \\ n &\to a_n \\ \infty &\to \alpha \end{aligned}$$

is continuous with respect to the topology on $\mathbb{N} \cup \{\infty\}$ given by open sets of the form $\{N, N+1, \cdots\} \cup \{\infty\}$ and arbitrary subsets of $\mathbb{N}$.

*In this case we say the sequence (a_n) is **convergent**. If this doesn't hold it is **divergent**.*

Proposition 5.4.12 (Uniqueness of Limits)
Let X be a normed vector space. Then it is Hausdorff and limits are unique.

Proposition 5.4.13 (Continuous Image of Convergent Sequence)
Let X, Y be normed vector spaces, $S \subset X$ and $f : S \to Y$ a map. Then the following are equivalent

a) f is continuous in the topological sense

b) For every convergent sequence $a_n \to \alpha$ we have $f(a_n) \to f(\alpha)$.

Proof. Recall that X is first-countable and so the equivalence follows from (4.1.50) and (4.1.52). ∎

Proposition 5.4.14 (Arithmetic of Convergent Sequences in an NVS)
Let X be a normed vector space and (a_n) and (b_n) be convergent sequences in X. Then

$$\lim_{n\to\infty} (a_n + b_n) = \lim_{n\to\infty} a_n + \lim_{n\to\infty} b_n$$

$$\lim_{n\to\infty} \lambda a_n = \lambda \lim_{n\to\infty} a_n$$

In particular $a_n \to \alpha \iff a_n - \alpha \to 0$.

Proposition 5.4.15 (Arithmetic of Convergent Sequences in a valued field)
Let K be a valued field and (a_n) and (b_n) be convergent sequences. Then

$$\lim_{n\to\infty} a_n b_n = \left(\lim_{n\to\infty} a_n\right)\left(\lim_{n\to\infty} b_n\right)$$

Further if both b_n and $\lim_{n\to\infty} b_n$ is non-zero then

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{\lim_{n\to\infty} a_n}{\lim_{n\to\infty} b_n}$$

Proposition 5.4.16 (Arithmetic of Convergent Sequences in an ordered field)
Let K be an ordered field and (b_n) a sequence of non-negative elements. Then $b_n \to \infty \iff b_n^{-1} \to 0$.

Proposition 5.4.17
Let K be a valued field and $x \in K$ such that $|x| < 1$. Then

$$\lim_{n\to\infty} x^n = 0$$

Proof. We may reduce to the case that $x \in \mathbb{R}$ and $0 < x < 1$. Let $y = \frac{1-x}{x} > 0$, then it is enough to show that $(1+y)^n \to \infty$ by (5.4.16). However from (5.1.30) $(1+y)^n \geq 1 + ny$, and the result follows from Archimedean property (...). ∎

Definition 5.4.18 (Cauchy Sequence)
*Let X be a normed vector space and $(a_n)_{n\in\mathbb{N}}$ a sequence in X. We say (a_n) is a **cauchy sequence** if for all $\varepsilon > 0$ there exists $N(\varepsilon)$ such that*

$$n, m \geq N(\varepsilon) \implies \|a_n - a_m\| < \varepsilon$$

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*Every **convergent sequence** is **cauchy**. If the converse holds then we say X is **complete**.*

5.4.4 Finite-Dimensional Normed Spaces

Definition 5.4.19 (Equivalent Norms)
*Let X be a vector space over a valued field K. Consider two norms $\|\cdot\|_1$ and $\|\cdot\|_2$. We say that they are **equivalent** if there exists constants $c_1, c_2 > 0$ such that*

$$c_1\|x\|_1 \leq \|x\|_2 \leq c_2\|x\|_1$$

We may prove easily that this is an equivalence relation on norms.

Proposition 5.4.20
Let X be a vector space over a valued field K. Suppose that $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent norms for X. Then the following derived quantities are the same

- *Induced topology*
- *Convergent sequences*
- *Cauchy sequences*

Proposition 5.4.21 (Supremum Norm)
Let X be a finite-dimensional vector space with basis $(e_1, \cdots, e_n)$. Then the supremum norm defined by

$$\left\|\sum_{i=1}^n \lambda_i e_i\right\|_\infty := \max(|\lambda_1|, \cdots, |\lambda_n|)$$

is well-defined (it is the product norm on $X \cong K \times \cdots \times K$). If K is complete then so is X.

Proposition 5.4.22 (Equivalence of Norms)
Let X be a non-zero finite-dimensional vector space over a complete valued field K. Then every norm is equivalent to the supremum norm (and therefore complete).

Proof. Fix a basis $(e_1, \cdots, e_n)$. Then for any norm $\|\cdot\|$ we have by the triangle inequality

$$\left\|\sum_{i=1}^n \lambda_i e_i\right\| \leq \sum_{i=1}^n |\lambda_i|\|e_i\| \leq \left(\sum_{i=1}^n \|e_i\|\right)\left\|\sum_{i=1}^n \lambda_i e_i\right\|_\infty$$

The reverse inequality we prove by induction. The case $n = 1$ we may take $c_1 = \|e_1\|$. Suppose $n > 1$ and there is no such constant C. Then there are vectors $v_k \in X$ such that

$$\|v_k\| < \frac{1}{k}\|v_k\|_\infty$$

Suppose

$$v_k = \sum_{i=1}^n \lambda_{ki} e_i$$

Consider the sets of integers

$$S_i := \left\{k \in \mathbb{N} \;\middle|\; \|v_k\|_\infty = |\lambda_{ki}|\right\} \quad i = 1 \cdots n$$

Then at least one of $S_1, \cdots, S_n$ is infinite, since the union is $\mathbb{N}$. Without loss of generality we assume it is S_n.

Therefore by passing to a subsequence, and scaling, we may assume that $\|v_k\|_\infty = \lambda_{kn} = 1$ and $\|v_k\| < \frac{1}{k}$. Define the vector $w_k := v_k - e_n$. Then by definition $w_k \to e_n$ with respect to $\|\cdot\|$.

Let W be the $(n-1)$-dimensional subspace spanned by $\{e_1, \cdots, e_{n-1}\}$. We prove also that $w_k \to w \in W$ with respect to $\|\cdot\|$, which contradicts uniqueness of limits. Evidently w_k is a cauchy sequence with respect to $\|\cdot\|$ restricted to W. By induction $\|\cdot\|_\infty$ is equivalent to $\|\cdot\|$ on W, and therefore w_k is cauchy with respect to $\|\cdot\|_\infty$. As W is complete (5.4.21) we see that $w_k \to w$ with respect to $\|\cdot\|_\infty$, and therefore with respect to $\|\cdot\|$. ∎

Corollary 5.4.23
Let X be a finite dimensional normed vector space with basis $\{x_1, \cdots, x_n\}$. The projection map

$$\pi_i : X \to \langle x_i \rangle \xrightarrow{\sim} K$$

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given by $\pi_i(x) = x_i^\vee(x)x_i$ is a continuous, open map.

5.4.5 Convergent Series

Definition 5.4.24 (Series)
Let X be a normed vector space and (a_n) a sequence in X. The formal expression

$$\sum_{n=1}^{\infty} a_n$$

*is called the **series** associated to this sequence. For such a sequence define the **partial sum** sequence as follows*

$$s_n := \sum_{k=1}^{n} a_k$$

*If the partial sum sequence converges then we say that the series **converges** and write*

$$\sum_{n=1}^{\infty} a_n := \lim_{n\to\infty} s_n$$

If the series

$$\sum_{n=1}^{\infty} \|a_n\|$$

*converges then we say the original series **converges absolutely**.*

The choice of terminology is justified in the case that X is complete.

Proposition 5.4.25 (Absolute Convergence $\implies$ Convergence)
Let X be a complete normed vector space and $\sum_{n=0}^{\infty} a_n$ a series which converges absolutely. Then the series converges in the usual sense.

Proof. The partial sums of the series $\sum_{n=1}^{\infty}\|a_n\|$ are cauchy by (5.4.18). By the triangle inequality we may demonstrate that the partial sums of the series $\sum_{n=1}^{\infty} a_n$ are cauchy. Then by completeness of X the partial sums must also converge. ∎

Proposition 5.4.26
Let X be a normed vector space and $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ two convergent sequences. Then the following series are convergent with limits

$$\sum_{n=1}^{\infty}(a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$$

$$\sum_{n=1}^{\infty} \lambda a_n = \lambda \sum_{n=1}^{\infty} a_n$$

where $\lambda \in K$.

Proposition 5.4.27
Let X be a normed vector space and $\sum_{n=1}^{\infty} a_n$ a convergent series. Then $\|a_n\| \to 0$.

Proof. Let $\alpha := \sum_{n=1}^{\infty} a_n$. Then by definition there exists N such that $n \geq N \implies \left\|\sum_{k=1}^{n} a_k - \alpha\right\| < \varepsilon$. As this holds for $n+1$ we may use the triangle inequality to find that $n \geq N+1 \implies \|a_n\| < 2\varepsilon$. As ε was arbitrary we find $\|a_n\| \to 0$ as required. ∎

Proposition 5.4.28 (Comparison Test)
Let X be a complete normed vector space. Suppose (a_n) is a sequence in X and (c_n) is a sequence in $\mathbb{R}$ such that

a) The series $\sum_{n=0}^{\infty} c_n$ converges
b) $\|a_n\| \leq c_n$ for $n \geq N_0$

then the series $\sum_{n=0}^{\infty} a_n$ converges absolutely.

Proof. By assumption *b)* (c_n) is non-negative, and by (5.4.29) the sequence of partial sums $\sum_{k=0}^{n} c_k$ is bounded. therefore so is the sequence $\sum_{k=0}^{n}\|a_k\|$, and by the same result the series $\sum_{n=0}^{\infty}\|a_n\|$ converges, in other words $\sum_{n=0}^{\infty} a_n$ converges absolutely. ∎

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Proposition 5.4.29 (Series of Positive Reals)
Let (a_n) be a sequence of non-negative real numbers. Then the following are equivalent

a) The sequence of partial sums $A_n := \sum_{k=0}^{n} a_k$ is bounded

b) The series $\sum_{n=0}^{\infty} a_n$ converges (absolutely)

Proof. By (5.1.23) we always have $A_n \to \sup A_n \in \mathbb{R}^\sharp$, and the limit is finite iff A_n is bounded. ∎

Proposition 5.4.30 (Geometric Series)
Suppose $0 \leq x < 1$ is a real number, then

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

and in particular the partial sums are bounded by $\frac{1}{1-x}$.

Proof. It's straightforward to show by induction that the partial sums are given explicitly by

$$\sum_{n=0}^{N} x^n = \frac{1-x^N}{1-x}\ .$$

Then the result follows from (5.4.17). ∎

Proposition 5.4.31 (Root Test)
Let X be a normed vector space. Given a series $\sum_{n=0}^{\infty} a_n$, define $\alpha := \limsup_n \sqrt[n]{\|a_n\|}$ Then

a) if $\alpha < 1$ then the series converges absolutely

b) if $\alpha > 1$ then the series diverges

Proof.

a) Define $\alpha := \limsup_n \sqrt[n]{\|a_n\|}$ and $\beta := \frac{1+\alpha}{2} < 1$. Then by definition there is an integer N such that

$$n \geq N \implies \sqrt[n]{\|a_n\|} \leq \alpha + \frac{1-\alpha}{2} = \beta \implies \|a_n\|^n \leq \beta^n$$

In particular for $n \geq N$

$$\sum_{k=0}^{n} \|a_k\| \leq \sum_{n=0}^{N-1} \|a_k\| + \sum_{k=N}^{n} \|a_k\| \leq \sum_{k=0}^{N-1} \|a_k\| + \sum_{k=N}^{n} \beta^k \leq \sum_{k=0}^{N-1} \|a_k\| + \sum_{k=0}^{\infty} \beta^k$$

where the last step follows from by (5.4.30). We conclude from (5.4.29) that the series converges absolutely.

b) By (5.1.35) there exists a subsequence a_{n_k} such that $\sqrt[n_k]{\left\|a_{n_k}\right\|} \to \alpha > 1$. Therefore we may find a subsequence such that $\sqrt[n_k]{\left\|a_{n_k}\right\|} \geq \frac{1+\alpha}{2} > 1 \implies \left\|a_{n_k}\right\| > 1$. This shows that $\left\|a_{n_k}\right\| \not\to 0$, whence $\|a_n\| \not\to 0$ by (...), and the series diverges by (5.4.27).

∎

Proposition 5.4.32 (Ratio Test)
Let K be a valued field and consider a series $\sum_{n=1}^{\infty} a_n$. If

$$\limsup_n \left|\frac{a_{n+1}}{a_n}\right| < 1$$

then the series is convergent.

Proof. Let $\alpha := \limsup \left|\frac{a_{n+1}}{a_n}\right|$ and $\beta = \frac{1+\alpha}{2}$. Then by assumption $\beta < 1$, and there exists N such that $n \geq N \implies \left|\frac{a_{n+1}}{a_n}\right| \leq \beta \implies |a_{N+k}| \leq |a_N|\beta^k$. We then see by Comparison Test (5.4.28) that the series is convergent. ∎

Proposition 5.4.33 (Cauchy Product of Series)
Let X be a complete valued field and $\sum_{n=1}^{\infty} a_n$, $\sum_{n=1}^{\infty} b_n$, $\sum_{n=1}^{\infty} c_n$ series such that

a) one of $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are absolutely convergent

b) $c_n := \sum_{k=0}^{n} a_k b_{n-k} = \sum_{k=0}^{n} a_{n-k} b_k$

then $\sum_{n=0}^{\infty} c_n$ is absolutely convergent and

$$\sum_{n=0}^{\infty} c_n = \left(\sum_{n=1}^{\infty} a_n\right)\left(\sum_{n=1}^{\infty} b_n\right)$$

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Proof. Denote the partial sums by A_n, B_n and C_n, and the limits by A, B respectively. Then evidently

$$C_n = \sum_{i=0}^{n} a_{n-i} B_i$$

therefore

$$C_n = \sum_{i=0}^{n} a_{n-i}(B_i - B) + A_n B$$

Then it's sufficient to show that the first term tends to 0. Observe that $\beta_n := B_n - B \to 0$. **TODO** ∎

5.4.6 Function Spaces

Definition 5.4.34 (Uniform Convergence)
*Let S be a metric space and Y a normed vector space. Suppose $\{f_n : S \to Y\}$ is a sequence of functions and $f : S \to Y$ is a function. We say that $f_n \to f$ **converges uniformly** if*

$$\forall \varepsilon > 0 \exists N \text{ s.t. } n \geq N \implies \|f_n(x) - f(x)\| < \varepsilon \forall x \in S$$

Note this implies, but is strictly stronger than, pointwise convergence that is $f_n(x) \to f(x)$ for all $x \in S$, as N may be chosen independently of x.

Proposition 5.4.35 (Uniform Limit of Continuous Functions is Continuous)
Let S be a metric space and Y a normed vector space. Suppose that $\{f_n : S \to Y\}$ is a sequence of continuous functions converging uniformly to a limit $f : S \to Y$. Then f is continuous.

Proof. Fix $\varepsilon > 0$. There exists N such that

$$n \geq N \implies \|f_n(x) - f(x)\| < \varepsilon \forall x \in S$$

Fix $x \in S$ then there exists $\delta > 0$ such that

$$\|y - x\| < \delta \implies \|f_n(y) - f_n(x)\| < \varepsilon$$

Therefore

$$\|y - x\| < \delta \implies \|f(x) - f(y)\| \leq \|f(x) - f_n(x)\| + \|f_n(x) - f_n(y)\| + \|f(y) - f_n(y)\| < 3\varepsilon$$

Consequently f is also continuous. ∎

Definition 5.4.36 (Bounded Function Space)
*Let S be a metric space and Y a normed vector space. Define the vector space of **bounded functions** by*

$$\mathcal{B}(S, Y) := \{f : S \to Y \mid \exists C > 0 \text{ s.t. } \|f(x)\| \leq C \text{ for all } x \in S\}$$

*This is a normed vector space under the **sup norm** given by*

$$\|f\|_\infty := \sup_{x \in S} \|f(x)\|$$

Evidently convergence in this norm is equivalent to uniform convergence.

Proposition 5.4.37 (Bounded Functions are Complete)
Suppose Y is a Banach Space, then $\mathcal{B}(S, Y)$ is also a Banach Space.

Proof. Let $\{f_n\}$ be a cauchy sequence of maps $S \to Y$ with respect to the sup norm. Then for every $\varepsilon > 0$ there exists N such that

$$n, m \geq N \implies \|f_n - f_m\| < \varepsilon$$

In particular for every $x \in S$ we have $\{f_n(x)\}$ is cauchy. As Y is complete it is a convergent sequence, for which we denote the limit by $f(x)$. For every $x \in S$ we may also choose some $m \geq N$ such that

$$\|f_m(x) - f(x)\| < \varepsilon$$

then

$$\|f(x) - f_n(x)\| \leq \|f(x) - f_m(x)\| + \|f_m(x) - f_n(x)\| < 2\varepsilon$$

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As this inequality is independent of m we find that $\|f - f_n\| \to 0$ as required. We may also show easily that it is bounded. ∎

5.4.7 Continuous Linear Maps

Proposition 5.4.38 (Continuous Linear Maps)
Let $f : X \to Y$ be a linear map of normed vector spaces. Then the following are equivalent

a) f is continuous

b) There exists a constant $C > 0$ such that

$$\|f(x)\| \leq C\|x\|$$

c) f is bounded on the unit sphere

$$\sup\{\|f(x)\|_Y \mid \|x\|_X = 1\} < \infty$$

When X is finite-dimensional then every linear map is continuous.

Proof. Suppose b) holds then f is even uniformly continuous because

$$\|x - y\| < \frac{\varepsilon}{C} \implies \|f(x) - f(y)\| < \varepsilon$$

Conversely given a) there exists δ such that

$$\|x\| \leq \delta \implies \|f(x)\| < 1$$

Therefore

$$\left\| f\left(\frac{\delta x}{\|x\|}\right) \right\| < 1 \implies \|f(x)\| < \frac{1}{\delta}\|x\|$$

and we may take $C = \frac{1}{\delta}$. Given b) evidently $\|f\| \leq C < \infty$, and conversely $\|f(x)\| \leq \|f\|\|x\|$. ∎

Proposition 5.4.39
Let X, Y be normed vector spaces. Denote by $L(X, Y)$ the set of continuous linear maps. It is a normed vector space, with norm given by

$$\|\theta\| := \sup\{\|\theta(x)\|_Y \mid \|x\|_X = 1\}$$

Furthermore if Y is complete, so is $L(X, Y)$.

Proof. Evidently $\|\lambda\theta\| = |\lambda|\|\theta\|$ and $\|\theta_1 + \theta_2\| \leq \|\theta_1\| + \|\theta_2\| < \infty$. Let $\{\theta_n\}$ be a cauchy sequence in $L(X, Y)$. Given $x \in X$ then

$$\|\theta_n(x) - \theta_m(x)\| \leq \|\theta_n - \theta_m\|\|x\|$$

whence $\{\theta_n(x)\}$ is a cauchy sequence, and therefore converges to $\theta(x)$. By uniqueness of limits we see that θ is linear. Furthermore by the reverse triangle inequality (...)

$$|\|\theta_n\| - \|\theta_m\|| \leq \|\theta_n - \theta_m\|$$

in otherwords $\|\theta_n\|$ is a cauchy sequence and converges to some limit C. Suppose that $\|x\| = 1$ then

$$\|\theta(x)\| \leq \|\theta(x) - \theta_n(x)\| + \|\theta_n(x)\| \leq \|\theta(x) - \theta_n(x)\| + \|\theta_n\|$$

Choose N_1 such that $n \geq N_1 \implies \|\theta_n\| \leq C + \varepsilon$ and N_2 such that $n \geq N_2 \implies \|\theta(x) - \theta_n(x)\| < \varepsilon$. As ε was arbitrary we see that $\|\theta\| \leq C$ and in particular is finite. ∎

Proposition 5.4.40 (Finite-Dimensional Linear Maps are continuous)
Suppose that $\theta : X \to Y$ is a linear map of normed vector spaces and X is finite-dimensional. Then θ is continuous.

Proof. Let $\{e_1, \cdots, e_n\}$ be a basis. By (5.4.22) we know that $\|\cdot\|_X$ is equivalent to the infinity norm $\|\cdot\|_\infty$. therefore it is sufficient to show that θ is continuous with respect to this norm. However

$$\left\| \theta\left(\sum_{i=1}^n \lambda_i e_i\right) \right\| \leq \sum_{i=1}^n |\lambda_i|\|\theta(e_i)\| \leq n\left\|\sum_{i=1}^n \lambda_i e_i\right\|_\infty \max_i \|\theta(e_i)\|$$

∎

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Proposition 5.4.41 (Continuous Multilinear Maps)
Let $\psi : X_1 \times \cdots \times X_p \to Z$ be a multilinear map of normed vector spaces (using the product norm). Then the following are equivalent

a) ψ is continuous

b) There exists a constant $C > 0$ such that

$$\|\psi(x_1, \cdots, x_p)\| \leq C\|x_1\| \cdots \|x_p\|$$

c) The following quantity is finite

$$\sup\{\|\psi(x_1, \cdots, x_p)\| \mid \|x_1\| = \cdots = \|x_p\| = 1\} < \infty$$

In this case $L(X_1, \cdots, X_p; Z)$ is a normed vector space with norm given by expression in c). Furthermore if Z is complete so is $L(X_1, \cdots, X_p; Z)$.

Finally if $X_1, \cdots X_p$ are finite-dimensional then every multilinear map is continuous.

Proof. **a) $\implies$ b)** Suppose there is no such constant C, then there exists $(x_n^1, \cdots, x_n^p) \in X_1 \times \cdots \times X_p$ such that

$$\|\psi(x_n^1, \cdots, x_n^p)\| > n^p \|x_n\| \|y_n\|$$

Define $\hat{x}_n^i := \frac{x_n^i}{n\|x_n\|}$. Then $\|(\hat{x}_n^1, \cdots, \hat{x}_n^p)\| = \frac{1}{n} \to 0$ but $B(0, \cdots, 0) = 0$ by linearity and $B(\hat{x}_n^1, \cdots, \hat{x}_n^p) > 1$, which contradicts continuity of limits (...).

b) $\implies$ a) Observe by linearity

$$\begin{aligned}\psi(x_1, \cdots, x_p) - \psi(y_1, \cdots, y_p) &= \psi(x_1 - y_1, x_2, \cdots, x_p) + \psi(y_1, x_2 - y_2, x_3, \cdots, x_p) + \cdots + \psi(y_1, \cdots, y_{p-1}, x_p - y_p) \\ &\leq C\|x_1 - y_1\|\|x_2\| \cdots \|x_p\| + C\|x_2 - y_2\|\|y_1\|\|x_3\| \cdots \|x_p\| + \cdots \\ &+ C\|x_p - y_p\|\|y_1\| \cdots \|y_{p-1}\|\end{aligned}$$

Fix any $\varepsilon > 0$ then it is enough to show that

$$\|x_i - y_i\| < \min(\|y_i\|, \varepsilon) \implies \|\psi(x_1, \cdots, x_p) - \psi(y_1, \cdots, y_p)\| < M\varepsilon$$

for some positive constant M depending only on y. For this implies that $\|x_i\| < 2\|y_i\|$ so that

$$\|\psi(x_1, \cdots, x_p) - \psi(y_1, \cdots, y_p)\| \leq \varepsilon \sum_{i=1}^{p} 2C\|y_1\| \cdots \widehat{\|y_i\|} \cdots \|y_p\|$$

as required.

b) $\iff$ c) Straightforward

The supremum is evidently a norm, and the verification of completeness follows as before. ∎

Definition 5.4.42 (Space of Continuous Multilinear Maps)
Let $X_1, \cdots, X_p$, Y be normed vector spaces then let $L(X_1, \cdots, X_p; Y)$ denote the K-vector space of all continuous multilinear maps.

Denote by $L^p(X, Y) := L(X^p; Y)$ the vector space of continuous multilinear maps on the p-fold product of X.

Proposition 5.4.43 (Alternating Multilinear Maps)
*Let $L_a(X_1, \cdots, X_p; Y)$ denote the subspace of continuous linear maps $L(X_1, \cdots, X_p; Y)$ which are **alternating** (3.8.5). Then this is a closed subset. In particular when Y is complete so is $L_a(X_1, \cdots, X_p; Y)$.*

As before define $L_a^p(X, Y) := L_a(X^p; Y)$.

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5.5 Differentiable Functions

Definition 5.5.1 (Differentiable Scalar Function)
*Let K be a complete valued field, $U \subset K$ an open subset, Y a normed vector space over K and $f : U \to Y$ a function. We say that f is **differentiable** at $x \in U$ if the following relation holds*

$$\forall \varepsilon > 0 \exists \delta > 0 \text{ s.t. } 0 < |y - x| < \delta \implies \left\| \frac{f(y) - f(x)}{y - x} - f'(x) \right\| < \varepsilon$$

for some $f'(x) \in K$.

*We say that f is simply **differentiable** if it is differentiable at every point of U.*

Trivially if f is differentiable then so is $f|_V$ for any open subset $V \subset U$.

Proposition 5.5.2
Let K be a valued field, $U \subset K$ an open subset and $f : U \to Y$ a function which is differentiable at $x \in U$. Suppose $x_n \to x$ is a sequence such that $x_n \neq x$. Then

$$\frac{f(x_n) - f(x)}{x_n - x} \to f'(x)$$

Proposition 5.5.3
Let K be a valued field, $U \subset K$ an open subset and $f : U \to Y$ a function which is differentiable at $x \in U$. Then f is continuous at x.

Proposition 5.5.4 (Arithmetic of Differentiable Functions)
Let K be a valued field, $U \subset K$ an open subset and $f, g : U \to Y$ be functions differentiable at $x \in U$. Then

a) $(f + g)'(x) = f'(x) + g'(x)$
b) $(\lambda f)'(x) = \lambda f'(x)$

If Y has continuous algebra structure (e.g. $Y = K$) then

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x)$$

and if $Y = K$ and f is non-zero on U then

$$\left(\frac{1}{f}\right)'(x) = -f' \frac{x}{f(x)^2}$$

Finally every linear (and therefore constant) function is trivially differentiable.

Proposition 5.5.5
Let K be a valued field, $U \subset K$ an open subset and $U = \bigcup_{\alpha \in \mathcal{A}} V_\alpha$ an open cover. Suppose $f : U \to Y$ is a function. Then f is (continuously) differentiable if and only if $f|_{V_\alpha}$ is (continuously) differentiable for all $\alpha \in \mathcal{A}$.

Corollary 5.5.6
Let K be a valued field of characteristic zero and n a positive integer. Then the derivative of x^n is nx^{n-1}.

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5.6 Power Series

Definition 5.6.1 (Convergent Power Series)
*Let K be a field. Then a **power series** is a formal expression of the form*

$$P(X) := \sum_{n=0}^{\infty} a_n X^n$$

where $a_n \in K$. We denote the ring of power series by $K[[X]]$. If K is a valued field then, we say this converges at $z \in K$ if the series

$$\sum_{n=0}^{\infty} a_n z^n$$

converges in K.

Proposition 5.6.2 (Radius of convergence)
Let K be a complete valued field and consider the formal power series

$$\sum_{n=0}^{\infty} a_n X^n \quad a_n \in K \ .$$

*Define the **radius of convergence** to be*

$$R := \left(\limsup \sqrt[n]{|a_n|}\right)^{-1}$$

where we understand $0^{-1} = \infty$ and $\infty^{-1} = 0$. Then

a) For every $|z| < R$ the series $\sum_{n=0}^{\infty} a_n z^n$ is absolutely convergent, and we denote this function by $f(z)$.
b) For every $|z| > R$ the series $\sum_{n=0}^{\infty} a_n z^n$ is divergent
c) R is the unique value such that a), b) hold.
d) The limit $f(z)$ is continuous and differentiable in the open set $B(0,R)$ with derivative

$$f'(z) = \sum_{n=1}^{\infty} n a_n z^{n-1}$$

which has radius of convergence at least R

Proof.

a) First consider the case $R < \infty$. Define $c_n := a_n z^n$ then $\sqrt[n]{|c_n|} = |z| \sqrt[n]{|a_n|}$. Then clearly $|z| < R \implies \limsup \sqrt[n]{|c_n|} < 1$ whence the result follows from (5.4.31). In the case $R = \infty$ we see that $\limsup \sqrt[n]{|c_n|} = 0$ for all z and the result follows similarly.
b) This follows similarly by (5.4.27).
c) This is evident for suppose $R' \neq R$ satisfies the same properties and consider z such that $|z| = \frac{R'+R}{2}$ for a contradiction.
d) Let R' be the radius of convergence for the given series, we claim that $R' \geq R$, or in otherwords that $|z| < R$ implies $\sum_{n=0}^{\infty}(n+1)a_{n+1}z^n$ converges. For we may choose w such that $|z| < |w| < R$. Then by definition

$$\sum_{n=0}^{\infty} a_n w^n$$

converges. In particular by (...) $|a_n w^n| \to 0$ and therefore $|a_n w^n|$ is bounded, by M. Therefore

$$\sum_{n=1}^{\infty} \left|n a_n z^{n-1}\right| = \sum_{n=1}^{\infty} n \left|a_n w^{n-1}\right| \left|\frac{z}{w}\right|^{n-1} \leq \frac{M}{|w|} \sum_{n=1}^{\infty} n \left|\frac{z}{w}\right|^{n-1}$$

The latter series converges by the Ratio Test (5.4.32) and therefore the series converges by the Comparison Test (5.4.28). Consider distinct $z, w \in B(0,R)$ then

$$\frac{f(w) - f(z)}{w - z} - f'(z) = \sum_{n=2}^{\infty} a_n \left[\frac{w^n - z^n}{w - z} - n z^{n-1}\right]$$

Observe that for $n \geq 1$

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$$\frac{w^n - z^n}{w - z} = w^{n-1} + w^{n-2}z + \cdots + wz^{n-2} + z^{n-1}$$

Therefore for $n \geq 2$

$$\begin{aligned}\frac{w^n - z^n}{w - z} - nz^{n-1} &= \big(w^{n-1} - z^{n-1}\big) + z\big(w^{n-2} - z^{n-2}\big) + \cdots + z^{n-2}(w - z) \\ &= (w - z)\big[w^{n-2} + 2w^{n-3}z + \cdots + (n-1)z^{n-2}\big]\end{aligned}$$

For any w such that $|w - z| < \frac{R-|z|}{2}$ we have $|w| \leq \frac{R+|z|}{2} =: r < R$. Then

$$\left|\frac{w^n - z^n}{w - z} - nz^{n-1}\right| \leq |w - z|\frac{n(n-1)}{2}r^{n-2}$$

which implies

$$\left|\frac{f(w) - f(z)}{w - z} - f'(z)\right| \leq |w - z|\sum_{n=2}^{\infty}|a_n|\frac{n(n-1)}{2}r^{n-2}$$

The sum is finite because the second formal derivative is absolutely convergent at r, by the part already demonstrated. Therefore we see that f is differentiable at z with derivative $f'(z)$. This also shows that f is continuous at z (...).

■

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5.7 Real Analysis

5.7.1 Closed Intervals

Proposition 5.7.1 (k-cells are closed)
Subsets of the form

$$[a_1, b_1] \times \cdots \times [a_k, b_k] \subset \mathbb{R}^k$$

are closed.

Proof. By definition of the product topology we may reduce to the case $k = 1$. Then $[a, b]$ is sequentially closed by (5.1.21), which is equivalent to being closed by (4.1.19) as $\mathbb{R}$ is first countable. ∎

5.7.2 Power Function

Proposition 5.7.2 (Power function)
Let $\alpha \in \mathbb{Q}$ and $x \in \mathbb{R}$ be non-negative numbers. Suppose $\alpha = \frac{m}{n}$ for integers m, n, where $n > 0$. Then the function

$$x^\alpha := \sqrt[n]{x^m} = \left(\sqrt[n]{x}\right)^m$$

is continuous and strictly increasing/decreasing (according to the sign of α). Furthermore

$$x^\alpha x^\beta = x^{\alpha+\beta}$$

Proof. We first consider the case $\alpha > 0$ and $m, n > 0$. The two definitions are equivalent because x^α is the unique function satisfying $(x^\alpha)^n = x^m$, the latter being an injective function. We may consider the cases $\alpha = m$ and $\alpha = \frac{1}{n}$ separately (since the composition of continuous and increasing functions is continuous and increasing). In light of (5.1.32) and (5.4.10) then it remains to show that $\sqrt[n]{x}$ is continuous. However by (5.1.31)

$$\left|\sqrt[n]{y} - \sqrt[n]{x}\right| \leq \frac{|y - x|}{n\sqrt[n]{x}^{n-1}} .$$

Suppose $\alpha = \frac{m}{n}$ and $\beta = \frac{p}{q}$ then $\alpha + \beta = \frac{mq+np}{nq}$ and

$$\left(x^\alpha x^\beta\right)^{nq} = (x^\alpha)^{nq}\left(x^\beta\right)^{nq} = x^{mq}x^{np} = x^{mq+np}$$

whence by uniqueness $x^\alpha x^\beta = x^{\alpha+\beta}$.

We may define $x^0 = 1$ and $x^{-\alpha} = (x^\alpha)^{-1}$, and verify that the addition formula still holds. ∎

5.7.3 Countability

Proposition 5.7.3 ($\mathbb{R}$ is separable and second countable)
The subset $\mathbb{Q} \subset \mathbb{R}$ is dense. Furthermore $\mathbb{R}$ is second countable, with a base formed by open balls with rational center and radius.

Similarly $\mathbb{Q}^k$ is dense in $\mathbb{R}^k$, which is second countable.

Proof. We only need to show that $\mathbb{Q} \subset \mathbb{R}$ is dense, and the rest follows from (5.3.15) because $\mathbb{Q}$ is countable.

By (5.1.26) for every $x \in \mathbb{R}$ there is a sequence $x_n \in \mathbb{Q}$ such that $x_n \to x$. Then (4.1.19) shows that $x \in \overline{\mathbb{Q}}$ as required. ∎

5.7.4 Bolzano-Weierstrass Theorem

Proposition 5.7.4 (Monotone Subsequence Theorem)
Let (a_n) be a sequence in an ordered field. Then it has a monotone subsequence.

Proof. Define the set of peaks to be

$$X = \{n \in \mathbb{N} \mid m \geq n \implies x_n \geq x_m\}$$

If X is infinite then the elements determine a decreasing subsequence. Suppose X is finite and define $n_1 := \max(X) + 1$. We claim inductively that there is a sequence of integers

$$n_1 < n_2 < \cdots < n_k < \cdots$$

such that a_{n_k} is increasing. Evidently $n_k \notin X$ whence there exists $n_{k+1} > n_k$ such that $a_{n_{k+1}} > a_{n_k}$. ∎

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Proposition 5.7.5 (Bolzano-Weierstrass Theorem)
Let (a_n) be a bounded sequence in $\mathbb{R}$. Then it contains a convergent subsequence.

Proof 1.. By (5.7.4) a_n has a monotone subsequence. By (5.1.23) this subsequence is convergent. ∎

Proof 2.. Define two sequences of real numbers (l_n), (u_n) recursively as follows

$$(l_n, u_n) := \begin{cases} (L, U) & n = 1 \\ \left(l_{n-1}, \frac{l_{n-1}+u_{n-1}}{2}\right) & \{a_n\} \cap \left[l_{n-1}, \frac{l_{n-1}+u_{n-1}}{2}\right] \text{ is infinite} \\ \left(\frac{l_{n-1}+u_{n-1}}{2}, u_{n-1}\right) & \text{otherwise} \end{cases}$$

Then by definition

a) (l_n) is increasing and (u_n) is decreasing
b) $l_n \leq u_m$ for all n, m
c) $|u_n - l_n| = \frac{U-L}{2^n} \to 0$
d) $[l_n, u_n]$ contains infinitely many elements of the sequence (a_n)

We claim that

$$\bigcap_{n\in\mathbb{N}} [l_n, u_n] = \{\sup l_n\} = \{\inf u_n\}$$

Suppose x, y lie in the intersection then by c) we have $|x - y| < \varepsilon$ for all ε, whence they are equal. So the set consists of at most one element. We claim $\sup l_n$ lies in the intersection. For $l_n \uparrow \sup l_n$ by (...) and $l_n \leq u_m$ whence $\sup l_n \leq u_m$ by (...) as required.

Finally let $\left(a_{n_k}\right)$ be a subsequence such that $a_{n_k} \in [l_k, u_k]$ consequently by c) we have $\left|a_{n_k} - \alpha\right| \to 0$ where $\alpha := \sup l_n$. ∎

Corollary 5.7.6 (Bolzano-Weierstrass in $\mathbb{R}^k$)
Let (a_n) be a bounded sequence in $\mathbb{R}^k$. Then it contains a convergent subsequence.

Proof. Observe that $(a_n) = \left(a_n^1, \cdots, a_n^k\right)$ where a^i is a bounded sequence in $\mathbb{R}$. By applying (5.7.5) to each coordinate in turn we may find a subsequence n_j such that $a^i_{n_j} \to a^i$ for $i = 1 \cdots k$. By definition of the product norm it is clear that $a_{n_j} \to \left(a^1, \cdots, a^k\right)$. ∎

Proposition 5.7.7 (Heine-Borel Theorem)
Let X be a subset of $\mathbb{R}^k$. Then the following are equivalent

a) X is closed and bounded
b) X is sequentially compact
c) X is compact

In particular every closed k-cell $[\boldsymbol{a}, \boldsymbol{b}]$ is compact.

Proof 1.. Recall from (5.7.3) that $\mathbb{R}^k$ is second countable, and therefore so is X. Therefore the equivalences $b) \iff c)$ follows from (4.1.106).

$b), c) \implies a)$ We may use (4.1.110) to show that X is closed, observing that $\mathbb{R}^k$ is both first countable and Kolmogorov. Alternatively we may observe $\mathbb{R}^k$ is Hausdorff and appeal to (4.1.107).

To show that it is bounded we may assume it is not to find a sequence x_n such that $\|x_n\| \geq n$. By assumption this has a convergent subsequence $x_{n_j} \to x$. Further there is some N_1 such that $N_1 > \|x\|$ and some N_2 such that $n \geq N_2 \implies \|x_n - x\| < N_1 - |x| \implies \|x_n\| < N_1$. Then taking $n = \max(N_1, N_2)$ implies $\|f(x_n)\|$ is both less and greater than N_1, a contradiction.

Alternatively we may reduce to the case $k = 1$ by noting that $\pi_i(X)$ is compact (4.1.109). The family $\{B(n; 1) \mid n \in \mathbb{N}\}$ forms an open cover, for which there must exist a finite subcover. This immediately shows that $\pi_i(X)$, and therefore X, is bounded.

$a) \implies b), c)$ By assumption $X \subset [-M, M] \times \cdots \times [-M, M]$ for some M, which by (5.7.6) is sequentially compact. We may then use either (4.1.108) or (4.1.111). ∎

Proof 2.. Firstly X is complete iff it is closed by (5.3.13). Evidently a subset of $\mathbb{R}$ is totally bounded iff it is bounded, and the same argument may be extended to $\mathbb{R}^k$. Therefore the equivalence follows from (5.3.17). ∎

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5.7.5 Boundedness Theorem

Proposition 5.7.8
Let X be a compact topological space and $f : X \to \mathbb{R}^k$ be a continuous function. Then f is bounded and attains its bounds.

Proof 1.. We may prove directly the case $X = [a, b]$ and $k = 1$.

Suppose f is not bounded, then there is some sequence (x_n) in X such that $|f(x_n)| \geq n$. By choosing a convergent subsequence (5.7.5) we may assume that $x_n \to x$. Further X is closed so $x \in X$ (4.1.19). By continuity (5.4.13) $f(x_n) \to f(x)$. Choose $N_1 > |f(x)|$ and N_2 such that $n \geq N_2 \implies |f(x_n) - f(x)| < N_1 - |f(x)| \implies |f(x_n)| < N_1$. Then $n \coloneqq \max(N_1, N_2) \implies |f(x_n)| < N_1$ and $|f(x_n)| > N_1$ a contradiction. ∎

Proof 2.. We may reduce to the case $k = 1$ by considering the composition with the norm $\|\cdot\|$.

By (4.1.109) (or (4.1.112)) $f(X)$ is (sequentially) compact. Then by (5.7.7) $f(X)$ is closed and bounded.

Let $y = \sup f(X)$. By (5.1.18) there exists $y_n \to y$ with $y_n \in f(X)$. Since $f(X)$ is closed we see $y \in f(X)$ by (4.1.19) as required. ∎

5.7.6 Intermediate Value Theorem

Proposition 5.7.9 (Intervals are connected)
Let $X \subset \mathbb{R}$. Then the following are equivalent
a) X is connected
b) For every $x < y < z$ such that $x, z \in X$, we have $y \in X$

In particular $(a, b], (a, b), [a, b)$ are connected.

Proof. $b) \implies a)$ Suppose X is disconnected, then $X = U \sqcup V$ with U, V non-empty open and closed subsets of X. Let $x \in U$ and $z \in V$, and assume wlog that $x < z$. Define

$$\alpha \coloneqq \sup(U \cap (x, z))$$

By (5.1.21) $\alpha \in [x, z] \subset X$. By (5.1.18) there exists $\alpha_n \in U \cap (x, z)$ such that $\alpha_n \uparrow \alpha$. Then by (4.1.20) we have $\alpha \in \mathrm{cl}_{\mathbb{R}}(U) \cap X \overset{(4.1.21)}{=} \mathrm{cl}_X(U) \overset{(4.1.19)}{=} U$. If $\alpha = z$ then we reach a contradiction as U and V were assumed to be disjoint. Similarly if $\alpha < z$, as U is open in X, we may choose $0 < \varepsilon < z - \alpha$ such that $(\alpha - \varepsilon, \alpha + \varepsilon) \cap X \subset U$. By assumption $\alpha + \varepsilon \in X$ whence $\alpha + \varepsilon \in U$, which contradicts maximality of α.

$a) \implies b)$ Suppose we have $x < y < z$ such that $x, z \in X$ but $y \notin X$. Then define the open subsets $U = (-\infty, y) \cap X$ and $V = (y, \infty) \cap X$. Then

$$U \coloneqq (-\infty, y) \cap X = (-\infty, y] \cap X$$

is both open and closed. ∎

Proposition 5.7.10 (Intermediate Value Theorem)
Let $a < b$ be real numbers and $f : [a, b] \to \mathbb{R}$ continuous such that $f(a) < f(b)$. Then for every $f(a) < c < f(b)$ there exists $x \in (a, b)$ such that $f(x) = c$.

Proof 1.. Consider

$$X \coloneqq \{x' \in [a, b] \mid f(x') < c\}$$

By definition X is bounded, and $a \in X$, so it is non-empty, so we may define $x \coloneqq \sup X$. We claim that $f(x) = c$. By (5.1.18) there is a sequence $x_n \in X$ such that $x_n \to x$. By (5.4.13) $f(x_n) \to f(x)$ and by (5.1.21) $x \leq b$ and $f(x) \leq c$. Further by assumption $x \neq b$.

To show $f(x) \geq c$ consider the sequence $x_n \coloneqq x + \min\left(\frac{1}{n}, b - x\right) \in [a, b]$. Evidently $x_n \to x$ so by continuity $f(x_n) \to f(x)$. By construction $f(x_n) \geq c$ whence $f(x) \geq c$.

Therefore we conclude $f(x) = c$ as required. ∎

Proof 2.. By (5.7.9) $[a, b]$ is connected, whence by (4.1.118) $f([a, b])$ is connected. The result follows from (5.7.9) again. ∎

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5.7.7 Mean-Value Theorem

Proposition 5.7.11 (Mean-Value Theorem)
Let $f : [a, b] \to \mathbb{R}$ be a continuous function which is differentiable on (a, b). Then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Proof 1.. Define the function

$$g(x) := f(x) - \frac{f(b) - f(a)}{b - a}(x - a)$$

Then g satisfies the same conditions with $g(a) = g(b) = f(a)$ and $g'(x) = f'(x) - \frac{f(b)-f(a)}{b-a}$. Evidently it is sufficient to find $c \in (a, b)$ such that $g'(c) = 0$.

By (5.7.8) g is bounded and attains its maximum and minimum, say at c, d respectively. If both $c, d \in \{a, b\}$ then g must be the constant function whence g' is exactly zero. Otherwise we may assume for example that $c \in (a, b)$. By (5.5.2)

$$\lim_{n\to\infty} \frac{g\left(c \pm \frac{1}{n}\right) - g(c)}{\pm\frac{1}{n}} = g'(c)$$

As c is an extremum we may deduce from (5.1.21) that both $g'(c) \leq 0$ and $g'(c) \geq 0$, whence $g'(c) = 0$. ∎

Corollary 5.7.12
Let $f : [a, b] \to \mathbb{R}$ a continuous function which is differentiable on (a, b). If $f'(x)$ is positive (resp. strictly positive, negative, strictly negative) then $f(x)$ is increasing (resp. strictly increasing, decreasing, strictly decreasing).

Proposition 5.7.13
Let $f : (a, b) \to \mathbb{R}$ be a strictly increasing (resp. decreasing) continuous function. Then f is a homeomorphism onto its image.

Proof. Evidently f is injective, therefore it is only required to show that the map

$$f^{-1} : f(U) \to U$$

is continuous. Without loss of generality we may assume that f is strictly increasing. Suppose $y = f(x)$ then we require to show for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$|y' - y| < \delta \implies \left|f^{-1}(y') - f^{-1}(y)\right| < \varepsilon$$

Define $\varepsilon' \leq \varepsilon$ such that $(x - \varepsilon', x + \varepsilon') \subset (a, b)$. Define $\delta := \min(f(x + \varepsilon') - f(x), f(x) - f(x - \varepsilon'))$. Then

$$|y' - y| < \delta \implies f(x - \varepsilon') < y' < f(x + \varepsilon') \implies x - \varepsilon' < f^{-1}(y') < x + \varepsilon' \implies \left|f^{-1}(y') - f^{-1}(y)\right| < \varepsilon' \leq \varepsilon$$

as f^{-1} is also monotonic. ∎

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5.8 Integration

5.8.1 Algebras of Sets

Definition 5.8.1
Let Ω be a set and $\mathcal{F}$ a family of subsets of Ω. We consider the following types of families

*a) $\mathcal{F}$ is a **semi-ring** if the following properties hold*
- *i) $\emptyset \in \mathcal{F}$*
- *ii) $A_1, ..., A_n \in \mathcal{F} \implies A_1 \cap ... \cap A_n \in \mathcal{F}$*
- *iii) For all $A, B \in \mathcal{F}$ there is a family of disjoint subsets $C_1, ..., C_n \in \mathcal{F}$ such that*

$$B \setminus A = C_1 \sqcup ... \sqcup C_n$$

*b) $\mathcal{F}$ is a **ring** (resp. **algebra**) if the following properties hold*
- *i) $\emptyset \in \mathcal{F}$ (resp. $\Omega \in \mathcal{F}$)*
- *ii) $A_1, ..., A_n \in \mathcal{F} \implies A_1 \cup ... \cup A_n \in \mathcal{F}$*
- *iii) $A, B \in \mathcal{F} \implies A \setminus B \in \mathcal{F}$*

*c) $\mathcal{F}$ is a **σ-ring** (resp. **σ-algebra**) if the following properties hold*
- *i) $\emptyset \in \mathcal{F}$ (resp. $\Omega \in \mathcal{F}$)*
- *ii) $A_1, ..., A_n, ... \in \mathcal{F} \implies \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$*
- *iii) $A, B \in \mathcal{F} \implies A \setminus B \in \mathcal{F}$*

Evidently σ-ring $\implies$ ring $\implies$ semi-ring.

Example 5.8.2
The family of half-open intervals $\mathcal{F} = \{[a, b) \mid a < b\}$ forms a semi-ring on $\mathbb{R}$.

More generally the family of products of half-open intervals $\{[a_1, b_1) \times ... \times [a_n, b_n) \mid a_i < b_i\}$ forms a semi-ring on $\mathbb{R}^n$.

Lemma 5.8.3
Let Ω be a set and $\{\mathcal{F}_\alpha\}_{\alpha \in \mathcal{A}}$ a family of rings (resp. algebras, σ-rings, σ-algebras). Then the family of subsets

$$\bigcap_{\alpha \in \mathcal{A}} \mathcal{F}_\alpha$$

is a ring (resp. algebra, σ-ring, σ-algebra). In particular every family of subsets is contained in a smallest ring (resp. algebra, σ-ring, σ-algebra).

Definition 5.8.4 (Borel Algebra)
*Let X be a topological space. We say that the family of **borel sets** $\mathcal{B}(X)$ is the smallest σ-algebra containing the family of open sets.*

Proposition 5.8.5
Let $\mathcal{R}$ be a semi-ring (resp. semi-algebra). Then the family of finite disjoint unions

$$\mathcal{F} := \{A_1 \sqcup ... \sqcup A_n \mid A_1, ..., A_n \in \mathcal{R} \text{ pairwise disjoint}\}$$

is a ring (resp. algebra), and indeed the smallest ring (resp. algebra) containing $\mathcal{R}$.

Lemma 5.8.6 (Semi-Ring Extended Set Difference)
Let $\mathcal{R}$ be a semi-ring and $A_0, A_1, ..., A_n \in \mathcal{R}$. Then

$$A_0 \setminus \bigcup_{i=1}^{n} A_i = \bigsqcup_{k=1}^{m} B_k$$

for some disjoint sets $B_k \in \mathcal{R}$.

Proof. We proceed by induction, the case $n = 1$ holding by definition. Suppose the relation holds for n. Then

$$A_0 \setminus \bigcup_{i=1}^{n+1} A_i = \left(\bigsqcup_{k=1}^{m} B_k \right) \setminus A_{n+1} = \bigsqcup_{k=1}^{m} (B_k \setminus A_{n+1})$$

and by definition of $\mathcal{R}$ we have

$$B_k \setminus A_{n+1} = \bigsqcup_{j=1}^{N} C_{kj} \quad k = 1...m$$

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and $C_{kj} \in \mathcal{R}$. As B_k are disjoint we see that the C_{kj} are disjoint for all j,k and the result follows immediately. ■

5.8.2 Set Functions

Definition 5.8.7 (Finitely Additive Set Function)
Let $\mathcal{F}$ be a family of subsets of Ω containing $\emptyset$. A ***set function***

$$\mu : \mathcal{F} \to [0, \infty]$$

is ***finitely additive*** *if it satisfies*

a) $\mu(\emptyset) = 0$

b) $\mu(A_1 \sqcup ... \sqcup A_n) = \sum_{i=1}^{n} \mu(A_i)$ *whenever this is well-defined*

We say a set $A \in \mathcal{F}$ is ***finite*** *if $\mu(A) < \infty$, and μ is* ***finite*** *if $\mu(\mathcal{F}) < \infty$.*

*We say that μ is σ-****finite*** *if for every set $A \in \mathcal{F}$ there exists a sequence A_i such that $A \subset \bigcup_{i=1}^{\infty} A_i$ and $\mu(A_i) < \infty$. If $\Omega \in \mathcal{F}$ then it is sufficient for this condition to hold for Ω.*

Proposition 5.8.8 (Extension Theorem I)
Let $\mathcal{R}$ be a semi-ring, and $\mathcal{F}$ the ring generated by $\mathcal{R}$. Given a finitely additive set function $\mu : \mathcal{R} \to [0, \infty]$ there exists a unique extension to $\mathcal{F}$ given by

$$\mu(A_1 \sqcup ... \sqcup A_n) = \sum_{i=1}^{n} \mu(A_i)$$

which is finitely additive.

Proof. In order to ensure the definition is well-defined suppose

$$A_1 \sqcup ... \sqcup A_n = B_1 \sqcup ... \sqcup B_m$$

Then in particular

$$A_i = (A_i \cap B_1) \sqcup ... \sqcup (A_i \cap B_m)$$

We may deduce by the additive property that

$$\sum_{i=1}^{n} \mu(A_i) = \sum_{i=1}^{n} \sum_{j=1}^{m} \mu\big(A_i \cap B_j\big)$$

Evidently this is symmetric in A and B so this also equals $\sum_{j=1}^{m} \mu\big(B_j\big)$. This shows that the measure μ is well-defined. The additivity property is straightforward. ■

Proposition 5.8.9 (Properties of finitely-additive set functions)
Let $(\Omega, \mathcal{R})$ be a ***semi-ring*** *and $\mu : \mathcal{R} \to [0, \infty]$ a finitely additive set function. Then*

a) μ *is* ***monotone****, that is for sets $A \subset B$ we have*

$$\mu(A) \leq \mu(B)$$

b) Suppose $A_1 \sqcup ... \sqcup A_n \subset A_0$ for $A_i \in \mathcal{R}$ then

$$\sum_{i=1}^{n} \mu(A_i) \leq \mu(A_0)$$

c) Suppose $A_0 \subset A_1 \cup ... \cup A_n$ for $A_i \in \mathcal{R}$ then

$$\mu(A_0) \leq \sum_{i=1}^{n} \mu(A_i)$$

Proof.

a) By definition $B \setminus A = C_1 \sqcup ... \sqcup C_n$ and therefore by finite additivity

$$\mu(B) = \mu(C_1 \sqcup ... \sqcup C_n \sqcup A) = \mu(C_1) + ... + \mu(C_n) + \mu(A) \geq \mu(A)$$

b) By (5.8.6)

$$A_0 \setminus \bigsqcup_{i=1}^{n} A_i = \bigsqcup_{k=1}^{m} B_k$$

for some $B_k \in \mathcal{R}$ disjoint. Observe they are by construction disjoint from A_i therefore

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$$A_0 = A_1 \sqcup ... \sqcup A_n \sqcup B_1 \sqcup ... \sqcup B_m$$

The result follows from finite additivity.

c) By replacing A_i with $A_i \cap A_0$ we may assume that $A_0 = A_1 \cup ... \cup A_n$ by a). Observe that by (5.8.6)

$$\hat{A}_i := A_i \setminus \bigcup_{j=1}^{i-1} A_j = \bigsqcup_{k=1}^{m} C_{ik}$$

for $C_{ik} \in \mathcal{R}$ (if necessary padding out by $\emptyset$). Therefore

$$A_0 = \bigsqcup_{i=1}^{n} \hat{A}_i = \bigsqcup_{i,k} C_{ik}$$

and by finite additivity

$$\mu(A_0) = \sum_{i=1}^{n} \sum_{k=1}^{m} \mu(C_{ik})$$

By b)

$$\sum_{k=1}^{m} \mu(C_{ik}) \leq \mu(A_i)$$

from which the result follows.

■

Definition 5.8.10 (Simple Functions)
Let $(\Omega, \mathcal{F})$ be a family of subsets and X a normed vector space over $\mathbb{R}$. We define the space of simple functions to be

$$\text{Simple}(\mathcal{F}, X) := \left\{ \sum_{i=1}^{n} \lambda_i \mathbf{1}_{A_i} \;\middle|\; A_1 \sqcup ... \sqcup A_n = \Omega, \lambda_1, ..., \lambda_n \in X \right\}$$

Proposition 5.8.11 (Integration of Simple Functions)
Let $(\Omega, \mathcal{F})$ be a ring and μ a finitely additive measure and X a vector space over $\mathbb{R}$. There is a well-defined linear map

$$\int \mathrm{d}\mu : \text{Simple}(\mathcal{F}, X) \to X$$

$$\sum_{i=1}^{n} \alpha_i \mathbf{1}_{A_i} \to \sum_{i=1}^{n} \alpha_i \mu(A_i)$$

Proof. Suppose that $\sum_{i=1}^{n} \alpha_i \mathbf{1}_{A_i} = \sum_{j=1}^{m} \beta_j \mathbf{1}_{B_j}$ where both $\{A_i\}$ and $\{B_j\}$ are *disjoint* partitions of X. Then

$$A_i = (A_i \cap B_1) \sqcup ... \sqcup (A_i \cap B_m)$$

whence

$$\mu(A_i) = \sum_{j=1}^{m} \mu(A_i \cap B_j)$$

Therefore

$$\sum_{i=1}^{n} \alpha_i \mu(A_i) = \sum_{i=1}^{n} \sum_{j=1}^{m} \alpha_i \mu(A_i \cap B_j)$$

By the same argument

$$\sum_{j=1}^{m} \beta_j \mu(B_j) = \sum_{i=1}^{n} \sum_{j=1}^{m} \beta_j \mu(A_i \cap B_j)$$

Evidently $\{A_i \cap B_j\}$ is a disjoint partition of X. If $A_i \cap B_j \neq \emptyset$ then the first equality requires that $\alpha_i = \beta_j$. This shows that the integrals are equal as required. ■

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Proposition 5.8.12
Let $(\Omega, \mathcal{F})$ be a ring, μ a finitely additive measure and X a normed vector space over $\mathbb{R}$. Then we have the following properties

a) $\int (f+g)\,\mathrm{d}\mu = \int f\,\mathrm{d}\mu + \int g\,\mathrm{d}\mu$
b) $\int \lambda f\,\mathrm{d}\mu = \lambda \int f\,\mathrm{d}\mu$
c) $\left\|\int f\,\mathrm{d}\mu\right\| \leq \int \|f\|\,\mathrm{d}\mu$

If μ is finite then

$$\left\|\int f\,\mathrm{d}\mu\right\| \leq \mu(\Omega)\|f\|_\infty$$

Proof. For the last part observe that

$$\left\|\sum_{i=1}^{n} \alpha \mathbf{1}_{A_i}\right\| \leq \sum_{i=1}^{n} \|\alpha\|_i \mu(A_i) \leq \|f\|_\infty \sum_{i=1}^{n} \mu(A_i) \leq \mu(\Omega)\|f\|_\infty$$

■

5.8.3 Integration of Regulated Functions

Definition 5.8.13 (SubBorel Measure)
Let $\Omega = \mathbb{R}$ and $\mathcal{F}$ the ring generated by the half-open subintervals. Then there is a finitely additive measure $\mu : \mathcal{F} \to \mathbb{R}$ determined by

$$\mu([a,b)) = (b-a)$$

By (...) this determines a bounded linear map

$$\int : \mathrm{Simple}(\mathbb{R}, X) \to X$$

for any real vector space X.

Definition 5.8.14 (Regulated Function)
*Let X be a Banach space. A **regulated function** is a map*

$$f : \mathbb{R} \to X$$

*of **compact** support for which there exists a sequence (f_n) of* simple *functions (with respect to the ring generated by the half-open subintervals) such that*

$$f_n \to f$$

converges uniformly. We denote this by

$$\mathrm{Reg}(\mathbb{R}, X)$$

Lemma 5.8.15
A uniform limit of bounded functions is bounded. In particular a regulated function is bounded.

Proof. There is some N such that $n \geq N \implies \|f_n(x) - f(x)\| < 1$. Then

$$\|f(x)\| \leq \|f_n(x) - f(x)\| + \|f_n(x)\| \leq 1 + \|f_n(x)\|$$

We may take supremum over x to show that f is bounded. ■

Proposition 5.8.16 (Continuous Functions are Regulated)
Let X be a Banach space and $f : [a,b] \to X$ a continuous function. Then f is regulated.

Proof. We may define

$$f_n(x) := \sum_{i=1}^{N} f(a_i)\mathbf{1}_{[a_i,b_i)}(x) + \mathbf{1}\{x = b\}f(b)$$

$$a_i := a_i + \frac{j-1}{N}(b-a)$$

$$b_i := a_i + \frac{j}{N}(b-a)$$

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By (...) f is uniformly continuous so that for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$|x - y| < \delta \implies \|f(x) - f(y)\| < \varepsilon$$

Choose N such that

$$\max_i \frac{b_i - a_i}{N} < \delta$$

For every $x \in [a, b]$ we have $x \in [a_i, b_i)$ for some i and by construction $\|f(x) - f(a_i)\| < \varepsilon$ whence $\|f(x) - f_n(x)\| < \varepsilon$. As this applies for any $x \in [a, b]$ we see that the convergence is uniform. ∎

Proposition 5.8.17 (Integration of Regulated Functions)
Let X be a Banach space and $f : [a, b] \to X$ a function for which $f_n \to f$ converges uniformly. Then the sequence

$$\int_a^b f_n$$

converges in X. Furthermore the limit is independent of the choice of f_n, and we denote the limit by $\int_a^b f$.

In particular the integration of continuous functions with compact support is well-defined.

Proof. Let $x_n := \int f_n$. Then by (...) we have

$$\|x_n - x_m\| \leq (b - a)\|f_n - f_m\|_\infty$$

By the uniform convergence of f_n we see that x_n is a cauchy sequence, and so by assumption $x_n \to x$. The uniqueness is straightforward. ∎

Lemma 5.8.18
Let X be a Banach space and $f \in \mathrm{Reg}([a, b], X)$. Then $\|f\| \in \mathrm{Reg}([a, b], \mathbb{R})$.

Proposition 5.8.19 (Properties of the Integral)
Let X be a Banach space. Then the following properties hold

a) $\int_a^b (\lambda f + \mu g) = \lambda \int_a^b f + \mu \int_a^b g$
b) $\int_a^b f + \int_b^c f = \int_a^c f$
c) $\left\| \int_a^b f \right\| \leq \int_a^b \|f\|$

In the case $X = \mathbb{R}$ then we also have the following properties

d) $f \geq 0 \implies \int_a^b f \geq 0$
e) $f \leq g \implies \int_a^b f \leq \int_a^b g$
f) f continuous, positive such that $\int_a^b f = 0$ implies $f \equiv 0$

Proposition 5.8.20 (Fundamental Theorem of Calculus)
Let $f \in \mathrm{Reg}([a, b], X)$ where X is a Banach space. Suppose $c \in (a, b)$ is a point where f is continuous. Define

$$F(x) := \int_a^x f$$

Then F is differentiable at c and $F'(c) = f(c)$

Proof. Observe that for $0 < h < b - c$

$$\begin{aligned} \left\| \frac{F(c + h) - F(c)}{h} - f(c) \right\| &= \frac{1}{h} \left\| \int_c^{c+h} (f(t) - f(c))\, \mathrm{d}t \right\| \\ &\leq \frac{1}{h} \int_c^{c+h} \|f(t) - f(c)\|\, \mathrm{d}t \\ &\leq \sup_{|t-c|<h} \|f(t) - f(c)\| \end{aligned}$$

and similarly for $a - c < h < 0$. As f is regulated it is bounded (...) and so the supremum is finite. As f is assumed continuous we find $F'(c) = f(c)$. ∎

Proposition 5.8.21
Suppose $f_i \in \mathrm{Reg}([a, b], X_i)$ for $i = 1...n$. Then $(f_1, ..., f_n) \in \mathrm{Reg}([a, b], X)$ where $X := X_1 \times ...X_n$ and

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$$\int_a^b f = \left(\int_a^b f_1, ..., \int_a^b f_n \right)$$

5.8.4 Measure Spaces

To create a robust theory of integration it is necessary to replace finite additivity with countably additivity, and further show that such set functions can be defined over σ-algebras and not just algebras.

Definition 5.8.22 (Countably Additive Set Function)
Let $\mathcal{F}$ be a family of subsets of Ω containing $\emptyset$. A function

$$\mu : \mathcal{F} \to [0, \infty]$$

*is a **countably additive set function** (or **measure**) if it satisfies*

a) $\mu(\emptyset) = 0$
b) $\mu\left(\bigsqcup_{n=1}^{\infty} A_n\right) = \sum_{i=1}^{\infty} \mu(A_i)$ whenever this is well-defined

*We say it is **countably subadditive** if it satisfies*

$$\mu\left(\bigcup_{i=1}^{\infty} A_i \right) \leq \sum_{i=1}^{\infty} \mu(A_i)$$

whenever this is well-defined. We show in (5.8.25) that these conditions are almost equivalent.

Definition 5.8.23 (Measurable Space)
*Let $(\Omega, \mathcal{F})$ be a measurable space and $\mu : \mathcal{F} \to [0, \infty]$ a measure. Then the triplet $(\Omega, \mathcal{F}, \mu)$ is a **measurable space**.*

The following are standard results which are somewhat more awkward to prove in the case of semi-rings. However the proofs follow precisely the same lines and semi-ring extension theorem may be more directly applicable.

Proposition 5.8.24 (Continuity of Measures)
Let $\mathcal{R}$ be a semi-ring of subsets of Ω and $\mu : \mathcal{R} \to [0, \infty]$ a countably additive measure. Then for any sequence of sets $A_1, A_2, ... \in \mathcal{R}$ we have

$$\lim_{n \to \infty} \mu\left(\bigcup_{i=1}^{n} A_i \right) = \mu\left(\bigcup_{n=1}^{\infty} A_n \right)$$

whenever this is well-defined.

Proof. Define the increasing sequence

$$B_n := \bigcup_{i=1}^{n} A_i$$

then wish to show

$$\lim_{n \to \infty} \mu(B_n) = \mu\left(\bigcup_{n=1}^{\infty} B_n \right)$$

By definition we have (with the convention $B_0 = \emptyset$)

$$B_i \setminus \bigcup_{j=1}^{i-1} B_j = B_i \setminus B_{i-1} = \bigsqcup_{k=1}^{N_i} C_{ik} \quad i = 1, 2, ...$$

for $C_{ik} \in \mathcal{R}$ whence

$$B_n = \bigsqcup_{i=1}^{n} B_i \setminus B_{i-1} = \bigsqcup_{i=1}^{n} \bigsqcup_{k=1}^{N_n} C_{ik}$$

for $n = 1, 2, ..., \infty$. Then by additivity

$$\lim_{n \to \infty} \mu(B_n) = \lim_{n \to \infty} \sum_{i=1}^{n} \sum_{k=1}^{N_i} \mu(C_{ik}) = \sum_{i=1}^{\infty} \sum_{k=1}^{N_i} \mu(C_{ik})$$

and

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$$\mu\left(\bigcup_{n=1}^{\infty} B_n\right) = \mu\left(\bigsqcup_{i=1}^{\infty}\bigsqcup_{k=1}^{N_i} C_{ik}\right) = \sum_{i=1}^{\infty}\sum_{k=1}^{N_i}\mu(C_{ik})$$

■

Proposition 5.8.25 (Finitely Additive + Countably Subadditive $\iff$ Countably Additive)
Let $\mathcal{R}$ be a semi-ring of subsets of Ω and $\mu : \mathcal{R} \to [0,\infty]$ a finitely additive set function. Then

$$\mu\left(\bigsqcup_{i=1}^{\infty} A_i\right) \geq \sum_{i=1}^{\infty}\mu(A_i)$$

whenever this is well-defined. In particular μ is a measure iff it is countably subadditive.

Proof. By (5.8.9).b) we have

$$\sum_{i=1}^{n}\mu(A_i) \leq \mu\left(\bigsqcup_{i=1}^{\infty} A_i\right)$$

and the result follows by taking $n \to \infty$ (5.1.21).

Therefore if μ is countably subadditive it must also be countably additive. Conversely if μ is countably additive then we aim to show

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty}\mu(A_i)$$

for a not necessarily disjoint sequence $A_i \in \mathcal{R}$. By assumption μ is finitely additive, so by (5.8.9).c)

$$\mu\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n}\mu(A_i) \leq \sum_{i=1}^{\infty}\mu(A_i)$$

Using continuity of measures (5.8.24) we deduce that

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \lim_{n\to\infty}\mu\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{\infty}\mu(A_i)$$

as required. ■

Proposition 5.8.26 (Extension Theorem II)
Let $\mathcal{R}$ be a semi-ring of sets over Ω and $\mu : \mathcal{R} \to [0,\infty]$ a measure. Suppose that Ω is denumerable union of sets in $\mathcal{R}$. Then there exists an extension

$$\tilde{\mu} : \sigma(\mathcal{R}) \to [0,\infty]$$

to the σ-algebra generated by $\mathcal{R}$. When μ is σ-finite then this extension is unique. More precisely it is given by

$$\tilde{\mu}(A) := \inf\left\{\sum_{i=1}^{\infty}\mu(A_i) \;\middle|\; A \subset \bigcup_{i=1}^{\infty} A_i, A_i \in \mathcal{R}\right\}$$

Proof. Define first the "outer measure"

$$\mu^\star : \mathcal{P}(\Omega) \to [0,\infty]$$
$$\mu^\star(A) := \inf\left\{\sum_{i=1}^{\infty}\mu(E_i) \;\middle|\; A \subset \bigcup_{i=1}^{\infty} E_i \quad E_i \in \mathcal{F}\right\}$$

Note that the infimum is well-defined precisely by the hypothesis. Further define the family of subsets

$$\hat{\mathcal{R}} := \{A \subset \Omega \mid \mu^\star(B) = \mu^\star(A\cap B) + \mu^\star(A^c \cap B) \quad \forall B \subset \Omega\}$$

We make a number of claims

a) $\mu^\star(A) = \mu(A)$ for all $A \in \mathcal{R}$
b) $\mu^\star$ is countably subadditive and monotone

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$$\mu^\star\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu^\star(A_i) \quad A_i \subset \Omega$$

c) $\mathcal{R} \subset \hat{\mathcal{R}}$

d) $\hat{\mathcal{R}}$ is a σ-algebra containing $\sigma(\mathcal{R})$ and $\mu^\star$ is a measure on $\hat{\mathcal{R}}$

and prove each in turn

a) Evidently $\mu^\star(A) \leq \mu(A)$ as $A \subset A \cup \emptyset \ldots \cup \emptyset \ldots$. Conversely suppose that $A \subset \bigcup_{i=1}^{\infty} A_i$ for $A_i \in \mathcal{R}$ and $\mu^\star(A) < \infty$ (otherwise the reverse inequality is clear). Then evidently

$$A = \bigcup_{i=1}^{\infty} (A_i \cap A)$$

As μ is countably subadditive (5.8.25) and monotone (5.8.9) we see

$$\mu(A) \leq \sum_{i=1}^{\infty} \mu(A_i \cap A) \leq \sum_{i=1}^{\infty} \mu(A_i)$$

By definition of the infimum we find $\mu(A) \leq \mu^\star(A)$.

b) Let $A_1, \ldots, A_n, \ldots$ be subsets of Ω such that $A \subset \bigcup_{i=1}^{\infty} A_i$. For a given $\varepsilon > 0$, choose $A_{ij} \in \mathcal{R}$ such that $A_i \subset \bigcup_{j=1}^{\infty} A_{ij}$ and

$$\sum_{j=1}^{\infty} \mu(A_{ij}) \leq \mu^\star(A_i) + \frac{\varepsilon}{2^i}$$

Then evidently $A \subset \bigcup_{i,j=1}^{\infty} A_{ij}$ and so by definition

$$\mu^\star(A) \leq \sum_{i,j=1}^{\infty} \mu(A_{ij}) \leq \sum_{i=1}^{\infty} \mu^\star(A_i) + \varepsilon$$

As ε was arbitrary we deduce that $\mu^\star$ is countably subadditive. Monotonicity is straightforward from the definition.

c) Suppose $A \in \mathcal{R}$ and $B \subset \Omega$ then we require to prove that

$$\mu^\star(B) = \mu^\star(A \cap B) + \mu^\star(A^c \cap B)$$

By subadditivity it is sufficient so show that

$$\mu^\star(B) \geq \mu^\star(A \cap B) + \mu^\star(A^c \cap B)$$

Let $B \subset \bigcup_{i=1}^{\infty} B_i$ be an arbitrary cover with $B_i \in \mathcal{R}$, then by definition of $\mu^\star$ it is sufficient to show that

$$\mu^\star(A \cap B) + \mu^\star(A^c \cap B) \leq \sum_{i=1}^{\infty} \mu(B_i)$$

Recall by definition of semi-ring that

$$A^c \cap B_i = \bigsqcup_{k=1}^{n_i} C_{ik}$$

for some disjoint sets $C_{ik} \in \mathcal{R}$ and evidently

$$A \cap B \subset \bigcup_{i=1}^{\infty} (A \cap B_i)$$

$$A^c \cap B \subset \bigcup_{i=1}^{\infty} \bigsqcup_{k=1}^{n_i} C_{ik}$$

Therefore by monotonicity and countable subadditivity of $\mu^\star$

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$$\begin{aligned}\mu^\star(A\cap B)+\mu^\star(A^c\cap B) &\leq \mu^\star\left(\bigcup_{i=1}^{\infty}(A\cap B_i)\right)+\mu^\star\left(\bigcup_{i=1}^{\infty}\bigsqcup_{k=1}^{n_i}C_{ik}\right)\\ &\leq \sum_{i=1}^{\infty}\mu^\star(A\cap B_i)+\sum_{i=1}^{\infty}\sum_{k=1}^{n_i}\mu(C_{ik})\\ &= \sum_{i=1}^{\infty}\mu(A\cap B_i)+\sum_{i=1}^{\infty}\sum_{k=1}^{n_i}\mu(C_{ik})\\ &= \sum_{i=1}^{\infty}\left(\mu(A\cap B_i)+\sum_{k=1}^{n_i}\mu(C_{ik})\right)\\ &= \sum_{i=1}^{\infty}\mu(B_i)\end{aligned}$$

where the last step follows from finite additivity of μ.

d) We prove the various properties in turn

i) Evidently $\mu^\star(\emptyset)=0$ and therefore $\emptyset,\Omega\in\hat{\mathcal{R}}$

ii) Given $A_1,A_2\in\hat{\mathcal{R}}$ and $B\subset\Omega$. Observe that we may apply the measurability condition to B and A_1, and then to $B\cap A_1$ and A_2, and $B\cap A_1^c$ and A_2 to find

$$\begin{aligned}\mu^\star(B) &= \mu^\star(B\cap A_1)+\mu^\star(B\cap A_1^c)\\ &= \mu^\star(B\cap A_1\cap A_2)+\mu^\star(B\cap A_1\cap A_2^c)+\mu^\star(B\cap A_1^c\cap A_2)+\mu^\star(B\cap A_1^c\cap A_2^c)\end{aligned}$$

We may also consider the relationship with B replaced with $B\cap(A_1\cup A_2)$ to find

$$\mu^\star(B\cap(A_1\cup A_2))=\mu^\star(B\cap A_1\cap A_2)+\mu^\star(B\cap A_1\cap A_2^c)+\mu^\star(B\cap A_1^c\cap A_2) \qquad (*)$$

Combining these two relations we deduce

$$\mu^\star(B)=\mu^\star(B\cap(A_1\cup A_2))+\mu^\star(B\cap(A_1\cup A_2)^c)$$

As B was arbitrary then we deduce $A_1\cup A_2\in\hat{\mathcal{R}}$. Evidently by definition $A^c\in\hat{\mathcal{R}}$ and therefore by de-morgan $A_1\cap A_2\in\hat{\mathcal{R}}$. Therefore $\hat{\mathcal{R}}$ is closed under finite unions and intersections.

iii) Let $A_1,...,A_n\in\hat{\mathcal{R}}$ be disjoint then we claim that

$$\mu^\star\left(B\cap\left(\bigsqcup_{i=1}^{n}A_i\right)\right)=\sum_{i=1}^{n}\mu^\star(B\cap A_i)$$

From (*) we have

$$\mu^\star(B\cap(A_1\sqcup A_2))=\mu^\star(B\cap A_1)+\mu^\star(B\cap A_2)$$

and the result follows by induction.

iv) Let $A_1,A_2,...\in\hat{\mathcal{R}}$ be a countable family and $B\subset\Omega$. We require to prove that

$$\mu^\star(B)=\mu^\star\left(B\cap\bigcup_{i=1}^{\infty}A_i\right)+\mu^\star\left(B\cap\left(\bigcup_{i=1}^{\infty}A_i\right)^c\right)$$

By (5.8.6) we may consider the decomposition

$$A_i\setminus\bigcup_{j=1}^{i-1}A_j=\bigsqcup_{k=1}^{n_i}C_{ik}$$

where $C_{ik}\in\mathcal{R}$ are disjoint. Observe

$$\bigcup_{i=1}^{n}A_i=\bigsqcup_{i=1}^{n}\bigsqcup_{k=1}^{n_i}C_{ik}\quad\text{for } n=1,2,...,\infty$$

We have shown in ii) that $\bigcup_{i=1}^{n}A_i\in\hat{\mathcal{R}}$, therefore

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$$\begin{aligned}\mu^\star(B) &= \mu^\star\left(B \cap \left(\bigcup_{i=1}^{n} A_i\right)\right) + \mu^\star\left(B \cap \left(\bigcup_{i=1}^{n} A_i\right)^c\right) \\ &= \mu^\star\left(B \cap \left(\bigsqcup_{i=1}^{n}\bigsqcup_{k=1}^{n_i} C_{ik}\right)\right) + \mu^\star\left(B \cap \left(\bigcup_{i=1}^{n} A_i\right)^c\right)\end{aligned}$$

Recall from iii) that $\mu^\star(B \cap -)$ is finitely additive on $\hat{\mathcal{R}}$. Additionally by monotonicity we may deduce that

$$\begin{aligned}\mu^\star(B) &= \sum_{i=1}^{n}\sum_{k=1}^{n_i} \mu^\star(B \cap C_{ik}) + \mu^\star\left(B \cap \left(\bigcup_{i=1}^{n} A_i\right)^c\right) \\ &\geq \sum_{i=1}^{n}\sum_{k=1}^{n_i} \mu^\star(B \cap C_{ik}) + \mu^\star\left(B \cap \left(\bigcup_{i=1}^{\infty} A_i\right)^c\right)\end{aligned}$$

Letting $n \to \infty$ we find by subadditivity

$$\begin{aligned}\mu^\star(B) &\geq \sum_{i=1}^{\infty}\sum_{k=1}^{n_i} \mu^\star(B \cap C_{ik}) + \mu^\star\left(B \cap \left(\bigcup_{i=1}^{\infty} A_i\right)^c\right) \\ &\geq \mu^\star\left(B \cap \left(\bigcup_{i=1}^{\infty} A_i\right)\right) + \mu^\star\left(B \cap \left(\bigcup_{i=1}^{\infty} A_i\right)^c\right)\end{aligned}$$

However by subadditivity the reverse inequality holds and we deduce that these are all equal. This shows that $\bigcup_{i=1}^{\infty} A_i \in \hat{\mathcal{R}}$ and $\hat{\mathcal{R}}$ is closed under countable unions. Consider further the case A_i are disjoint and $B \coloneqq \bigsqcup_{j=1}^{\infty} A_j$. Then we may assume that $n_i = 1$ and $C_{i1} = A_i$. Using the above relationship

$$\mu^\star\left(\bigsqcup_{j=1}^{\infty} A_j\right) = \sum_{i=1}^{\infty} \mu^\star\left(\left(\bigsqcup_{j=1}^{\infty} A_j\right) \cap A_i\right) = \sum_{i=1}^{\infty} \mu^\star(A_i)$$

This shows that $\mu^\star$ is countably additive.

■

Definition 5.8.27 (Complete Set Function)
*Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. We say it is **complete** if for every null set N (such that $\mu(N) = 0$) and $N' \subseteq N$ we have $N' \in \mathcal{F}$.*

Proposition 5.8.28 (Completion of a Measure Space)
*Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and define the **completion** by*

$$\begin{aligned}\overline{\mathcal{F}} &\coloneqq \{A \cup Z \mid A \in \mathcal{F}, Z \subseteq N, \mu(N) = 0\} \\ \overline{\mu}(A \cup Z) &\coloneqq \mu(A)\end{aligned}$$

Then $\left(\Omega, \overline{\mathcal{F}}, \overline{\mu}\right)$ is a well-defined complete measure space. Furthermore $\overline{\mathcal{F}}$ is the smallest complete σ-algebra containing $\mathcal{F}$, and $\overline{\mu}$ is the unique extension.

5.8.5 Borel Measure on $\mathbb{R}$

Proposition 5.8.29
Let $\mathcal{R}$ be the semi-ring of half-open intervals over $\mathbb{R}$. Then the set function

$$\mu : \mathcal{R} \to [0, \infty]$$

given by

$$\mu([a, b)) = b - a,$$

is countably additive and σ-finite.

Proof. Evidently μ is finitely additive, therefore by (5.8.25) it is sufficient to show that it is countably subadditive. Suppose that

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$$\bigcup_{i=1}^{\infty}[a_i, b_i) = [a, b)$$

Let $\varepsilon < b - a$. Then $[a, b-\varepsilon]$ is compact by (5.7.7) and $U_i := \left(a_i - \frac{\delta}{2^i}, b_i\right)$ is open for every $\delta > 0$. By definition of compactness there is some finite integer n such that

$$\left[a, \tilde{b}\right] := [a, b-\varepsilon] \subset \bigcup_{i=1}^{n}\left(a_i - \frac{\delta}{2^i}, b_i\right) =: (\tilde{a}_i, b_i)$$

By reordering we assume that $\tilde{a}_1 < ... < \tilde{a}_n$. As $\left[a, \tilde{b}\right]$ is an interval we must have $\tilde{a}_{i+1} < b_i$. Further we may assume that $b_i < b_{i+1}$ (otherwise we may omit the interval $(\tilde{a}_i, b_i)$). By construction we must also have $\tilde{a}_1 < a$ and $\tilde{b} < b_n$.

Consequently

$$b - a - \varepsilon = \tilde{b} - a \leq b_n - \tilde{a}_1 = b_1 - \tilde{a}_1 + \sum_{i=1}^{n-1}(b_{i+1} - b_i) \leq \sum_{i=1}^{n}(b_i - \tilde{a}_i) \leq \sum_{i=1}^{n}(b_i - a_i) + \delta \leq \sum_{i=1}^{\infty}(b_i - a_i) + \delta$$

This shows

$$\mu([a, b)) = \mu\left(\bigcup_{i=1}^{\infty}[a_i, b_i)\right) \leq \sum_{i=1}^{\infty}\mu([a_i, b_i))$$

as required. ∎

Proposition 5.8.30
Let $D \subset \mathbb{R}$ be an additive subgroup containing $\mathbb{Q}$. Then $\mathcal{B}(\mathbb{R})$ is generated by any of the following families of intervals

1. $(a, b]$ $a < b \in D$
2. (a, b) $a < b \in D$
3. $[a, b)$ $a < b \in D$
4. (a, ∞) $a \in D$
5. $[a, \infty)$ $a \in D$
6. $(-\infty, a]$ $a \in D$
7. $(-\infty, a)$ $a \in D$

Further there is a unique measure

$$\mu : \mathcal{B}(\mathbb{R}) \to [0, \infty]$$

such that

$$\mu([a, b)) = b - a$$

Proof. Denote the corresponding σ-algebras by $\mathcal{F}_1, ..., \mathcal{F}_7$. Observe that by the Archimedean property

$$(a, b) = \bigcup_{n=1}^{\infty}\left(a, b - \frac{1}{n}\right] = \bigcup_{n=1}^{\infty}\left[a + \frac{1}{n}, b\right)$$

$$[a, b) = \bigcap_{n=1}^{\infty}\left(a - \frac{1}{n}, b\right)$$

$$(a, b] = \bigcap_{n=1}^{\infty}\left(a, b + \frac{1}{n}\right)$$

so that $\mathcal{F}_1 = \mathcal{F}_2 = \mathcal{F}_3$. We may argue similarly $\mathcal{F}_4 = \mathcal{F}_5 = \mathcal{F}_6 = \mathcal{F}_7$. Finally

$$(a, b) = (a, \infty) \cap (-\infty, b)$$

and

$$(a, \infty) = \bigcup_{n=1}^{\infty}(a, n)$$

to show these are equal. The measure μ exists by (5.8.29) and (5.8.26). ∎

Proposition 5.8.31 (Borel Measure on the Extended Real Line)
Let $\mathcal{B}(\mathbb{R}^\sharp)$ be the Borel Measure on the extended real line. Then $\mathcal{B}(\mathbb{R}^\sharp)$ consists of sets of the form

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$$\left\{A \subset \mathbb{R}^\sharp \mid A \cap \mathbb{R} \in \mathcal{B}(\mathbb{R})\right\}$$

or equivalently

$$\{A \cup B \mid A \in \mathcal{B}(\mathbb{R}), B \in \mathcal{P}(\{-\infty, \infty\})\}$$

Furthermore it is generated by any of the following families

1. $(a, \infty]$
2. $[a, \infty]$
3. $[-\infty, a)$
4. $[-\infty, a]$

for $a \in D$ where $D \subset \mathbb{R}$ is an additive subgroup containing $\mathbb{Q}$.

Proof. If $U \subset \mathbb{R}$ is open, then it is also open in $\mathbb{R}^\sharp$. Therefore $\mathcal{B}(\mathbb{R}) \subset \mathcal{B}(\mathbb{R}^\sharp)$. Evidently $\mathbb{R}$, $\{\infty\}$, $\{-\infty\} \in \mathcal{B}(\mathbb{R}^\sharp)$ as they are open and closed respectively. This shows that the given family equals $\mathcal{B}(\mathbb{R}^\sharp)$.

Denote by $\mathcal{F}_1, ..., \mathcal{F}_4$ the σ-algebras generated by the respective families of rays. Note that $\mathcal{F}_1 = \mathcal{F}_4$ and $\mathcal{F}_2 = \mathcal{F}_3$ by taking complements. Further $\{\infty\} \in \mathcal{F}_1 \cap \mathcal{F}_2$ by taking intersections as $a \to \infty$ and similarly $\{-\infty\} \in \mathcal{F}_4 \cap \mathcal{F}_3$. This shows that they also contain either the open rays (a, ∞) or the semi-open rays $[a, \infty)$. Therefore they contain $\mathcal{B}(\mathbb{R})$ by (5.8.30). Then by the characterisation in the first part we see that this equals $\mathcal{B}(\mathbb{R}^\sharp)$. ∎

5.8.6 Product Measure

Definition 5.8.32 (Product of σ-algebra)
*Let $(\Omega_1, \mathcal{F}_1), ..., (\Omega_n, \mathcal{F}_n)$ be a family of measurable spaces. Then define the **product measurable space** $(\Omega_1 \times ... \times \Omega_n, \mathcal{F}_1 \otimes ... \otimes \mathcal{F}_n)$ to be given by*

$$\mathcal{F}_1 \otimes ... \otimes \mathcal{F}_n := \sigma(\mathcal{F}_1 \times ... \times \mathcal{F}_n)$$

and

$$\mathcal{F}_1 \times ... \times \mathcal{F}_n := \{A_1 \times ... \times A_n \mid A_i \in \mathcal{F}_i\}$$

Definition 5.8.33 (Product Measure)
Let $(\Omega_i, \mathcal{F}_i, \mu_i)$ be a family of σ-finite measures for $i = 1...n$. Then we say a measure

$$\mu_1 \otimes ... \otimes \mu_n : \mathcal{F}_1 \otimes ... \otimes \mathcal{F}_n \to \mathbb{R}^\sharp$$

such that

$$\mu(A_1 \times ... \times A_n) = \mu_1(A_1) \cdot ... \cdot \mu_{n(A_n)}$$

*is called the **product measure space**.*

We defer proof of existence and uniqueness as it relies on theory of integration, but nevertheless prove the finitely additive case and then the case of $\mathbb{R}^n$ in the next section.

Lemma 5.8.34
Let $\mathcal{R}_1, ..., \mathcal{R}_n$ be semi-rings. Then the family

$$\mathcal{R}_1 \times ... \times \mathcal{R}_n = \{A_1 \times ... \times A_n \mid A_i \in \mathcal{R}_i\}$$

is a semi-ring.

Proof. We may reduce to the case of $n = 2$. Then observe that

$$(A_1 \times B_1) \setminus (A_2 \times B_2) = (A_1 \setminus A_2) \times (B_1 \setminus B_2) \sqcup (A_1 \cap A_2) \times (B_1 \setminus B_2) \sqcup (A_1 \setminus A_2) \times (B_1 \cap B_2)$$

∎

Lemma 5.8.35
Let $\mathcal{R}$ be a semi-ring and $A_1, ..., A_n \in \mathcal{R}$ a family of subsets. Then there exists a family of disjoint subsets $\mathcal{S} := \{B_1, ..., B_N\} \subset \mathcal{R}$ such that $\mathcal{S} = \mathcal{S}_1 \cup ... \cup \mathcal{S}_n$ and

$$A_i = \sqcup_{B \in \mathcal{S}_i} B$$

Proof. For a tuple $\varepsilon := (\varepsilon_1, ..., \varepsilon_n)$ for which $\varepsilon_i = \pm 1$ define the family of disjoint sets

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$$A^\varepsilon := A_1^{\varepsilon_1} \cap ... \cap A_n^{\varepsilon_n}$$

where $A_i^1 := A_i$ and $A_i^{-1} := A_i^c$. By (5.8.6) A^ε may be expressed as the disjoint union of some sets in $\mathcal{R}$, say $\mathcal{S}^\varepsilon$. Then $\mathcal{S} = \cup_\varepsilon \mathcal{S}^\varepsilon$ consists of pairwise disjoint sets, where we consider tuples ε for which at least one entry is positive. Further

$$A_i = \sqcup_{\varepsilon:\varepsilon_i=1} A^\varepsilon$$

therefore we may define $\mathcal{S}_i := \cup_{\varepsilon:\varepsilon_i=1} \mathcal{S}^\varepsilon$. Note throughout we may omit ε for which $A^\varepsilon = \emptyset$. ■

Lemma 5.8.36
Let $\{(\Omega_i, \mathcal{R}_i, \mu_i)\}_{i=1...n}$ be a collection of finitely additive set functions on semi-rings. The set function

$$\mu : \mathcal{R}_1 \times ... \times \mathcal{R}_n \to \mathbb{R} \cup \{\infty\}$$
$$A_1 \times ... \times A_n \to \mu_1(A_1)...\mu_{n(A_n)}$$

is finitely additive, and we write $\mu =: \mu_1 \times ... \times \mu_n$. Moreover if each μ_i is σ-finite then so is μ.

Proof. By induction it's enough to consider the case $n = 2$. Suppose that

$$A_i = B_i \times C_i \quad i = 0, ..., n$$
$$A_0 = \sqcup_{i=1}^n A_i$$

for $B_i \in \Omega_1$ and $C_i \in \Omega_2$. By (5.8.35) we may write

$$B_i := \sqcup_{\hat{B} \in \mathcal{S}_i} \hat{B}$$
$$C_i := \sqcup_{\hat{C} \in \mathcal{T}_i} \hat{C}$$

where $\mathcal{S}_i$ and $\mathcal{T}_i$ are finite subsets of $\mathcal{R}_1$ and $\mathcal{R}_2$ respectively. Evidently

$$B_i \times C_i = \sqcup_{(\hat{B},\hat{C}) \in \mathcal{S}_i \times \mathcal{T}_i} \hat{B} \times \hat{C}$$

and $\mathcal{S}_0 = \cup_{i=1}^n \mathcal{S}_i$ and $\mathcal{T}_0 := \cup_{i=1}^n \mathcal{T}_i$. As μ_1 and μ_2 are finitely additive then

$$\mu_1(B_i) = \sum_{\hat{B} \in \mathcal{S}_i} \mu_1\big(\hat{B}\big)$$
$$\mu_2(C_i) = \sum_{\hat{C} \in \mathcal{T}_i} \mu_2\big(\hat{C}\big)$$
$$\sum_{i=1}^n \mu(A_i) = \sum_{i=1}^n \mu_1(B_i)\mu_2(C_i) = \sum_{i=1}^n \sum_{(\hat{B},\hat{C}) \in \mathcal{S}_i \times \mathcal{T}_i} \mu_1\big(\hat{B}\big)\mu_2\big(\hat{C}\big)$$
$$\mu(A_0) = \mu_1(B_0)\mu_2(C_0) = \sum_{(\hat{B},\hat{C}) \in \mathcal{S}_0 \times \mathcal{T}_0} \mu_1\big(\hat{B}\big)\mu_2\big(\hat{C}\big)$$

As the sets $B_i \times C_i$ are disjoint we may deduce that

$$\mathcal{S}_0 \times \mathcal{T}_0 = \sqcup_{i=1}^n \mathcal{S}_i \times \mathcal{T}_i$$

from which the result follows. ■

5.8.7 Borel Measure on $\mathbb{R}^n$

Proposition 5.8.37
Let $\mathcal{R}$ be the semi-ring of half-open intervals over $\mathbb{R}^n$. Then the set function

$$\mu : \mathcal{R} \to [0, \infty]$$

given by

$$\mu([a_1, b_1) \times ... \times [a_n, b_n)) = \prod_{i=1}^n (b_i - a_i),$$

is countably additive and σ-finite.

Proof. By (5.8.36) μ is finitely additive. So by (5.8.25) it is sufficient to show that μ is countably subadditive. Suppose that

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$$[\boldsymbol{a}_0, \boldsymbol{b}_0) = \bigcup_{i=1}^{\infty} [\boldsymbol{a}_i, \boldsymbol{b}_i)$$

Then for any positive constants $\boldsymbol{\varepsilon}$, $\boldsymbol{\delta}_1, \boldsymbol{\delta}_2, \ldots$ we have

$$[\boldsymbol{a}_0, \boldsymbol{b}_0 - \boldsymbol{\varepsilon}] \subset \bigcup_{i=1}^{\infty} (\boldsymbol{a}_i - \boldsymbol{\delta}_i, \boldsymbol{b}_i)$$

By the Heine-Borel theorem (5.7.7) closed intervals are compact, so we have

$$[\boldsymbol{a}_0, \boldsymbol{b}_0 - \boldsymbol{\varepsilon}) \subset \bigcup_{i=1}^{N(\boldsymbol{\delta})} [\boldsymbol{a}_i - \boldsymbol{\delta}_i, \boldsymbol{b}_i)$$

Therefore by (5.8.9)

$$\begin{aligned}
\prod_{j=1}^{n} (b_{0j} - a_{0j} - \varepsilon) &\leq \sum_{i=1}^{N(\boldsymbol{\delta})} \prod_{j=1}^{n} (b_{ij} - a_{ij} + \delta_{ij}) \\
&\leq \sum_{i=1}^{N(\boldsymbol{\delta})} \prod_{j=1}^{n} (b_{ij} - a_{ij}) \left(1 + \frac{\delta}{2^i}\right) \\
&\leq \sum_{i=1}^{N(\boldsymbol{\delta})} \prod_{j=1}^{n} (b_{ij} - a_{ij}) \sum_{i=1}^{N(\boldsymbol{\delta})} \left(1 + \frac{\delta}{2^i}\right) \\
&\leq (1 + \delta) \sum_{i=1}^{\infty} \prod_{j=1}^{n} (b_{ij} - a_{ij})
\end{aligned}$$

where we have defined $\delta_{ij} := \frac{b_{ij} - a_{ij}}{n} \left(\frac{\delta}{2^i}\right)^{\frac{1}{n-1}}$ and used the inequality $(1 + x)^n \leq 1 + nx^{n-1}$. As this holds for arbitrary $\delta > 0$ then it holds for $\delta = 0$. We may consider the sequence $\varepsilon := \frac{1}{m}$ and letting $m \to \infty$ use (4.1.50) and (5.1.21) to deduce the case of $\varepsilon = 0$. This shows that

$$\mu([\boldsymbol{a}_0, \boldsymbol{b}_0)) \leq \sum_{i=1}^{\infty} \mu([\boldsymbol{a}_i, \boldsymbol{b}_i))$$

i.e. μ is countably subadditive. ∎

Proposition 5.8.38 (Existence of Borel Measure)
Let $\left(\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k)\right)$ be the Borel measurable space. Then there exists a unique σ-finite measure

$$\mu : \mathcal{B}(\mathbb{R}^k) \to [0, \infty]$$

such that

$$\mu([\boldsymbol{a}, \boldsymbol{b})) = \prod_{i=1}^{k} (b_i - a_i)$$

Proof. This is a consequence of (5.8.26) and (5.8.37). ∎

5.8.8 Measurable Functions

Definition 5.8.39
*Let $(\Omega_1, \mathcal{F}_1)$ and $(\Omega_2, \mathcal{F}_2)$ be measurable spaces. A function $f : \Omega_1 \to \Omega_2$ is said to be **measurable** if*

$$E \in \mathcal{F}_2 \implies f^{-1}(E) \in \mathcal{F}_1$$

We may write $\mathcal{F}_1/\mathcal{F}_2$-measurable in order to make the σ-algebras explicit.

Similarly a function $f : \Omega \to X$ from a measurable space to a topological space is said to be measurable if it is $\mathcal{F}/\mathcal{B}(X)$ measurable.

Proposition 5.8.40
The following properties hold

a) The composition of measurable functions is measurable
b) Suppose $\mathcal{F}_2 = \sigma(\mathcal{C})$ then a function $f : \Omega_1 \to \Omega_2$ is $\mathcal{F}_1/\mathcal{F}_2$-measurable iff

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$$E \in \mathcal{C} \implies f^{-1}(E) \in \mathcal{F}_1$$

c) A function $f : \Omega \to X$ is measurable precisely when the inverse image of an open set (resp. closed set) is measurable.

Corollary 5.8.41 (Criteria for measurablity of a real-valued function)
Let $(\Omega, \mathcal{F})$ be a measurable space and $f : \Omega \to \mathbb{R}^\sharp$ a function. Then the following are equivalent

a) f is $\mathcal{F}/\mathcal{B}(\mathbb{R}^\sharp)$-measurable
b) $f^{-1}((x, \infty]) \in \mathcal{F}$ for all $x \in D$
c) $f^{-1}([x, \infty]) \in \mathcal{F}$ for all $x \in D$
d) $f^{-1}([-\infty, x)) \in \mathcal{F}$ for all $x \in D$
e) $f^{-1}([-\infty, x]) \in \mathcal{F}$ for all $x \in D$

where $D \subset \mathbb{R}$ is an additive subgroup containing $\mathbb{Q}$.

Proof. We may combine (5.8.40).b) and (5.8.31). ∎

Proposition 5.8.42 (Measurable Space Isomorphism)
Let $(\Omega_1, \mathcal{F}_1)$ and $(\Omega_2, \mathcal{F}_2)$ be measurable spaces and $f : \Omega_1 \to \Omega_2$ a bijective function. Then the following are equivalent

a) f and f^{-1} are measurable
b) f induces a bijection $\mathcal{F}_1 \to \mathcal{F}_2$
c) f^{-1} induces a bijection $\mathcal{F}_2 \to \mathcal{F}_1$

*In this case we say that $f : (\Omega_1, \mathcal{F}_1) \xrightarrow{\sim} (\Omega_2, \mathcal{F}_2)$ is a **measurable space isomorphism**.*

Proposition 5.8.43 (Limits of Measurable Functions)
Let $f_n : (\Omega, \mathcal{F}) \to (\mathbb{R}^\sharp, \mathcal{B}(\mathbb{R}^\sharp))$ be a sequence of measurable functions. Define the functions $\sup f_n, \inf f_n, \limsup_n f_n, \liminf_n f_n$ as follows

$$(\sup f_n)(x) := \sup_n f_n(x)$$
$$(\inf f_n)(x) := \inf_n f_n(x)$$
$$\left(\limsup_n f_n\right)(x) := \limsup_n f_n(x)$$
$$\left(\liminf_n f_n\right)(x) := \liminf_n f_n(x)$$

Then these functions are $\mathcal{F}/\mathcal{B}(\mathbb{R}^\sharp)$-measurable.

Proof. Observe that

$$\left(\inf_n f_n\right)^{-1}([x, \infty]) = \bigcap_{n=1}^{\infty} f_n^{-1}([x, \infty])$$

so $\inf f_n$ is measurable by (5.8.41). Further $\sup_n f_n = -\inf_n(-f_n)$. ∎

Proposition 5.8.44
Let $f, g : (\Omega, \mathcal{F}) \to (\mathbb{R}^\sharp, \mathcal{B}(\mathbb{R}^\sharp))$ be measurable functions. Then the following functions are measurable

a) $f \pm g$
b) λf is measurable for $\lambda \in \mathbb{R}$
c) $f \cdot g$
d) $\max(f, g)$, $\min(f, g)$

Note when $f \pm g, fg$ is not well-defined (e.g. $\infty - \infty$ or $-\infty \times -\infty$) then we choose an arbitrary, but fixed, sentinel value.

Proof. We make use of (5.8.41) throughout.

a) Define $h(\omega) := f(\omega) + g(\omega)$ and let A be the set on which the sum is well-defined. Then it is evidently measurable and

$$\begin{aligned} h^{-1}((x, \infty]) &= \{\omega \in B_x^c \mid f(\omega) + g(\omega) > x\} \cup B_x \\ &= \{\omega \in A \mid f(\omega) + g(\omega) > x\} \cup B_x \\ &= \left(\bigcup_{r \in \mathbb{Q}} \{\omega \in A \mid f(\omega) > r\} \cap \{\omega \in A \mid g(\omega) > x - r\}\right) \cup B_x \end{aligned}$$

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where $B_x = \emptyset$ or A^c according to if the sentinel value of $f+g$ is $\leq x$. For the final equality the reverse inclusion is immediate. Conversely suppose $f(\omega)+g(\omega) > x$ then by (5.1.29) there exists some $r \in \mathbb{Q}$ such that $f(\omega) > r > x - g(\omega)$. The case $f-g$ follows immediately from b).

b) Observe that

$$(\lambda f)^{-1}((x,\infty]) = \begin{cases} \left(\frac{x}{\lambda}, \infty\right] & \lambda > 0 \\ \left[-\infty, \frac{x}{\lambda}\right) & \lambda < 0 \\ \Omega & \lambda = 0, x < 0 \\ \emptyset & \lambda = 0, x \geq 0 \end{cases}$$

c) First consider the case f^2 and observe

$$(f^2)^{-1}((x,\infty]) = \begin{cases} f^{-1}(\{-\infty,\infty\}) \cup f^{-1}\left(\left(\sqrt{x},\infty\right]\right) \cup f^{-1}\left(\left[-\infty,-\sqrt{x}\right)\right) & x \geq 0 \\ \Omega & x < 0 \end{cases}$$

Then we may deduce the general case because

$$(fg)(\omega) = \begin{cases} \frac{(f(\omega)+g(\omega))^2 - (f(\omega)-g(\omega))^2}{4} & f(\omega), g(\omega) \notin \{-\infty,\infty\} \\ \infty & (f(\omega), g(\omega)) \in \{(\infty,\infty),(-\infty,-\infty)\} \\ -\infty & (f(\omega), g(\omega)) \in \{(\infty,-\infty),(-\infty,\infty)\} \end{cases}$$

d) We may see this as a special case of (5.8.43)

■

5.8.9 Integration over a Measure

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5.9 Differential Calculus on Banach Spaces

The main reference for this section is [Car71], along with [Lan12], in a generality which is almost certainly not required for the purposes of these notes.

5.9.1 Total Derivatives

Definition 5.9.1
*Let $U \subset X$ be an open neighbourhood of 0 and $\varphi : U \to Y$ a function. We say φ is **sublinear** if*

$$\forall \varepsilon > 0 \exists \delta \text{ s.t. } \|h\| < \delta \implies (h \in U) \wedge (\|\varphi(h)\| < \varepsilon \|h\|)$$

Note sublinear functions are closed under linear combinations, and the property is unchanged under equivalent norms.

Definition 5.9.2 (Total Derivative)
*Let $U \subset X$ be an open subset of a Banach space, Y a Banach space and $x \in U$. We say that a map $\varphi : U \to Y$ is differentiable at x if there exists a continuous linear map $D\varphi(x) \in L(X, Y)$ (**derivative** at x) such that*

$$\varphi(x + h) - \varphi(x) - D\varphi(x)(h) \text{ is sublinear in } h$$

*We say it is simply **differentiable** if this holds for every point $x \in U$ and write $D\varphi$ for the corresponding function $U \to L(X, Y)$.*

*If $D\varphi$ is also continuous then we say that φ is **continuously differentiable** or C^1.*

The following result shows that the total derivative is necessarily unique.

Proposition 5.9.3
Let $\alpha : X \to Y$ be a linear map which is also sublinear. Then $\alpha = 0$.

Proof. Consider $x \in X$ then for every $\varepsilon > 0$ there exists δ such that $\|h\| < \delta \implies \|\alpha(h)\| < \varepsilon \|h\|$. Define $\lambda := \frac{\delta}{2\|x\|}$. Then evidently $\|\alpha(\lambda x)\| < \varepsilon \|\lambda x\|$ whence $\|\alpha(x)\| < \varepsilon \|x\|$. As this holds for every ε we see $\|\alpha(x)\| = 0$ and $\alpha(x) = 0$. ∎

Proposition 5.9.4
Let $f : U \to Y$ be a differentiable map and $\alpha : Y \to Z$ a linear map. Then

$$D(\alpha \circ f)(x) = \alpha \circ Df(x)$$

Proposition 5.9.5
Let $U \subset K$ an open set and $\varphi : U \to Y$ a function. Then the following are equivalent

a) φ is (continuously) differentiable in the sense of (5.9.2)
b) φ is (continuously) differentiable in the sense of (5.5.1)

Furthermore we may identify the derivatives as follows

$$D\varphi(t)(\lambda) = \varphi'(t)\lambda$$

Proposition 5.9.6 (Chain Rule)
Let $U \subset X$ and $V \subset Y$ be open sets and $\varphi : U \to Y$ and $\psi : V \to Z$ (continuously) differentiable functions such that $\varphi(U) \subset V$. Then $\psi \circ \varphi$ is (continuously) differentiable and

$$(D(\psi \circ \varphi))(x) = (D\psi)(\varphi(x)) \circ (D\varphi)(x)$$

In the case $X = K$ we have

$$(\psi \circ \varphi)'(t) = (D\psi)(\varphi(t))(\varphi'(t))$$

Proposition 5.9.7 (Product Rule I)
Let $U \subset X$ and $V \subset Y$ open sets, $\varphi : U \to X'$, $\psi : V \to Y'$ (continuously) differentiable maps and $\Psi : X' \times Y' \to Z$ a C^1 bilinear map. Then define

$$(\varphi \times_\Psi \psi)(x, y) := \Psi(\varphi(x), \psi(y))$$

is (continuously) differentiable with derivative

$$D(\varphi \times_\Psi \psi)(x, y)(h, k) = \Psi(D\varphi(x)(h), \psi(y)) + \Psi(\varphi(x), D\psi(y)(k))$$

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Corollary 5.9.8
Let X, Y be normed vector spaces, $U \subset X$ an open subset and $\varphi, \psi : X \to Y$ (continuously) differentiable at x. Then both $\varphi + \psi$ and $\lambda\varphi$ are (continuously) differentiable at x and

$$D(\varphi + \psi)(x) = D(\varphi)(x) + D(\psi)(x)$$
$$D(\lambda\varphi)(x) = \lambda D(\varphi)(x)$$

If Y is also a Banach algebra then $\varphi \cdot \psi$ is (continuously) differentiable at x and

$$D(\varphi \cdot \psi)(x)(h) = D(\varphi)(x)(h) \cdot \psi(x) + \varphi(x) \cdot D(\psi)(x)(h)$$

Corollary 5.9.9 (Product Rule II)
Let $U \subset X$ be an open set, and $\varphi : U \to K$ and $\psi : U \to Y$ differentiable maps. Then

$$D(\varphi \cdot \psi)(x)(h) = D\varphi(x)(h)\psi(x) + \varphi(x)D\psi(x)(h)$$

If φ, ψ are C^1 then so is $\varphi \cdot \psi$.

Definition 5.9.10 (Higher-Order Derivatives)
*Let $f : U \to Y$ we say that f is n-times **continuously differentiable** (or C^n) if it is differentiable and Df is C^{n-1} (where C^0 means simply continuous). In this case for $k \leq n$ we may define the k-fold derivative inductively*

$$D^k f := D\big(D^{k-1} f\big) : U \longrightarrow L(X, \cdots, L(X, Y))$$

which is by definition continuous.

Proposition 5.9.11
Let $U \subset K$ be an open subset and $f : U \to Y$ a function. Then f is n-times differentiable in the sense of (...) if and only if it is n-times differentiable in the sense of (5.9.10). In this case

$$(D^n f)(x)(h_1, \cdots, h_n) = f^{(n)}(x) \cdot h_1 \cdots h_n$$

Proposition 5.9.12 (Mean Value Theorem)
Let $U \subset X$ be an open subset and $f : U \to Y$ a C^1 map. Suppose that $h \in B(x; r) \subset U$. Then

$$f(x + h) = f(x) + \int_0^1 Df(x + th)(h)\,\mathrm{d}t$$

Proof. Let $G(t) := f(x + th)$ be defined as a function on $[0, 1] \to Y$. Then by (5.9.6) $G'(t) = Df(x + th)(h)$ is continuous, because evidently $t \to x + th$ is continuously differentiable with scalar derivative h. The result follows by applying (5.8.20) to $G(t)$. ∎

5.9.2 Partial Derivatives

Lemma 5.9.13
Let $X_1, \cdots, X_n$ be Banach spaces and for every $x \in X$ define the injection

$$\lambda_j(x) : X_j \longrightarrow X_1 \times \cdots \times X_n$$
$$x' \to \ (x_1, \cdots, x_{j-1}, x', x_{j+1}, \cdots, x_n)$$

Then λ_j is differentiable at every $x_j \in X_j$ and $\big(D\lambda_j(x)\big)\big(x_j\big) = \mu_j$ where

$$\mu_j(h) = (0, \cdots, h, \cdots, 0)$$

Proof. We have $\lambda_j(x)(x' + h') - \lambda_j(x) = \mu_j(h')$ and μ_j is linear. ∎

Proposition 5.9.14 (Criteria for Total Differentiability)
Let $X_1, \cdots, X_n, Y$ be Banach spaces and $U \subset X := X_1 \times \cdots \times X_n$ an open subset. Consider a continuous map

$$\varphi : U \to Y$$

The following are equivalent

a) The map φ is continuously differentiable (with respect to the sup norm on the product)
b) For every $x \in U$ define $V_j := \lambda_j(x)^{-1}(U) \subset X_j$ an open neighbourhood of x_j
i) The map

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$$\varphi_j(x) := \varphi \circ \lambda_j(x) : V_j \to Y$$

is differentiable for all $j = 1 \cdots n$. *Denote the* j*-th partial derivative at* $x \in U$ *by*

$$D_j\varphi : U \to L\big(X_j, Y\big)$$
$$x_j \to h \to \big(D\varphi_j(x)\big)\big(x_j\big)(h)$$

ii) The map D_j *is continuous for all* $j = 1 \cdots n$

Furthermore in this case

$$D\varphi(x)(h) = \sum_{j=1}^{n} \big(D_j\varphi\big)(x)\big(h_j\big)$$

Proof. **a) $\implies$ b)** By the chain rule (5.9.6) and (5.9.13) we have φ_j is differentiable and

$$\big(D\varphi_j(x)\big)\big(x_j\big) = (D\varphi)(x) \circ \big(D\lambda_j(x)\big)\big(x_j\big) = (D\varphi)(x) \circ \mu_j$$

b) $\implies$ a) We consider the case $n = 2$. Observe

$$\begin{aligned}
\varphi(x_1 + h_1, x_2 + h_2) - \varphi(x_1, x_2) &= \varphi(x_1 + h_1, x_2 + h_2) - \varphi(x_1 + h_1, x_2) + \varphi(x_1 + h_1, x_2) - \varphi(x_1, x_2) \\
&= \int_0^1 D_2\varphi(x_1 + h_1, x_2 + th_2)(h_2)\,\mathrm{d}t + \int_0^1 D_1\varphi(x_1 + th_1, x_2)(h_1)\,\mathrm{d}t \\
&= D_2\varphi(x_1, x_2)(h_2) + D_1\varphi(x_1, x_2)(h_1) \\
&+ \int_0^1 [D_2\varphi(x_1 + h_1, x_2 + th_2)(h_2) - D_2\varphi(x_1, x_2)(h_2)]\,\mathrm{d}t \\
&+ \int_0^1 [D_1\varphi(x_1 + th_1, x_2)(h_1) - D_1\varphi(x_1, x_2)(h_1)]\,\mathrm{d}t
\end{aligned}$$

Then

$$\begin{aligned}
\left\| \int_0^1 [D_2\varphi(x_1 + h_1, x_2 + th_2)(h_2) - D_2\varphi(x_1, x_2)(h_2)]\,\mathrm{d}t \right\| &\leq \int_0^1 \|D_2\varphi(x_1 + h_1, x_2 + th_2)(h_2) - D_2\varphi(x_1, x_2)(h_2)\|\,\mathrm{d}t \\
&\leq \int_0^1 \|D_2\varphi(x_1 + h_1, x_2 + th_2) - D_2\varphi(x_1, x_2)\|\|h_2\|\,\mathrm{d}t \\
&\leq \int_0^1 \|D_2\varphi\|\|(h_1, th_2)\|\|h_2\|\,\mathrm{d}t \\
&\leq \|D_2\varphi\|\|h\|^2
\end{aligned}$$

and similarly for the second integral. Therefore φ is differentiable at (x_1, x_2) with derivative

$$D\varphi(x_1, x_2)(h) = D_1\varphi(x_1, x_2)(h_1) + D_2\varphi(x_1, x_2)(h_2)$$

as required.

We have $h = \sum_{j=1}^{n} \mu_j\big(h_j\big)$ and so by the first part

$$\begin{aligned}
D\varphi(x)(h) &= \sum_{j=1}^{n} \big(D\varphi(x)\big)\big(\mu_j\big(h_j\big)\big) \\
&= \sum_{j=1}^{n} \big(D\varphi_j(x)\big)\big(x_j\big)\big(h_j\big) \\
&= \sum_{j=1}^{n} \big(D_j\varphi\big)(x)\big(h_j\big)
\end{aligned}$$

■

Proposition 5.9.15
Let $U \subset X$ *be an open set and*

$$\varphi : U \to Y_1 \times \cdots \times Y_m$$

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Then φ is differentiable (resp. C^1) iff $\pi_i \circ \varphi$ is differentiable (resp. C^1) for $i = 1 \cdots m$.

Proposition 5.9.16 (Jacobian Matrix)
Let X, Y be finite-dimensional Banach spaces with bases $\{e_1, \cdots, e_n\}, \{f_1, \cdots, f_m\}$ such that $U \subset X$ is open and $\varphi : U \to Y$ a map. Then the following are equivalent

a) φ is C^1
b) For every $x \in U$, $j = 1 \cdots n$, $i = 1 \cdots m$ the map

$$g_{ij}(x) : V \to K$$
$$t \to (f_i^\vee \circ \varphi)(x + te_j)$$

is continuously differentiable on the open set $V = \{t \in K \mid x + te_j \in U\}$.

In this case define $\frac{\partial \varphi_i}{\partial e_j}(x) \coloneqq g_{ij}(x)'(0)$. Then the linear map $D\varphi(x)$ has the following matrix representation

$$J_\varphi(x) \coloneqq [D\varphi(x)] = \left(\frac{\partial \varphi_i}{\partial e_j}(x)\right)$$

*which we refer to as the **Jacobian Matrix**.*

When $Y = K$ then $\{1_K\}$ is a basis and we may simply write $\frac{\partial \varphi}{\partial e_j} : U \to K$ as the partial derivative.

Proof. Let $\psi_j : K \xrightarrow{\sim} \langle e_j \rangle$ be the canonical linear isomorphism and $\lambda_j(x) : \langle e_j \rangle \to X$ by the function defined in (5.9.13). Then

$$f_i^\vee \circ \varphi \circ \lambda_j(x) = g_{ij}(x) \circ \psi_j^{-1}$$

Suppose *b*) is satisfied. Because ψ_j^{-1} is continuously differentiable the hypotheses of (5.9.14) b) are satisfied and we conclude that $f_i^\vee \circ \varphi$ is C^1. Then by (5.9.15) so is φ.

Conversely if φ is C^1, then so is $f_i^\vee \circ \varphi$. And we may use the same identity and the chain rule to deduce $g_{ij}(x)$ is continuously differentiable.

By the same result we have

$$(f_i^\vee \circ D\varphi(x))(x_j) = D(f_i^\vee \circ \varphi)(x)(e_j) = D(f_i^\vee \circ \varphi \circ \lambda_j(x))(e_j)(e_j) = g_{ij}(x)'(0)e_j$$

which shows that $J_\varphi(x)$ has the required form (...). ∎

Proposition 5.9.17
Let X, Y, Z be finite-dimensional Banach spaces with bases $\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3$ respectively. Suppose $U \subset X, V \subset Y$ are open sets and $\varphi : U \to Y, \psi : V \to Z$ are C^1 maps such that $\varphi(U) \subset V$. Then $\psi \circ \varphi$ is C^1 and

$$J_{\psi \circ \varphi}(x) = J_\psi(\varphi(x)) J_\varphi(x)$$

5.9.3 Second Derivative

Definition 5.9.18
*For a map $\alpha : X \times X \to Y$ we say that $\alpha(v, w)$ is **sublinear** if for all $\varepsilon > 0$ there exists $\delta > 0$ such that*

$$\|(v, w)\| < \delta \implies \|\alpha(v, w)\| < \varepsilon \|v\| \|w\|$$

where the norm on $X \times X$ is the product norm.

As before we have the following uniqueness property

Lemma 5.9.19
Let $\alpha : X \times X \to Y$ be a bilinear map which is also sublinear. Then $\alpha = 0$.

Proposition 5.9.20 (Second derivative is symmetric)
Let $f : U \to Y$ be a C^2 map. Then for all $x \in U$ we have $D^2 f(x)$ is the unique bilinear map $X \times X \to Y$ such that

$$f(x + v + w) - f(x + w) - f(x + v) + f(x) - D^2 f(x)(v, w) \text{ is sublinear}$$

Furthermore it is symmetric.

Proof. Uniqueness follows from (5.9.19) and from this it immediately follows that $D^2 f(x)$ is symmetric. Therefore we only need to show the required property. Let $v, w \in X$ be such that $x + tv + sw \in U$ for all $0 \leq s, t \leq 1$. Then define

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$$g(x) := f(x+v) - f(x)$$

By (...) applied to g and Df we find

$$\begin{aligned} g(x+w) - g(x) &= \int_0^1 [Df(x+v+tw)(w) - Df(x+tw)(w)]\,\mathrm{d}t \\ &= \int_0^1 \int_0^1 D^2 f(x+sv+tw)(v)\,\mathrm{d}s(w)\,\mathrm{d}t \\ &= D^2 f(x)(v,w) + \int_0^1 \int_0^1 [D^2 f(x+sv+tw)(v) - D^2 f(x)(v)]\,\mathrm{d}s(w)\,\mathrm{d}t \end{aligned}$$

Let $\psi(v,w)$ denote the integral then

$$\begin{aligned} \|\psi(v,w)\| &\le \int_0^1 \int_0^1 \left\|D^2 f(x+sv+tw)(v) - D^2 f(x)(v)\right\| \mathrm{d}s\|w\|\,\mathrm{d}t \\ &\le \int_0^1 \int_0^1 \left\|D^2 f(x+sv+tw) - D^2 f(x)\right\| \mathrm{d}s\,\mathrm{d}t \cdot \|v\|\|w\| \\ &\le \sup_{0\le s,t\le 1} \left\|D^2 f(x+sv+tw) - D^2 f(x)\right\| \|v\|\|w\| \end{aligned}$$

where the estimate is finite by continuity of $D^2 f$ and the Boundedness Theorem (5.7.8). Note that $\|sv+tw\| \le \|v\| + \|w\| \le 2\|(v,w)\|$. Therefore we may use the continuity of $D^2 f$ to show that ψ is sublinear as required. ∎

Proposition 5.9.21 (Symmetry of partial derivatives)
Let $X = X_1 \times \cdots \times X_n$, $\varphi : U \to Y$ be a C^2 map and $x \in U$. Then

$$D^2\varphi(x)\big((0,\cdots,h_i,\cdots,0),(0,\cdots,h_j,\cdots,0)\big) = D_i D_j \varphi(x)\big(h_i,h_j\big)$$

In particular

$$D_i D_j \varphi(x)\big(h_j,h_i\big) = D_j D_i \varphi(x)\big(h_i,h_j\big)$$

Proof. The first equation follows from (5.9.14) and the second from (5.9.20). ∎

Proposition 5.9.22 (Hessian Matrix)
Let X be a finite-dimensional Banach spaces with basis $\{e_1,\cdots,e_n\}$, $U \subset X$ an open subset and $\varphi : U \to K$ a C^2 map. Then the second order partial derivatives $\frac{\partial^2\varphi}{\partial e_i \partial e_j} := \frac{\partial}{\partial e_j}\frac{\partial\varphi}{\partial e_i} : U \to K$ exist for all $1 \le i,j \le n$.

Denote by

$$H_\varphi := \left(\frac{\partial^2\varphi}{\partial e_i \partial e_j}\right)_{1\le i,j\le n}$$

*the **Hessian Matrix** which is the matrix of the bilinear form $D^2\varphi$ with respect to the given basis, and in particular symmetric.*

5.9.4 Differentiation under Integral Sign

Proposition 5.9.23
Let $A \subset X$ be a compact subset of a normed vector space and $f : A \to Y$ a continuous map. Then f is uniformly continuous. That is for all $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\|x-y\| < \delta \implies \|f(x) - f(y)\| < \varepsilon \quad \forall x,y \in A$$

Proof. Given $\varepsilon > 0$ for each $x \in A$ by continuity there exists a real number $\delta(x) > 0$ such that $\|y-x\| < \delta(x) \implies \|f(y) - f(x)\| < \varepsilon$. Evidently the balls $B\big(x, \frac{\delta(x)}{2}\big)$ cover A, so we may find a finite subcover of balls centered at $x_1,\cdots,x_n$. Let $\delta := \min\Big(\frac{\delta(x_1)}{2},\cdots,\frac{\delta(x_n)}{2}\Big)$.

TODO ∎

Proposition 5.9.24
Let $I = [a,b]$ be a real interval, $U \subset X$ an open subset of a finite-dimensional normed vector space and $f : I \times U \to Y$ be a continuous map such that $D_2 f$ exists and is continuous. Define

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$$g(x) = \int_a^b f(t,x)\,\mathrm{d}t$$

Then g is differentiable on U with derivative given by

$$Dg(x)(h) = \int_a^b D_2 f(t,x)(h)\,\mathrm{d}t$$

Proof. Consider a point $x \in U$ with a compact neighbourhood $N \subset U$ (...). Consider $\lambda \in L(X,Y)$ determined by

$$\lambda(h) = \int_a^b (D_2 f)(t,x)(h)\,\mathrm{d}t$$

Then by (5.9.12)

$$\begin{aligned} g(x+h) - g(x) - \lambda(h) &= \int_a^b [f(t,x+h) - f(t,x) - (D_2 f)(t,x)(h)]\,\mathrm{d}t \\ &= \int_a^b \left[\int_0^1 (D_2 f)(t,x+uh)(h)\,\mathrm{d}u - (D_2 f)(t,x)(h)\right]\mathrm{d}t \\ &= \int_a^b \left[\int_0^1 [(D_2 f)(t,x+uh)(h) - (D_2 f)(t,x)(h)]\,\mathrm{d}u\right]\mathrm{d}t \end{aligned}$$

For some δ_1 we have $B(x,\delta_1) \subset N$. Therefore if $\|h\| < \delta_1$ then $x + uh \in N$ for all $u \in [0,1]$. Observe that

$$\|(t,x+uh) - (t,x)\| = \|(0,uh)\| = |u|\|h\|$$

By uniform continuity (5.9.23) on the compact set $[a,b] \times N$ (4.1.115) there is some δ_2 such that

$$(x+uh) \in N \wedge |u|\|h\| < \delta_2 \implies \|(D_2 f)(t,x+uh) - (D_2 f)(t,x)\| < \varepsilon$$

Therefore for $\|h\| < \min(\delta_1, \delta_2)$ we have

$$\begin{aligned} \|g(x+h) - g(x) - \lambda(h)\| &\leq \int_a^b \int_0^1 \|(D_2 f)(t,x+uh)(h) - (D_2 f)(t,x)(h)\|\,\mathrm{d}u\,\mathrm{d}t \\ &\leq (b-a)\|h\|\varepsilon \end{aligned}$$

By definition this means $g(x)$ is differentiable with the given derivative. ∎

Corollary 5.9.25
Let $I = [a,b]$ be a real interval, $U \subset K$ an open subset of a complete valued field K and $f(t,x) : I \times U \to Y$ a continuous map such that $\frac{\partial f}{\partial x} : I \times U \to K$ exists and is continuous. Define

$$g(x) = \int_a^b f(t,x)\,\mathrm{d}t$$

Then g is differentiable on U (in the sense of ....) and

$$g'(x) = \int_a^b \frac{\partial f}{\partial x}(t,x)\,\mathrm{d}t$$

Proof. This follows because $\frac{\partial f}{\partial x} = (D_2 f)(x)(1_K)$ and $g'(x) = (Dg)(x)(1_K)$. ∎

5.9.5 Taylor's Theorem

Lemma 5.9.26 (Integration by Parts)
Let $U : [0,1] \to K$ and $V : [0,1] \to X$ be C^1 functions. Then

$$\int_0^1 U'(t)V(t)\,\mathrm{d}t = U(1)V(1) - U(0)V(0) - \int_0^1 U(t)V'(t)\,\mathrm{d}t$$

Proof. Let $H(t) := U(t)V(t)$ then by (5.9.9) $H'(t) = U'(t)V(t) + U(t)V'(t)$ is continuous. The result then follows from (5.8.20). ∎

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Corollary 5.9.27 (Integration by Parts II)
Let $G : [0,1] \to X$ be a C^1 function. Then

$$\int_0^1 G(t)\,\mathrm{d}t = G(0) + \int_0^1 (1-t)G'(t)\,\mathrm{d}t$$

Proposition 5.9.28 (Taylor's Theorem)
Let $U \subset X$ be an open set and $f : U \to Y$ a C^2 map. Suppose $x \in U$, $B(x;r) \subset U$ and $\|h\| < r$. Then

$$f(x+h) = f(x) + Df(x)(h) + \int_0^1 (1-t)D^2 f(x+th)(h,h)\,\mathrm{d}t$$

Proof. Let $G(t) = Df(x+th)(h) : [0,1] \to Y$ then

$$G(t) = \mathrm{ev}_h \circ Df \circ (x+th)$$

By (5.9.4)

$$D(\mathrm{ev}_h \circ Df)(x) = \mathrm{ev}_h \circ D^2 f(x)$$

whence by (5.9.6) $G(t)$ is C^1 with scalar derivative

$$G'(t) = (\mathrm{ev}_h \circ D^2 f(x+th))(h) = D^2 f(x+th)(h,h)$$

Recall by the Mean Value Theorem (5.9.12) and Integration by Parts (5.9.27)

$$\begin{aligned} f(x+h) &= f(x) + \int_0^1 G(t)\,\mathrm{d}t \\ &= f(x) + G(0) + \int_0^1 (1-t)G'(t)\,\mathrm{d}t \end{aligned}$$

which is the required result. ∎

Corollary 5.9.29 (Scalar form of Taylor's Theorem)
Let X be a finite-dimensional Banach space with basis $\{e_1, \cdots, e_n\}$, $U \subset X$ an open subset and $f : U \to K$ a C^2 map. Suppose $x \in U$ and $y \in B(x;r) \subset U$ then

$$f(y) = f(x) + \sum_{i=1}^n (y_i - x_i)\frac{\partial f}{\partial e_i}(x) + \sum_{i,j=1}^n (y_i - x_i)(y_j - x_j)R_{ij}$$

with

$$R_{ij}(y;x) = \int_0^1 (1-t)\frac{\partial^2 f}{\partial e_i \partial e_j}(x + t(y-x))\,\mathrm{d}t$$

If f is C^{2+p} (resp. C^∞) then R_{ij} is C^p (resp. C^∞) on $B(x;r)$.

5.9.6 Differential Forms on Banach Spaces

Definition 5.9.30
Let X, Y be Banach spaces and $U \subset X$ an open subset. A continuous mapping

$$\omega : U \longrightarrow L_a^p(X,Y)$$

*is called a **differential form of degree** p or a **differential p-form***

We say that ω is n-times continuously differentiable if it is so with respect to the canonical norm on $L_a^p(X,Y)$.

Proposition 5.9.31
Let X be a finite-dimensional vector space with basis $e_1, \cdots, e_k$ and $U \subset X$ an open subset. For each $i = 1 \cdots k$ define the mapping

$$\mathrm{d}x_i : U \to L_a^1(X,K)$$

which assigns the dual map $e_i^\vee$. Then for every differential p-form ω we have

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$$\omega(x) = \sum_{i=1}^{n} c_i(x)\,\mathrm{d}x_i$$

for some unique functions $c_i : U \to K$.

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5.10 Complex Analysis

5.10.1 Cauchy-Riemann Equations

Proposition 5.10.1 (Isometry between CC and RR^2)
The canonical map $\chi : \mathbb{C} \to \mathbb{R}^2$ is a Banach space isometry (using the square norm $\|\cdot\|_2$). Furthermore define

$$\begin{aligned}\mathrm{Re}(z) &:= \pi_1\chi(z)\\ \mathrm{Im}(z) &:= \pi_2\chi(z)\end{aligned}$$

then

$$z = \mathrm{Re}(z)\cdot 1 + \mathrm{Im}(z)\cdot i$$

where $\mathbb{C}$ is viewed as an $\mathbb{R}$ vector space. One may also verify that complex multiplication may be represented in matrix form

$$\chi(z\cdot w) = \begin{pmatrix}\mathrm{Re}(z) & -\mathrm{Im}(z)\\ \mathrm{Im}(z) & \mathrm{Re}(z)\end{pmatrix}\chi(w)^T$$

Proposition 5.10.2 (Cauchy-Riemann Equations)
Let $U \subset \mathbb{C}$ and $f : U \to \mathbb{C}$ a function. There exists unique functions $u, v : \chi(U) \to \mathbb{R}$ such that

$$f(x+yi) = u(x,y) + v(x,y)i$$

Further the following are equivalent

a) $f(z)$ is continuously differentiable (as a complex function)
b) u, v are C^1 and satisfy the relations

$$\begin{aligned}\frac{\partial u}{\partial x}(x,y) &= \frac{\partial v}{\partial y}(x,y)\\ \frac{\partial u}{\partial y}(x,y) &= -\frac{\partial v}{\partial x}(x,y)\end{aligned}$$

Proof. We may define $F : \chi(U) \to \mathbb{R}^2$ by $F = \chi \circ f \circ \chi^{-1}$ and $u = \pi_1 \circ F$ and $v = \pi_2 \circ F$.

a) $\implies$ b) Fix $z = x + yi \in U$ and $f'(z) = a + bi$. Then for sufficiently small $w = h + ki$

$$f(z+w) - f(z) - f'(z)w$$

is sublinear in w. By assumption F is C^1 and has Jacobian matrix (...) at (x,y) equal to

$$J_F(x,y) := \begin{pmatrix}\frac{\partial u}{\partial x}(x,y) & \frac{\partial u}{\partial y}(x,y)\\ \frac{\partial v}{\partial x}(x,y) & \frac{\partial v}{\partial y}(x,y)\end{pmatrix}$$

and $Df(x,y)(h,k) = J_F(x,y)\begin{pmatrix}h\\k\end{pmatrix}^T$. However by conjugating the above equation by χ we find that

$$F(x+h, y+k) - F(x,y) - \begin{pmatrix}a & -b\\ b & a\end{pmatrix}\begin{pmatrix}h\\k\end{pmatrix}^T$$

(where $f'(z) = a + bi$) is sublinear (and using the fact χ is an isometry). Therefore the result then follows from the uniqueness of the total derivative $Df(x,y)$, and hence Jacobian J_F.

b) $\implies$ a) Conversely if the conditions hold then we may define $f'(z) := \frac{\partial u}{\partial x}(\chi(z)) + \frac{\partial v}{\partial x}(\chi(z))i$ which is evidently continuous. Then by using matrix representation of complex multiplication (5.10.1) we find

$$\chi[f(z+w) - f(z) - f'(z)w] = F(\chi(z+w)) - F(\chi(z)) - J'(\chi(z))\chi(w)^T$$

where

$$J'(x,y) = \begin{pmatrix}\frac{\partial u}{\partial x}(x,y) & -\frac{\partial v}{\partial x}(x,y)\\ \frac{\partial v}{\partial x}(x,y) & \frac{\partial u}{\partial x}(x,y)\end{pmatrix}$$

The conditions mean precisely that $J'(x,y) = J(x,y)$. As χ is an isometry we find that $f(z+w) - f(z) - f'(z)w$ is sublinear in w as required. ∎

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5.10.2 Holomorphic Functions

Definition 5.10.3 (Holomorphic Function)
*Let $U \subset \mathbb{C}$ be an open subset and $f : U \to \mathbb{C}$ a function. Then we say that f is **holomorphic** if it is infinitely (complex) differentiable (5.10.2). For a holomorphic function every derivative is automatically continuous (5.5.3).*

*If f is injective and $f(U)$ is open we say that f is **biholomorphic** if its inverse is also holomorphic. A biholomorphic functions is a homeomorphism onto its image (5.5.3).*

Trivially if f is holomorphic then so is $f|_V$ for any open subset $V \subset U$. Conversely if $U = \bigcup_{\alpha\in\mathcal{A}} V_\alpha$ and $f|_{V_\alpha}$ are holomorphic for all $\alpha \in \mathcal{A}$, then so is f.

Proposition 5.10.4
Let $U \subset \mathbb{C}$ be an open subset. Then the set of holomorphic functions on U forms a $\mathbb{C}$-algebra.

For $f : U \to \mathbb{C}$ non-constant then $D(f) := \{x \in U \mid f(x) \neq 0\}$ is an open subset and $\left(f|_{D(f)}\right)^{-1}$ is a holomorphic function.

Proof. This follows from (5.5.4), provided we show that $D(f)$ is open. For $\mathbb{C}$ is Hausdorff (5.4.5) so $\{0\}$ is closed (4.1.48). Therefore by (4.1.27) $D(f) = U \setminus f^{-1}(\{0\})$ is open. ∎

5.10.3 Exponential and Trigonometric Functions

Proposition 5.10.5 (Exponential Function)
Let K be a complete valued field. Then the series

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

has infinite radius of convergence. Furthermore it satisfies the following properties

a) e^z is infinitely differentiable with derivative e^z
b) $e^{z+w} = e^z e^w$
c) $e^z e^{-z} = 1$

In the case $K = \mathbb{R}$ then we also have the following property

d) e^x is positive, strictly increasing and bijective : $(-\infty, \infty) \to (0, \infty)$

Proof. We have

$$\left|\frac{a_{n+1}z^{n+1}}{a_n z^n}\right| = \frac{z}{n+1} \to 0$$

so the series converges unconditionally. Therefore by (5.6.2) it has infinite radius of convergence and is infinitely differentiable. Observe formally the product of power series is given by

$$\begin{aligned} e^z e^w &= \left(\sum_{n=0}^{\infty} \frac{z^n}{n!}\right)\left(\sum_{n=0}^{\infty} \frac{w^n}{n!}\right) \\ &= \sum_{n=0}^{\infty}\left(\sum_{k=0}^{n} \frac{z^k w^{n-k}}{k!(n-k)!}\right) \\ &= \sum_{n=0}^{\infty} \frac{(z+w)^n}{n!} \\ &= e^{z+w} \end{aligned}$$

which by (5.4.33) converge to the same value.

Evidently $x \geq 0 \implies e^x \geq 1 + x$. Then $x > y \geq 0 \implies e^x = e^{(x-y)+y} = e^{x-y}e^y \geq (1 + x - y)e^y > 1 \cdot e^y$. Similarly suppose $x < y \leq 0$ then $-y > -x \geq 0 \implies e^{-y} > e^{-x} \implies e^x < e^y$. Finally $x \leq 0 \implies e^{-x} \geq 1 - x \implies e^x \leq \frac{1}{1+x}$. This shows that e^x is strictly increasing and therefore injective on $(0, \infty)$. Further using $e^x e^{-x} = 1$ we see it is a positive function and $x \leq 0 \implies e^x \leq 1$.

As e^x is unbounded the intermediate value theorem (5.7.10) shows it achieves every value in $[1, \infty)$. Using $e^x e^{-x} = 1$ we see it also achieves every value in $(0, \infty)$. Therefore it is a bijection $(-\infty, \infty) \to (0, \infty)$ as required. ∎

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Proposition 5.10.6 (Trigonometric Functions)
The power series

$$\sin(z) = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots$$

$$\cos(z) = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \cdots$$

have infinite radius of convergence in $\mathbb{C}$*. Furthermore*

a) $\sin(0) = 0$ *and* $\cos(0) = 1$
b) $e^{iz} = \cos(z) + i\sin(z)$
c) $\sin(z) = \frac{e^{iz} - e^{-iz}}{2i}$ *and* $\cos(z) = \frac{e^{iz} + e^{-iz}}{2}$*.* $\sin(-z) = -\sin(z)$ *and* $\cos(-z) = \cos(z)$
d) $\sin'(z) = \cos(z)$ *and* $\cos'(z) = -\sin(z)$ *are infinitely differentiable*
e) $\sin^2(z) + \cos^2(z) = 1$
f) $\sin(w + z) = \sin(w)\cos(z) + \cos(w)\sin(z)$
g) $\cos(w + z) = \cos(w)\cos(z) - \sin(w)\sin(z)$

Proof. It is simpler to define $\sin(z)$ and $\cos(z)$ by *c*). The power series expansion then follows from (5.4.26).

a) Immediate from the series expansion
b) Immediate from *c*)
c) Already shown
d) Immediate from *c*) and the chain rule (5.9.6).
e) Use *b*) and $e^{iz}e^{-iz} = 1$

f)-g) Using *b*) we find

$$\begin{aligned}\cos(z + w) + i\sin(z + w) &= e^{i(z+w)} = e^{iz}e^{iw} \\ &= (\cos(z) + i\sin(z))(\cos(w) + i\sin(w)) \\ &= (\cos(z)\cos(w) - \sin(z)\sin(w)) + i(\sin(z)\cos(w) + \cos(z)\sin(w))\end{aligned}$$

and the result follows from equating real and imaginary parts ∎

Proposition 5.10.7 (cos and sin are periodic)
There is a unique positive real number $\pi \in \mathbb{R}$ *such that the real roots of* $\sin(x)$ *are precisely*

$$\{n\pi \mid n \in \mathbb{Z}\}$$

Furthermore we have the period formulae

a) $\sin(x) = \cos(x - \frac{\pi}{2}) = \cos(\frac{\pi}{2} - x)$
b) $\cos(x) = \sin(\frac{\pi}{2} - x) = -\sin(x - \frac{\pi}{2})$
c) $\sin(x) = -\sin(x + \pi)$ *and* $\sin(x + 2\pi) = \sin(x)$
d) $\cos(x) = -\cos(x + \pi)$ *and* $\cos(x + 2\pi) = \cos(x)$

and the real roots of $\cos(x)$ *are precisely*

$$\left\{\left(n + \frac{1}{2}\right)\pi \;\middle|\; n \in \mathbb{Z}\right\}$$

Proof. By (5.10.6).e) we know that $0 \leq \sin^2(x) \leq 1$ which means $-1 \leq \sin(x) \leq 1$, and similarly for $\cos(x)$. We first prove that $\cos(x)$ has a positive root. Suppose not then by the intermediate value theorem (5.7.10) we must have $\cos(x) > 0$ for all x. Then by the Mean-Value Theorem (5.7.12) $\sin(x)$ must be strictly increasing and therefore positive for all $x > 0$. We claim that for $x \geq a > 0$

$$\cos(x) \leq \cos(a) + \cos'(a)(x - a) = \cos(a) - \sin(a)(x - a)$$

For define $g(x) \coloneqq \cos(a) - \sin(a)(x - a) - \cos(x)$ then $g(a) = \cos(a) > 0$ and $g'(x) = \sin(x) - \sin(a) > 0$. By the Mean-Value Theorem $g(x)$ is increasing and the claim is proven. However it is evident that this implies $\cos(x)$ is unbounded below contradicting $-1 \leq \cos(x)$. Therefore $\cos(x)$ has a strictly positive root.

Define this to be

$$\frac{\pi}{2} \coloneqq \inf\{x \in \mathbb{R}^+ \mid \cos(x) = 0\}$$

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By (5.1.18) there exists a sequence x_n such that $\cos(x_n) = 0$ and $x_n \uparrow \frac{\pi}{2}$. By continuity and (5.3.6) we have $\cos(\frac{\pi}{2}) = 0$, and evidently $\pi \neq 0$. We must have $\cos(x)$ positive on the interval $[0, \frac{\pi}{2}]$, for otherwise by the intermediate value theorem we would have a smaller root. This shows that $\sin(x)$ is increasing on the interval $[0, \frac{\pi}{2}]$. By (5.10.6) we have $\sin^2(\frac{\pi}{2}) = 1$ and therefore $\sin(\frac{\pi}{2}) = 1$.

The period formula $a)$ then follows from the addition formula (5.10.6).g)

$$\cos\Big(\frac{\pi}{2} - x\Big) = \cos\Big(x - \frac{\pi}{2}\Big) = \cos(x)\cos\Big(-\frac{\pi}{2}\Big) - \sin(x)\sin\Big(-\frac{\pi}{2}\Big) = \sin(x)$$

and $b)$ follows directly from $a)$. Then $c), d)$ follow from combining $a), b)$. In particular $\sin(n\pi) = 0$. We claim that π is the smallest positive root of $\sin(x)$. For suppose $0 < x < \pi$ satisfies $\sin(x) = 0$ then by $a)$ we have $\cos\big(\big|x - \frac{\pi}{2}\big|\big) = 0$ and $0 \leq \big|x - \frac{\pi}{2}\big| < \frac{\pi}{2}$ which contradicts the choice of π. Suppose $\sin(y) = 0$ with $y > 0$. Then consider

$$n := \max\{m \in \mathbb{N} \mid y - m\pi \geq 0\}$$

Then $\sin(y - n\pi) = 0$ by $c)$. By choice of m we have $0 \leq y - n\pi < \pi$. As π is the smallest positive root we have $y - n\pi = 0$. By symmetry then the roots of $\sin(x)$ are precisely $\{n\pi\}$. Evidently any such π is unique because $\pi' = n\pi = nm\pi'$ whence $n = m = 1$. Using $a)$ yields the precise set of roots of $\cos(x)$. ∎

Proposition 5.10.8
There is a continuous, strictly decreasing function

$$\arccos : [-1, 1] \to [0, \pi]$$

such that

$$\cos(\arccos(x)) = x$$

Proof. By (5.10.7) $\sin(x)$ is positive on the interval $(0, \pi)$ (by the intermediate value theorem). By the Mean Value Theorem then $\cos(x)$ is strictly decreasing on this interval. Furthermore $\cos(0) = 1$ and $\cos(\pi) = -\cos(\pi - \pi) = -1$. By the intermediate value theorem then $\cos : [0, \pi] \to [-1, 1]$ is surjective and injective. Therefore the map $\arccos$ is well-defined and it is continuous by (5.7.13). ∎

Proposition 5.10.9 (Polar Coordinates)
There is a unique function

$$\theta : \mathbb{R}^2 \setminus \{(0, 0)\} \to [0, 2\pi)$$

such that

$$r(x, y)\cos(\theta(x, y)) = x \text{ and } r(x, y)\sin(\theta(x, y)) = y$$

where $r(x, y) := \sqrt{x^2 + y^2}$. *Explicitly it is given by*

$$\theta(x, y) := \begin{cases} \arccos\big(\frac{x}{r(x,y)}\big) & y \geq 0 \\ 2\pi - \arccos\big(\frac{x}{r(x,y)}\big) & y < 0 \end{cases}$$

and it is continuous on $\mathbb{R}^2 \setminus \{(x, 0) \mid x > 0\}$, *and* $\theta(x, 0-) = 2\pi + \theta(x, 0+)$ *when* $x > 0$.

Proof. Observe that $0 \leq \frac{x}{\sqrt{x^2+y^2}} \leq 1$ and so $r(x, y)\cos(\theta(x, y)) = x$. Further $\sin^2(\theta(x, y)) = 1 - \cos^2(\theta(x, y)) = \frac{y^2}{x^2+y^2}$, and so $r(x, y)\sin(\theta(x, y)) = \pm y$. When $y \geq 0$ we have $\theta(x, y) \in [0, \pi] \implies \sin(\theta(x, y)) \geq 0$. Similarly from the case $y \leq 0$ we find $r(x, y)\sin(\theta(x, y)) = y$ as required. ∎

Proposition 5.10.10
Let $r, r' > 0$ *and* $\theta, \theta' \in \mathbb{R}$. *Then*

$$(r\cos(\theta), r\sin(\theta)) = (r'\cos(\theta'), r'\sin(\theta'))$$

if and only if

$$r = r' \text{ and } \theta = \theta' + 2n\pi$$

Proof. We may use (5.10.6) to show that

$$r^2 = (r\cos(\theta))^2 + (r\sin(\theta))^2 = (r'\cos(\theta'))^2 + (r'\sin(\theta'))^2 = (r')^2$$

which means $r = r'$. Similarly we may use the addition formula to find

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$$\sin(\theta - \theta') = \sin(\theta)\cos(\theta') - \cos(\theta')\sin(\theta) = 0$$

which means $\theta = \theta' + n\pi$ by (5.10.7). Using the fact $\cos(\theta + n\pi) = (-1)^n \cos(\theta)$ and $\sin(\theta + n\pi) = (-1)^n \sin(\theta)$ we may deduce that n is even (since $\sin(\theta) = \cos(\theta) = 0$ is impossible). ■

5.10.4 Polar Coordinates of a Path

We would like to show that any path $\gamma : [a, b] \to \mathbb{C} \setminus \{0\}$ has a *continuous* parameterisation $(r(t), \theta(t))$ in polar coordinates, and that this is essentially unique.

Definition 5.10.11

*Suppose X is a Hausdorff space. A continuous surjective map $p : \tilde{X} \to X$ is a **covering map** if every $x \in X$ has a neighbourhood U_x such that*

a) $p^{-1}(U_x) = \bigsqcup_{i \in I_x} V_{x,i}$

b) $p|_{V_{x,i}} : V_{x,i} \to U_x$ is a homeomorphism for all $i \in I_x$

We may show that $\tilde{X}$ is necessarily Hausdorff.

Proposition 5.10.12 (Punctured Plane Covering Map)

The map

$$\begin{aligned} \mathbb{R}_{>0} \times \mathbb{R} &\to \mathbb{R}^2 \setminus \{(0,0)\} \\ (r, \theta) &\to (r\cos(\theta), r\sin(\theta)) \end{aligned}$$

is a covering map.

Proof. Define

$$\begin{aligned} U_1 &:= \mathbb{R}^2 \setminus \{(x, 0) \mid x \geq 0\} \\ U_2 &:= \mathbb{R}^2 \setminus \{(x, 0) \mid x \leq 0\} \end{aligned}$$

We claim that

$$\begin{aligned} p^{-1}(U_1) &= \bigsqcup_n \mathbb{R}_{>0} \times (2n\pi, (2n+2)\pi) \\ p^{-1}(U_2) &= \bigsqcup_n \mathbb{R}_{>0} \times ((2n-1)\pi, (2n+1)\pi) \end{aligned}$$

Denote the open sets by $V_{i,n}$ for $i = 1, 2$. Then the maps

$$p|_{V_{i,n}} \to U_i$$

are injective by (5.10.10). We may define explicit continuous inverses by

$$\begin{aligned} U_1 &\to V_{1,n} \\ (x, y) &\to \left(\sqrt{x^2 + y^2}, \theta(x, y) + 2n\pi\right) \\ U_2 &\to V_{2,n} \\ (x, y) &\to \left(\sqrt{x^2 + y^2}, -\theta(-x, y) + 2(n-1)\pi\right) \end{aligned}$$

■

Proposition 5.10.13 (Path lifting)

Let $p : \tilde{X} \to X$ be a covering map and $\gamma : [a, b] \to X$ a continuous path. Consider any $\tilde{x}_0 \in \tilde{X}$ such that $p(\tilde{x}_0) = \gamma(a)$. Then there exists a unique lifting $\tilde{\gamma} : [a, b] \to \tilde{X}$ such that $p \circ \tilde{\gamma} = \gamma$ and $\tilde{\gamma}(a) = \tilde{x}_0$.

Proof. We first prove uniqueness. Consider two liftings $\tilde{\gamma}_1, \tilde{\gamma}_2 : [a, b] \to \tilde{X}$ and define

$$X := \{t \in [a, b] \mid \tilde{\gamma}_1(t) = \tilde{\gamma}_2(t)\}$$

Fix $t \in X$ and let U be a neighbourhood of $\gamma(t)$ such that $p^{-1}(U) = \bigsqcup_{i \in I} V_i$. Suppose $\tilde{\gamma}_1(t) = \tilde{\gamma}_2(t) \in V_i$. By continuity there exists a neighbourhood W of t such that $\tilde{\gamma}_1(W) \subseteq V_i$ and $\tilde{\gamma}_2(W) \subseteq V_i$. By definition of X this means $W \subseteq X$. Therefore X is open. By (4.1.99) X is also closed. By (5.7.9) then X is either empty or $[a, b]$. However we have $a \in X$ and the liftings must be equal.

Define

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$$L := \left\{ x \in [a,b] \;\middle|\; \exists \tilde{\gamma} : [a,x] \to \tilde{X} \text{ s.t. } p \circ \tilde{\gamma} = \gamma \text{ and } \tilde{\gamma}(a) = \tilde{x}_0 \right\}$$

To show that L is closed consider a sequence $x_n \in L$ such that $x_n \to x \in [a,b]$. Choose $\gamma(x) \in U$ such that $p^{-1}(U) = \bigsqcup_i V_i$ and $p|_{V_i}$ is a homeomorphism. For some ε we have $(x-\varepsilon, x+\varepsilon) \cap [a,b] \subset \gamma^{-1}(U)$, and for some N, $x_N \in (x-\varepsilon, x+\varepsilon) \cap [a,b]$. In particular there is a lift

$$\tilde{\gamma}' : [a, x_N] \to \tilde{X}.$$

and $\tilde{\gamma}'(x_N) \in V_i$ for some i. If $x_N \geq x$ then we are done, otherwise assume $x_N < x$ and define the lift

$$\tilde{\gamma}(t) := \begin{cases} \tilde{\gamma}'(t) & t \leq x_N \\ \left(p|_{V_i}\right)^{-1}(\gamma(t)) & x_N \leq t \leq x \end{cases}$$

Therefore $x \in L$, and L is closed. By (5.7.9) we conclude $L = [a,b]$ and we are done. ∎

Proposition 5.10.14 (Almost uniqueness of polar coordinates)
Let $p : \mathbb{R}_{>0} \times \mathbb{R} \to \mathbb{R}^2 \setminus \{(0,0)\}$ be the covering map of the punctured plane (5.10.12) and $\gamma : [a,b] \to \mathbb{R}^2 \setminus \{(0,0)\}$ a continuous path. Suppose (r_1, θ_1) and (r_2, θ_2) are liftings. Then

$$r_1(t) = r_2(t)$$

$$\theta_1(t) = \theta_2(t) + 2n\pi$$

for some $n \in \mathbb{Z}$.

Proof. By (5.10.10) we have $\theta_2(a) - \theta_1(a) = 2n\pi$. Then $(r_2(t), \theta_2(t) - 2n\pi)$ is also a lifting and so by uniqueness equals $(r_1(t), \theta_1(t))$ as required. ∎

Definition 5.10.15 (Winding Number)
*Let $\gamma : [a,b] \to \mathbb{C} \setminus \{0\}$ be a continuous path. We define the **angle traversed** by the path γ to be*

$$\theta(b) - \theta(a)$$

*where $(r(t), \theta(t))$ is any lifting of γ. Note when γ is a closed path then this equals $2n\pi$ for a unique integer n. We call this the **winding number** of the closed path γ, which we denote by $W(\gamma; 0)$. In general we define*

$$W(\gamma; z_0) := W(\gamma - z_0; 0)$$

for any $z_0 \notin \gamma([a,b])$.

5.10.5 Path Integrals

Definition 5.10.16
*A **curve** is a continuous path $\gamma : [a,b] \to \mathbb{C}$, for which there exists a partition*

$$a = x_0 < x_1 < \cdots < x_{n+1} = b$$

such that

a) γ is C^1 on (x_i, x_{i+1}) for $i = 0 \cdots n-1$
b) γ is left and right differentiable at x_i and

$$\lim_{h \downarrow 0} \frac{\gamma(x_i + h) - \gamma(x_i)}{h} = \lim_{h \downarrow 0} \gamma'(x_i + h)$$

$$\lim_{h \uparrow 0} \frac{\gamma(x_i + h) - \gamma(x_i)}{h} = \lim_{h \uparrow 0} \gamma'(x_i + h)$$

We say γ is closed if in addition $\gamma(b) = \gamma(a)$.

Definition 5.10.17 (Path Integral)
Let $\gamma : [a,b] \to \mathbb{C}$ be a curve supported on $\{x_0, \cdots, x_{n+1}\}$. Define the integral

$$\int_\gamma f(z)\,\mathrm{d}z := \sum_{i=0}^{n} \int_{x_i}^{x_{i+1}} f(\gamma(t))\gamma'(t)\,\mathrm{d}t$$

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Proposition 5.10.18 (Path Integral of a function with primitive)
Let $f : U \to \mathbb{C}$ be a continuous function and $g : U \to \mathbb{C}$ a differentiable function such that $g'(z) = f(z)$. Then for any curve $\gamma : [a, b] \to U$ we have

$$\int_\gamma f(z)\,\mathrm{d}z = g(b) - g(a)$$

In particular the integral over a closed curve is zero.

Proposition 5.10.19
Let $\gamma : [0, 1] \to \mathbb{C}$ the path given by $\gamma(t) \coloneqq Re^{2\pi i t}$. Then evidently $W(\gamma, 0) = 1$. Further

$$\int_\gamma z^n\,\mathrm{d}z = \begin{cases} 2\pi i & n = -1 \\ 0 & n \neq -1 \end{cases}$$

Proof. When $n \neq -1$ then z^n has primitive $\frac{z^{n+1}}{n+1}$ and we are done by the previous result. In the case $n = -1$ we may calculate explicitly

$$\int_\gamma z^{-1}\,\mathrm{d}z = \int_0^1 \frac{\gamma'(t)}{\gamma(t)}\,\mathrm{d}t = 2\pi i$$

■

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5.11 Riemann Surfaces

I develop the elementary theory of Riemann surface, starting with the more standard concept approach of maximal atlases (see [War13]) and then relating to the notion sheaves (see [Wed16] (Section 4.2)). The language of sheaves is well suited to complex analytic spaces and aligns well with the approach in the algebraic case.

Definition 5.11.1
*Let X be a topological space. By definition a **chart** on X is a pair (U, φ) where $U \subset X$ is an open set and φ is a homeomorphism onto an open subset $\varphi(U)$ of $\mathbb{C}$.*

*We say that two charts (U, φ) and (V, ψ) are **compatible** if either $U \cap V = \emptyset$ or the composite map $(\psi \circ \varphi^{-1}) : \varphi(U \cap V) \to \psi(U \cap V)$ is **biholomorphic**.*

*An **atlas** $\mathcal{A}$ on X is a collection of charts $\{(U_\alpha, \varphi_\alpha)\}$ for which $X = \bigcup_\alpha U_\alpha$ and every pair of charts is compatible.*

*We say two atlases $\mathcal{A}, \mathcal{A}'$ are **equivalent** if $\mathcal{A} \cup \mathcal{A}'$ is an atlas.*

Proposition 5.11.2
The notion of compatible charts is an equivalence relation. Similarly for the notion of equivalent atlases.

A union of equivalent atlases is again an atlas, and every atlas is contained in a unique maximal atlas.

We may then use the charts to define what it means for a function on a manifold to be holomorphic.

Lemma 5.11.3
Let X be a topological space, $U \subset X$ an open subset, $f : U \to \mathbb{C}$ a function and (V, φ), (V', ψ) two compatible charts such that $U \cap V \cap V' \neq \emptyset$.

Then $f \circ \varphi^{-1}|_{\varphi(U \cap V \cap V')}$ is holomorphic if and only if $f \circ \psi^{-1}|_{\psi(U \cap V \cap V')}$ is holomorphic.

Lemma 5.11.4
Let X be a topological space, $U \subset X$ an open subset, $f : U \to \mathbb{C}$ a function and $(V_\alpha, \varphi_\alpha)_{\alpha \in \mathcal{A}}$ a family of charts such that $U \subset \bigcup_{\alpha \in \mathcal{A}} V_\alpha$. Further suppose that (V, φ) is a chart such that $U \cap V \neq \emptyset$ and is compatible with every $(V_\alpha, \varphi_\alpha)$. Then the following are equivalent

a) $f \circ \varphi^{-1}|_{\varphi(V \cap U)}$ is holomorphic
b) $f \circ \varphi_\alpha^{-1}|_{\varphi_\alpha(V_\alpha \cap U \cap V)}$ is holomorphic for all $\alpha \in \mathcal{A}$ for which $V_\alpha \cap V \neq \emptyset$

Proposition 5.11.5 (Holomorphic Function)
Let X be a topological space, $U \subset X$ an open subset, $f : U \to \mathbb{C}$ a function and $\mathcal{A}$ a given atlas. Then the following are equivalent

a) there exists a collection of charts $\{(V_\alpha, \varphi_\alpha)\} \subset \mathcal{A}$ such that $V_\alpha \cap U \neq \emptyset$, $f \circ \varphi_\alpha^{-1}|_{\varphi_\alpha(V_\alpha \cap U)}$ is holomorphic and $U \subset \bigcup_\alpha V_\alpha$
b) for every chart $(V, \varphi) \in \mathcal{A}$ such that $U \cap V \neq \emptyset$ the function $f \circ \varphi^{-1}|_{\varphi(U \cap V)}$ is holomorphic.

*When this holds we say that f is **holomorphic** with respect to $\mathcal{A}$.*

Proof. $b) \implies a)$ Straightforward

$a) \implies b)$ As every chart (V, φ) is compatible with the collection $\{(V_\alpha, \varphi_\alpha)\}$ this follows from (5.11.4). ∎

Proposition 5.11.6
A holomorphic function $f : U \to \mathbb{C}$ is continuous.

Proof. For every chart (V, φ) the function $f \circ \varphi^{-1}|_{\varphi(U \cap V)}$ is holomorphic (5.11.5) and therefore continuous. Further by definition the map $\varphi|_{U \cap V}$ is continuous whence $f|_{U \cap V}$ is continuous (4.1.28). As the charts cover X we are done by (4.1.29). ∎

We first prove that this definition does not depend on which atlas is chosen.

Proposition 5.11.7
Let X be a topological space, $U \subset X$ an open subset, $f : U \to \mathbb{C}$ a function and $\mathcal{A}, \mathcal{A}'$ two equivalent atlases.

Then f is holomorphic with respect to $\mathcal{A}$ if and only if it is holomorphic with respect to $\mathcal{A}'$.

Proof. Since equivalence of atlases is symmetric we only need to show one direction. So suppose that f is holomorphic with respect to $\mathcal{A}$, and let $(W, \psi) \in \mathcal{A}'$ be a chart. Then by (5.11.4) $f \circ \psi^{-1}$ is holomorphic, so that f is holomorphic with respect to $\mathcal{A}'$. ∎

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Further we show that just as in the usual case this is a local condition.

Proposition 5.11.8 (Structure Sheaf associated to an Atlas)
Let X be a topological space together with an atlas $\mathcal{A}$. Let $U = \bigcup_{i\in I} U_i$ an open cover and $f : U \to \mathbb{C}$ a function. Then f is holomorphic (with respect to $\mathcal{A}$) if and only if $f|_{U_i}$ is holomorphic for all $i \in I$.

In particular for every atlas the mapping

$$\mathcal{O}_X(U) \coloneqq \{f : U \to \mathbb{C} \mid f \text{ holomorphic } \}$$

*determines a sheaf of $\mathbb{C}$-algebras, and the pair $(X, \mathcal{O}_X)$ constitutes a space of $\mathbb{C}$-functions, which we call the **structure sheaf**. It contains an isomorphic copy of $\mathbb{C}$ consisting of the constant functions.*

Two atlases are equivalent if and only if they determine the same structure sheaf.

A space of functions $(X, \mathcal{O}_X)$ arises in this way if and only if there is an open cover $X = \bigcup_{i\in I} U_i$ such that $\left(U_i, \mathcal{O}_X|_{U_i}\right) \xrightarrow{\sim} \left(V_i, \mathcal{O}_{V_i}\right)$ for $V_i \subset \mathbb{C}$ an open subset with the usual sheaf of holomorphic functions.

Proof. Suppose that $(V, \varphi) \in \mathcal{A}$ is a chart such that $V \cap U_i \neq \emptyset$. Then by assumption $f \circ \varphi^{-1}|_{\varphi(V\cap U)}$ is holomorphic whence by (...) $f|_{U_i} \circ \varphi^{-1}|_{\varphi(V\cap U_i)}$ is holomorphic. Since the chart (V, φ) was arbitrary we conclude $f|_{U_i}$ is holomorphic with respect to the atlas $\mathcal{A}$.

Conversely suppose $f|_{U_i}$ is holomorphic with respect to $\mathcal{A}$ for all $i \in I$. Consider an arbitrary chart $(V, \varphi) \in \mathcal{A}$ such that $V \cap U \neq \emptyset$. Then $V \cap U = \bigcup_{i\in I} V \cap U_i$, where we may restrict to open subsets with $V \cap U_i \neq \emptyset$. Then by assumption $f|_{U_i} \circ \varphi^{-1}|_{\varphi(V\cap U_i)} = \left(f \circ \varphi^{-1}|_{\varphi(V\cap U)}\right)|_{\varphi(V\cap U_i)}$ is holomorphic. By (...) $f \circ \varphi^{-1}|_{\varphi(V\cap U)}$ is holomorphic. As the chart was arbitrary we conclude that f is holomorphic with respect to $\mathcal{A}$.

The holomorphic functions form a $\mathbb{C}$-algebra by (5.5.4), and in particular contain the constant functions. Further $D(f) = X \setminus f^{-1}(\{0\})$ which is an open subset of X, and $\left(f|_{D(f)}\right)^{-1}$ is holomorphic whenever f is not identically zero.

By definition then $(X, \mathcal{O}_X)$ is a space of $\mathbb{C}$-functions. By (5.11.7) an equivalent atlas determines the same structure sheaf. Conversely suppose $\mathcal{A}, \mathcal{A}'$ determine the same structure sheaves. Consider two atlases $(V, \varphi) \in \mathcal{A}$ and $(V', \varphi') \in \mathcal{A}'$. Then by definition we see that $\varphi \circ (\varphi')^{-1}$ is holomorphic and $(\varphi') \circ \varphi^{-1}$ is holomorphic, and therefore they are compatible as required. ∎

Definition 5.11.9 (Riemann Surface)
Let $(X, \mathcal{O}_X)$ be a space of $\mathbb{C}$-functions, which is Hausdorff and second countable. Further suppose there is an open cover $X = \bigcup_{i\in I} U_i$ such that there exists isormorphisms

$$\varphi_i : \left(U_i, \mathcal{O}_X|_{U_i}\right) \xrightarrow{\sim} \left(V_i, \mathcal{O}_{V_i}\right)$$

*for $V_i \subset \mathbb{C}$ an open subset. Then we say X is a **Riemann Surface**.*

We say $\mathcal{A}$ is a compatible atlas if it determines the same structure sheaf. Further there exists a unique compatible maximal atlas (5.11.8).

Then $\varphi : U \to \mathbb{C}$ is a chart for X if it lies in at least one compatible atlas (equivalently is in the unique maximal atlas).

5.11.1 Morphisms of Riemann Surfaces

Proposition 5.11.10 (Holomorphic maps of Riemann Surfaces)
Let $f : X \to Y$ be a continuous map of Riemann surfaces with given compatible atlases $\mathcal{A}_1, \mathcal{A}_2$. Then the following are equivalent

a) $g \in \mathcal{O}_Y(V) \implies g \circ f \in \mathcal{O}_X(f^{-1}(V))$
b) For all charts $(V, \psi) \in \mathcal{A}_2$ the function $\psi \circ f|_{f^{-1}(V)}$ is holomorphic
c) For all charts $(U, \varphi) \in \mathcal{A}_1$, $(V, \psi) \in \mathcal{A}_2$ for which $U \cap f^{-1}(V) \neq \emptyset$ we have

$$\psi \circ f \circ \varphi^{-1} : \varphi(U \cap f^{-1}(V)) \to \mathbb{C}$$

is holomorphic

*This definition is independent of the choice of atlases, and we say in this case that f is **holomorphic**.*

Proof. *a)* $\implies$ *c)* As $\psi \in \mathcal{O}_Y(V)$ is trivially holomorphic we have by assumption $\psi \circ f \in \mathcal{O}_X(f^{-1}(V)) \implies (\psi \circ f)|_{U\cap f^{-1}(V)} \in \mathcal{O}_X(U \cap f^{-1}(V))$. By definition this means $\psi \circ f \circ \varphi^{-1}$ is holomorphic.

c) $\implies$ *b)* By (5.11.5) $\psi \circ f|_{f^{-1}(V)}$ is holomorphic

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b) $\Longrightarrow$ *a)* Let $(V_\alpha, \psi_\alpha) \in \mathcal{A}_2$ be charts for which $V \subset \bigcup_\alpha V_\alpha$ and $V_\alpha \cap V \neq \emptyset$. Then

$$(g \circ f)|_{f^{-1}(V_\alpha)} = (g \circ \psi_\alpha^{-1}) \circ (\psi_\alpha \circ f)|_{f^{-1}(V_\alpha)}$$

is by definition holomorphic. Since this is a local condition (5.11.8) we conclude that $g \circ f$ is holomorphic as required.

The last statement follows because *a)* is independent of the choice of compatible atlases (5.11.8). ∎

5.11.2 Tangent and Cotangent Space

Definition 5.11.11 (Local Ring)
Let X be a Riemann surface and $x \in X$ a point. Then define the local ring

$$\mathcal{O}_{X,x} \coloneqq \varinjlim_{x \in U} \mathcal{O}_X(U)$$

to be the $\mathbb{C}$-algebra of germs at x. Every germ $\boldsymbol{f}$ has a well-defined value at x which we denote by $\boldsymbol{f}(x)$. The unique maximal ideal $\mathfrak{m}_{X,x}$ consists of those germs which vanish at x. Further the residue field $\mathbb{C}(\mathfrak{m}_{X,x})$ is isomorphic to $\mathbb{C}$ (that is, $\mathbb{C} \to \mathcal{O}_{X,x} \to \mathcal{O}_{X,x}/\mathfrak{m}_{X,x}$ is an isomorphism).

Proposition 5.11.12 (Cotangent Space is Finite-Dimensional)
The local ring $\mathcal{O}_{X,x} = \mathbb{C} \oplus \mathfrak{m}_{X,x}$ and there is a canonical isomorphism

$$\mathrm{Der}(\mathcal{O}_{X,x}, \mathbb{C}) \xrightarrow{\sim} \mathrm{Hom}_{\mathbb{C}}(\mathfrak{m}_{X,x}/\mathfrak{m}_{X,x}^2, \mathbb{C})$$

*For every $f \in \mathcal{O}_X(U)$ and $x \in U$ denote by $\mathrm{d}f(x)$ the image of $(f - f(x))$ in the **cotangent space** $T_x^\star X \coloneqq \mathfrak{m}_{X,x}/\mathfrak{m}_{X,x}^2$.*

This has dimension 1, more explicitly for any open chart (U, φ) around x, $0 \neq \mathrm{d}\varphi(x)$ is a basis for $\mathfrak{m}_{X,x}/\mathfrak{m}_{X,x}^2$.

We denote the inverse image of the dual basis $(\mathrm{d}\varphi(x))^\vee$ by $\left(\frac{\partial}{\partial \varphi}\right)|_x$ (which is therefore a basis for $\mathrm{Der}(\mathcal{O}_{X,x}, \mathbb{C})$).

Proof. Every germ $\boldsymbol{f}$ may be written as $\boldsymbol{f} = \boldsymbol{f}(x) + (\boldsymbol{f} - \boldsymbol{f}(x))$. This is unique decomposition by considering evaluation at x, which gives the first statement. The isomorphism follows the same lines as were proved for affine algebras that tangent space is dual to cotangent space (see (3.32.50)). Explicitly the map is given by $D(\mathrm{d}f(x)) \coloneqq Df_x$.

Let (U, φ) be a chart and define the holomorphic function $z(x') \coloneqq \varphi(x') - \varphi(x)$. For every $f \in \mathcal{O}_X(U)$ define $\hat{f} \coloneqq f \circ \varphi^{-1}$. Then by Taylor's theorem (5.9.29)

$$\hat{f}(z') = \hat{f}(\varphi(x)) + (z' - \varphi(x))\hat{f}'(\varphi(x)) + (z' - \varphi(x))^2 R_2(z')$$

where $R_2(z')$ is a holomorphic function on $\varphi(U)$. Therefore

$$f(x') = f(x) + z(x')(f \circ \varphi^{-1})'(\varphi(x)) + z(x')^2 (R_2 \circ \varphi)(x')$$

where $R_2 \circ \varphi$ is a holomorphic function on U. Therefore if $f(x) = 0$ we have $f_x - (f \circ \varphi^{-1})'(\varphi(x)) z_x \in \mathfrak{m}_{X,x}^2$ and $\mathrm{d}f(x) = (f \circ \varphi^{-1})'(x)\,\mathrm{d}z(x)$. In other words $\mathrm{d}z(x) = \mathrm{d}\varphi(x)$ is spanning.

Observe that $\hat{z}'(x) = 1$. We claim that for any f such that $f_x \in \mathfrak{m}_{X,x}^2$ we have $\hat{f}'(x) = 0$, which will show that $z_x \notin \mathfrak{m}_{X,x}^2$ and therefore $\mathrm{d}z(x) = \mathrm{d}\varphi(x) \neq 0$.

For in this case $\hat{f} = \sum_{i=1}^n \hat{g}_i \hat{h}_i$ when restricted to some open neighbourhood V, with $\hat{g}_i(\varphi(x)) = \hat{h}_i(\varphi(x)) = 0$. By the product rule $\hat{f}'(\varphi(x)) = 0$ as required. ∎

Definition 5.11.13
*Let X be a Riemann Surface and $x \in X$. We define the **tangent space** to be the $\mathbb{C}$-vector space of derivations*

$$T_x X \coloneqq \mathrm{Der}(\mathcal{O}_{X,x}, \mathbb{C})$$

By (5.11.12) it has dimension 1, more explicitly given any chart (U, φ) around x it has basis $\left(\frac{\partial}{\partial \varphi}\right)|_x$ the unique derivation such that

$$\left(\frac{\partial}{\partial \varphi}\right)|_x \ \varphi_x = 1$$

Lemma 5.11.14 (Derivations are Directional Derivatives)
Let $f \in \mathcal{O}_X(U)$, $x \in U$ and (V, φ) a chart around x, then

$$\left(\frac{\partial}{\partial \varphi}\right)|_x \ f_x = (f \circ \varphi^{-1})'(\varphi(x))$$

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In particular

$$\mathrm{d}f(x) = \left(f \circ \varphi^{-1}\right)'(\varphi(x))\,\mathrm{d}\varphi(x)$$

Proof. As the derivative of a function depends only on its germ, the right hand side is a well-defined derivation on $\mathcal{O}_{X,x}$, by the usual rules of differentiation (5.5.4). Further $(\varphi \circ \varphi^{-1})(z) = z$ which has derivative 1 and so both sides are equal when evaluated at $f_x = \varphi_x$. By uniqueness (5.11.13) then they are equal. ∎

Corollary 5.11.15 (Change of Variables Formula)
Let X be a Riemann Surface and (U, φ), (V, ψ) two charts around x. Then

$$\mathrm{d}\varphi(x) = \left(\varphi \circ \psi^{-1}\right)'(\psi(x))\,\mathrm{d}\psi(x)$$

and dually

$$\left(\frac{\partial}{\partial \psi}\right)|_x = \left(\varphi \circ \psi^{-1}\right)'(\psi(x))\left(\frac{\partial}{\partial \varphi}\right)|_x$$

Proof. By (5.11.12) both $\mathrm{d}\varphi(x)$ and $\mathrm{d}\psi(x)$ form a basis for $T_x^\star X$, whence $\mathrm{d}\varphi(x) = \lambda\,\mathrm{d}\psi(x)$ for some $\lambda \in K^\star$. Therefore $\varphi_x - \lambda\psi_x \in \mathbb{C} + \mathfrak{m}_{X,x}^2$. By (...)

$$0 = \left(\frac{\partial}{\partial \psi}\right)|_x\ (\varphi_x - \lambda\psi_x) = \left(\varphi \circ \psi^{-1}\right)'(\psi(x)) - \lambda$$

as required. ∎

5.11.3 Holomorphic 1-forms

Lemma 5.11.16
Let X be a Riemann surface, $U \subset X$ an open subset and $\omega : U \to \coprod_{x\in U} T_x^\star X$ a function satisfying $\omega(x) \in T_x^\star X$. Then for every chart (V, φ) such that $U \cap V \neq \emptyset$

$$\omega(x) = \omega_\varphi(x)\,\mathrm{d}\varphi(x) \quad \forall x \in U \cap V$$

$$\omega_\varphi(x) := \left(\frac{\partial}{\partial \varphi}\right)|_x\ \omega(x)$$

Suppose that (W, ψ) is another chart such that $U \cap V \cap W \neq \emptyset$ then

$$\omega_\psi(x) = \left(\varphi \circ \psi^{-1}\right)'(\psi(x))\omega_\varphi(x) \quad \forall x \in U \cap V \cap W$$

Definition 5.11.17 (Holomorphic 1-form)
*Let X be a Riemann surface, $U \subset X$ an open subset and $\mathcal{A}$ a compatible atlas. We say a function $\omega : U \to \coprod_{x\in U} T_x^\star X$ is a **holomorphic 1-form** if*

a) $\omega(x) \in T_x^\star X$ for all $x \in U$
b) for every chart $(V, \varphi) \in \mathcal{A}$ such that $U \cap V \neq \emptyset$, the function $\omega_\varphi : U \cap V \to \mathbb{C}$ is holomorphic

Lemma 5.11.18
The definition of holomorphic 1-form (5.11.17) is independent of the choice of atlas.

Proof. Suppose ω is a holomorphic 1-form on U with respect to the atlas $\mathcal{A}$. Let $(V_i, \varphi_i) \in \mathcal{A}$ and (W, ψ) be compatible charts such that $U \cap V_i \cap W \neq \emptyset$. Then by (5.11.16) and (5.11.15) for all $x \in U \cap V_i \cap W$

$$\omega_\psi(x) = \left(\varphi_i \circ \psi^{-1}\right)'(\psi(x))\omega_{\varphi_i}(x)$$

whence by definition $\omega_\psi \circ \varphi_i^{-1} : \varphi_i(U \cap V_i \cap W) \to \mathbb{C}$ is holomorphic. Then by (5.11.5) ω_ψ is holomorphic on $U \cap W$. Therefore if $\mathcal{A}'$ is a compatible atlas we find ω is holomorphic with respect to $\mathcal{A}'$. By symmetry we see that the definition is independent of the choice of atlas. ∎

Proposition 5.11.19
Let X be a Riemann surface and $f \in \mathcal{O}_X(U)$. Then the function

$$\begin{aligned} \mathrm{d}f : U &\longrightarrow \coprod_{x\in U} T_x^\star U \\ x &\longrightarrow \mathrm{d}f(x) \end{aligned}$$

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is a holomorphic 1*-form on* U*. Moreover for any chart* (U, φ) *we have*

$$\mathrm{d}f(x) = \left(\partial \frac{f}{\partial \varphi}\right)|_x \, \mathrm{d}\varphi(x) = \left(f \circ \varphi^{-1}\right)'(\varphi(x)) \, \mathrm{d}\varphi(x)$$

Proposition 5.11.20
Let X *be a Riemann surface. For every open subset* $U \subset X$ *define*

$$\Omega^1_X(U) \coloneqq \left\{ \omega : U \to \coprod_{x \in U} T^\star_x X \;\middle|\; \omega \text{ holomorphic } \ 1\text{-form} \right\}$$

Then Ω^1_X *is a locally-free* $\mathcal{O}_X$*-module of rank* 1*, more precisely for every chart* (U, φ) *we have* $\Omega^1_X|_U$ *has global basis* $\mathrm{d}\varphi|_U$*.*

Proof. Evidently $\Omega^1_X(U)$ is a vector space over $\mathbb{C}$, as $\mathcal{O}_X(U)$ forms a $\mathbb{C}$-algebra. Further for $f \in \mathcal{O}_X(U)$ we find $f\omega$ is a holomorphic 1-form and $(f\omega)|_V = f|_V \, \omega|_V$, so that Ω^1_X is an $\mathcal{O}_X$-module. For every chart (U, φ) and open subset $V \subset U$ we have an isomorphism of $\mathcal{O}_X(V)$-modules

$$\begin{aligned} \mathcal{O}_X(V) &\xrightarrow{\sim} \Omega^1_X(V) \\ f &\to f \, \mathrm{d}\varphi|_U \\ \omega_\varphi &\leftarrow \omega \end{aligned}$$

whence an isomorphism

$$\mathcal{O}_X|_U \xrightarrow{\sim} \Omega^1_X|_U$$

■

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